THE DRASIL FRAMEWORK

SUCCINCTLY VERBOSE: THE DRASIL FRAMEWORK

BY

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Lay Abstract

Scientific software is supported by more than source code. Requirements, design documents, equations, tests, and build instructions often repeat the same facts in different forms. When people update each artifact separately, the copies can disagree or important details can be omitted. This thesis presents Drasil, a framework that stores scientific and engineering knowledge once and uses it to generate consistent software and documentation. Drasil represents concepts, symbols, units, equations, and relationships as reusable "chunks," then uses "recipes" to select and arrange them for different audiences. Reimplementing six scientific-computing case studies showed that corrections made in one place propagated through generated artifacts, hidden assumptions and inconsistencies became easier to detect, and closely related software variants could reuse shared knowledge. The approach also has costs: knowledge must be encoded carefully, and new contributors face a learning curve. These results motivate better authoring tools and broader empirical validation.

Abstract

Scientific software projects rely on collections of related artifacts, including requirements specifications, design documents, source code, tests, and build instructions. These artifacts present overlapping knowledge to different audiences, but maintaining independently editable representations introduces unnecessary redundancy and creates opportunities for omissions, inconsistency, and loss of traceability. This thesis investigates how a knowledge-centric approach can reduce that maintenance burden while preserving the distinct forms that software artifacts require.

The work first analyzes artifacts from scientific-computing case studies to identify recurring structures, shared domain knowledge, and repeated projections of the same information. These observations inform Drasil, a framework that represents this knowledge (concepts, definitions, symbols, units, equations, and their relationships) as reusable, typed chunks. Artifact recipes select and organize those chunks, while deterministic generators render the resulting structures as documentation, source code, and supporting files.

Reimplementations of six case studies provide observational evidence that this single-source approach enforces consistency across generated artifacts, preserves explicit provenance, and supports reproducible generation. Encoding knowledge explicitly also exposed missing definitions, inconsistent symbols, implicit conversions, and

undocumented assumptions in the original artifacts. Related systems reused shared knowledge to produce software-family variants with targeted changes. These benefits require greater up-front modeling effort and impose an onboarding cost on new contributors. The thesis therefore establishes the feasibility and practical value of knowledge-centric artifact generation while identifying improved authoring support, broader domain coverage, and controlled empirical validation as priorities for future work.

Your Dedication

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Acknowledgements

Acknowledgements go here.

Contents

List of Figures

List of Tables

Notation, Definitions, and Abbreviations

Notation

K, L, M	Sets used illustratively to denote chunks of related knowledge. This set notation supports the conceptual discussion in Chapter 3; it is not a formal model of Drasil's chunk types.
$p(\cdot)$	A projection function that selects and presents knowledge for a particular artifact, audience, or context.
$p(K) \equiv K$	An identity projection, which presents the full knowledge represented by K, possibly using different notation or language.
$p(K) \subset K$	A non-identity projection, which intentionally omits some details represented by K.
input $\rightarrow$ ***process*** $\rightarrow$ ***output***	The computational pattern that defines the primary class of scientific software considered in this thesis.

Definitions

Knowledge Information that has been structured and contextualized to be meaningful and useful. It includes facts, concepts, definitions, theories, and relationships relevant to a domain or problem.

Knowledge chunk

A reusable, uniquely identifiable representation of related knowledge. Chunks may encode names, definitions, symbols, units, expressions, or relationships and may be composed into richer structures.

Knowledge projection

A context-specific presentation of selected knowledge for a particular artifact or audience.

Recipe An intermediate specification that selects knowledge and defines how it is organized in one or more generated softifacts.

Softifact A portmanteau of "software" and "artifact." The term refers to any artifact produced during software development that conveys knowledge, including documentation, source code, test cases, and build instructions.

Unnecessary redundancy

Multiple independently editable, authoritative representations of the same knowledge. Repeated renderings derived from one source are not considered unnecessary redundancy.

Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
AST	Abstract Syntax Tree
CBSE	Component-Based Software Engineering
CI/CD	Continuous Integration and Continuous Delivery
CORBA	Common Object Request Broker Architecture
CSS	Cascading Style Sheets
DD	Data Definition
DSL	Domain-Specific Language
GOOL	Generic Object-Oriented Language
HGHC	Drasil's minimal heat-transfer case study involving the coefficients h_g and h_c
HTML	Hypertext Markup Language
IDE	Integrated Development Environment
IM	Instance Model
IPO	Input–Process–Output
JSON	JavaScript Object Notation

LLM	Large Language Model
LOC	Lines of Code
LP	Literate Programming
LSD	Literate Software Development
MDE	Model-Driven Engineering
MG	Module Guide
MIS	Module Interface Specification
ML	Machine Learning
NFL	Non-Factored Load
NoPCM	Solar Water Heating System with No Phase-Change Material
ODE	Ordinary Differential Equation
OS	Operating System
PCM	Phase-Change Material
PDF	Portable Document Format
QA	Quality Assurance
SC	Scientific Computing
SI	International System of Units
SPL	Software Product Line

SRS	Software Requirements Specification
SSP	Slope Stability Analysis Program
SWHS	Solar Water Heating System
TM	Theoretical Model
WYSIWYG	What You See Is What You Get

Declaration of Academic Achievement

This thesis was completed by Daniel M. Szymczak under the supervision of Jacques Carette and Spencer Smith. The principal original contributions of the author are the analysis, design, implementation, and evaluation presented here for the Drasil framework as a knowledge-centric approach to generating software artifacts from a centralized representation of domain knowledge. In particular, the author carried out the systematic analysis of redundancy and inconsistency across scientific software artifacts, developed the thesis account of Drasil's chunk- and recipe-based architecture, implemented substantial portions of the framework and its supporting representations, and reimplemented the case studies used in this thesis within Drasil to evaluate the approach.

This thesis also draws on prior and collaborative work. The broader Drasil project is part of an ongoing research effort, and the author's work should not be taken to mean that all ideas, infrastructure, or repository contents originated solely with the author. In addition, most of the case studies discussed in this thesis were originally developed by other teams at McMaster University; the author's contribution in this thesis is their analysis and re-expression within Drasil, rather than authorship of all original source artifacts for those systems. Although Drasil uses GOOL as an intermediate layer for code generation, GOOL itself was not developed by the author

and is not claimed as a contribution of this thesis. Where prior templates, existing artifacts, supervisory guidance, or contributions from other project members informed the work, they are acknowledged as part of the broader research context so as to avoid misattribution.

Chapter 1

Introduction

There is near universal agreement on the value of documentation [1, 3, 14, 23, 24, 25, 30, 31, 32, 36, 37, 38, 39, 40, 41, 42, 43, 45, 57, 58, 59, 64, 68, 69, 71, 72, 75, 76, 78, 79, 80, 81, 85, 90, 95]. Despite this, it is not often prioritized on software projects. Many projects focus heavily on source code, treating it as the primary artifact, while documentation is often neglected despite its necessary role. Documentation serves as a critical bridge between stakeholders, enabling effective communication, knowledge transfer, and long-term maintainability. It helps clarify requirements, supports onboarding of new team members, facilitates regulatory compliance, and provides a reference for future maintenance and evolution. Well-maintained documentation reduces the risk of misunderstandings, errors, and knowledge loss, ultimately contributing to higher quality and more reliable software systems.

To simplify our discussion of both source code and documentation, this thesis introduces the term softifact to refer to any artifact produced during software development that conveys knowledge, including source code, requirements, design documents, test cases, user manuals, and any other relevant materials. Code and other

software artifacts (softifacts) by their nature must say the same thing, albeit to different audiences (i.e. developers, maintainers, regulators, or end users); if they did not, they would be describing different systems. This principle applies regardless of the context of the softifact. Take this thesis for example: it is written for an audience familiar with classical software engineering, including those who have experience with requirements documents, design documents, and the challenges of maintaining large, long-lived software systems. The examples and case studies are drawn from scientific computing, but the pain points and solutions discussed are relevant to wider software engineering contexts as a whole.

Speaking broadly about softifacts, let us consider a few different documents. A software requirements document is a human-readable abstraction of *what* the software is supposed to do. Whereas a design document is a human-readable version of *how* the software is supposed to fulfill its requirements. The source code itself is a computer-readable list of instructions combining *what* must be done and, in many languages, *how* that is to be accomplished. Importantly, this multiplicity of views is not limited to separate softifacts; even within a single artifact the same information may appear in multiple forms. For example, a requirements document may include natural language descriptions, mathematical equations, and other relationships expressed in a multitude of ways (tables, graphs, diagrams, etc.).

Table 1.1: The same information as natural language, equation, and code

Description	**Equation**	**Code (Python)**
P_b is the Probability of Breakage	$P_b = 1 - e^{-B}$	`def func_P_b(B):` `  return 1.0 - math.exp(-B)`

Table 1.1 shows how the same information (the definition of P_b) can be represented in several different views: as a natural language description, equation, and code definition. We aim to take advantage of the inherent redundancy across these views to distill a single source of information, thus removing the need to manually duplicate information across softifacts. This example illustrates a broader challenge: when the same knowledge must be expressed in multiple forms for different audiences or purposes, the effort to keep these representations consistent becomes significant.

Manually writing and maintaining a full range of softifacts (i.e. multiple documents for different audiences plus the source code) is redundant and tedious. Factor in deadlines, changing requirements, and other common issues faced during development and you have a perfect storm for inter-artifact syncronization issues.

How can we avoid having our artifacts fall out of sync with each other? Some would argue “just write code!" And that is exactly what a number of other approaches have tried. Documentation generators like Doxygen, Javadoc, Pandoc, and more take a code-centric view of the problem. Typically, they work by having natural-language descriptions and/or explanations written as specially delimited comments in the code which are later automatically compiled into a human-readable document.

While these approaches definitely have their place and can come in quite handy, they do not solve the underlying redundancy problem. The developers are still forced to manually write descriptions of systems in both code and comments. They also do not generate all softifacts—commonly, they are used to generate only application programming interface (API) documentation targeted toward developers or user manuals.

If we can truly understand the differences and commonalities between softifacts,

we can develop tools to capitalize on them and streamline the process of creating and maintaining them. To accomplish this, we first need to break down common softifacts to develop an understanding of their contents, intended audience, and how they differ whether on intra- or inter-project scales.

We propose a new framework, Drasil[1], alongside a knowledge-centric view of software, to help take advantage of inherent redundancy while reducing manual duplication and synchronization problems. Our approach looks at what underlies the problems we solve using software and capturing that “common" or “core" knowledge. We then use that knowledge to generate our softifacts, thus gaining the benefits inherent to the generation process: lack of manual duplication, one source to maintain, and “free" traceability of information.

1.1 Problem Statement and Motivation

Despite the recognized value of documentation and softifacts, the process of creating and maintaining them is fraught with redundancy, tedium, and potential risks of inconsistency. Change is inevitable in software projects: requirements evolve, bugs are fixed, and features are added. As software evolves, artifacts (requirements documents, design documents, source code, etc.) often fall out of sync, leading to errors, increased maintenance costs, and reduced trust in documentation. This thesis addresses the challenge: *How can we systematically reduce unnecessary redundancy and improve consistency across softifacts, especially in domains where correctness and traceability are critical?*

[1]The name Drasil comes from a shortening of the name Yggdrasil from Norse mythology, as we believe Drasil can span the many worlds (domains) of software engineering.

The consequences of these challenges are particularly acute in scientific computing and other domains where correctness, traceability, and regulatory compliance are essential. In such contexts, even minor inconsistencies can undermine confidence in the software and its documentation, potentially leading to costly errors or rework.

This thesis therefore investigates a knowledge-centric approach that captures the core information underlying a system and uses it to generate the relevant softifacts. The goal is to reduce manual duplication while improving consistency, maintainability, and traceability across the software lifecycle.

1.2 The Mundane and Why It Matters

What we are doing is not big and flashy; instead, it is focused on the tedium of day-to-day software engineering work. Much of this thesis is motivated by the small but recurring annoyances that consume time and attention: updating multiple artifacts after a single change, tracking down mismatches between documents and code, or fixing errors caused by duplicated information.

These little pain points, including having to manually update softifacts after modifying code, especially in regulated industries, accumulate over time and can have a significant impact on productivity and morale. While each instance may seem minor, together they represent a substantial barrier to efficient and reliable software development, particularly in domains where correctness and traceability are essential.

The contrast here is important. By "mundane" we mean the recurring work of keeping softifacts aligned, propagating changes consistently, maintaining interfaces, and preserving traceability. By contrast, the "non-mundane" work often lies in the domain-specific numerical methods themselves: designing models, selecting solvers, or

implementing and optimizing specialized algorithms. Drasil does not need to generate every line of that code. In many cases, it is more appropriate to rely on existing numerical libraries and use Drasil to generate the surrounding softifacts and the code that interfaces with those libraries.

The value of this approach is most apparent in the small details: the hours saved by not having to update documentation by hand, the confidence that comes from knowing all softifacts are synchronized, and the ability to trace the origin of any piece of information. These improvements may not be as immediately dramatic as solving a major technical challenge, but they accumulate to create a more reliable, maintainable, and satisfying development experience. In this way, Drasil demonstrates that addressing the mundane is not only worthwhile, but essential for advancing the practice of software engineering.

1.3 Scope

We are well aware of the ambitious nature of attempting to solve the problem of manual duplication and unnecessary redundancy across all possible software systems. Frankly, it would be highly impractical to attempt to solve such a broad spectrum of problems. Each software domain poses its own challenges, alongside specific benefits and drawbacks.

Accordingly, this thesis focuses on the generation of source code, requirements, and build instructions. Although Drasil is extensible and can be adapted to generate additional artifact types, those artifacts are outside the scope of the present work.

Our work on Drasil is most relevant to software that is well-understood [13] and undergoes frequent change (maintenance). Good candidates for development using

Drasil are long-lived (10+ years) software projects with artifacts of interest to multiple stakeholders. With that in mind, we have decided to focus on Scientific Computing (SC) software. Specifically, we are looking at software that follows the pattern *input* → *process* → *output*. This focus allows us to make the initial implementation of Drasil feasible and tractable, while still capturing a broad class of scientific computing problems.

SC software has a strong fundamental underpinning of well-understood concepts. It also has the benefit of seldomly changing, and when it does, existing models are not necessarily invalidated. For example, rigid-body problems in physics are well-understood and the underlying modeling equations are unlikely to change. However, should they change, the current models will likely remain as good approximations under a specific set of assumptions. For instance, who hasn't heard "assume each body is a sphere" during a physics lecture?

SC software could also benefit from buy-in to good software development practices as many SC software developers put the emphasis on science and not development [38]. Rather than following rigid, process-heavy approaches deemed unfavourable [14], developers of SC software choose to use knowledge acquisition driven [37], agile [1, 14, 23, 73], or amethododical [36] processes instead.

Finally, while recent advances in machine learning (ML), large language models (LLMs), and other artificial intelligence (AI)-based tooling have shown promise in automating aspects of software engineering, Drasil intentionally does not employ these approaches. This is because our focus is on achieving determinism and reproducibility: properties that are essential in scientific computing. The artifacts generated by Drasil must be consistent and repeatable, which is not guaranteed by current AI-based

methods.

1.4 Roadmap

The remainder of this thesis is organized as follows:

- **Chapter 2: Background** reviews relevant literature on softifacts, redundancy, reuse, literate programming, and generative programming.
- **Chapter 3: Our Process** details the methods used to analyze softifacts, identify redundancy, and motivate our knowledge-centric approach.
- **Chapter 4: The Drasil Framework** presents the design, architecture, and implementation of Drasil, including its realization as a collection of lightweight domain-specific languages (DSLs) embedded in Haskell for knowledge capture and artifact generation.
- **Chapter 5: Results** presents observations and analysis from reimplementing existing case studies using Drasil, highlighting practical benefits.
- **Chapter 6: Discussion** interprets the results, explores the implications of the knowledge-centric approach, and discusses limitations and challenges encountered.
- **Chapter 7: Conclusion** summarizes the findings, contributions, and implications of this work and outlines potential extensions, open questions, and directions for further research.

1.5 Contributions

The main contributions of this thesis are:

- **Systematic analysis of redundancy in softifacts and its operationalization as knowledge-centric generation.** Through detailed breakdowns of softifacts across multiple scientific computing case studies, this work identifies patterns of unnecessary duplication and inconsistency and formalizes a process for capturing, organizing, and projecting domain knowledge to enable automated artifact generation from a single source of truth.

- **Design and implementation of the Drasil framework.** Drasil is presented as a novel, extensible framework for knowledge-centric software artifact generation. Its architecture includes robust mechanisms for knowledge capture (chunks), projection (recipes), and rendering (multi-format output), supporting traceability and maintainability by construction.

- **Empirical evidence that single-source generation enforces consistency, traceability, and reproducibility by construction.** Through reimplementation of several scientific computing systems using Drasil, the thesis shows that local changes to domain knowledge propagate automatically across generated artifacts, that generated content retains explicit provenance, and that artifact generation is deterministic and reproducible.

- **Increased error visibility and clarification of tacit knowledge.** By forcing explicit encoding of domain knowledge, Drasil surfaces hidden assumptions, semantic inconsistencies, and other defects, enabling single-point correction and consistent documentation across all generated artifacts.

- **Separation of knowledge from presentation through the recipe DSL.** Drasil's recipe-based generation approach allows the structure of generated artifacts to be modified without altering the underlying knowledge base, improving maintainability and adaptability.

- **Framework for rapid generation and maintenance of software families.** The single-source approach enables efficient creation and evolution of software family variants through selective extension of the knowledge base and adaptation of recipes, minimizing duplication and ensuring consistency across related products.

- **Lessons learned and open questions for knowledge-centric development.** The thesis distills practical lessons from iterative development, onboarding, and refinement, highlighting the importance of centralized knowledge, improved authoring tools, and further validation for extending Drasil to new domains.

Drasil allows developers to focus on the system being built as a whole rather than the individual artifacts. We can still produce a "rational design" [58] without the explicit tedium of manually keeping everything in sync.

Chapter 2

Background

This chapter surveys foundational concepts and approaches that underpin the thesis. Rather than attempting an exhaustive survey of software engineering techniques, we focus on approaches relevant to a particular problem: software systems are represented through multiple artifacts, often created for different purposes and audiences, that nevertheless describe aspects of the same underlying system. Maintaining the relationships between these artifacts is an established concern in software engineering, particularly in requirements traceability, where requirements are related to downstream development artifacts throughout the software lifecycle [29, 99].

We first examine software artifacts and their role in communicating requirements, design decisions, and implementation details. We then examine software reuse and software families, focusing on approaches that exploit commonality within and across related systems. Reproducible research is also considered because it emphasizes the importance of being able to recreate computational results. Literate approaches are considered because they demonstrate how executable and explanatory representations can be developed from a common source [39]. Finally, generative programming and

related techniques demonstrate how higher-level representations can be transformed into concrete software artifacts [19, 92].

Together, these areas provide several pieces of the problem addressed by this thesis: maintaining related representations, reusing common information, and automatically deriving artifacts from higher-level descriptions. The remainder of this chapter examines those pieces individually before Chapter 3 considers how they interact in the software systems used as case studies in this thesis.

2.1 Software Artifacts

Software artifacts (or softifacts) come in a wide variety of forms. In the broadest sense, we can think of softifacts as anything and everything produced during the creation of a piece of software that serves some purpose. Any document detailing what the software should do, how it was designed, how it was implemented, how to test it, and so on would be considered a softifact, as would the source code whether as a text file, stack of punched cards, magnetic tapes, or other media. Softifacts also include items such as build instructions, Continuous Integration/Continuous Delivery (CI/CD) pipeline definitions, automated test cases, READMEs, etc.

Softifacts beyond just the source code are important to us for a number of reasons. They serve as a means of communication between different stakeholders involved in the software development process, or even different generations of developers involved in the production of a piece of software. They provide a common understanding of what the software is supposed to do, how it will be built, and how it will be tested. Softifacts also provide a record of the decisions that were made during the development process, which is important for future maintenance/modification of the

software or for compliance reasons. Combined with the former, this gives us a means to verify and validate the software against its original requirements and design [17, 57, 58].

Because these softifacts describe related aspects of the same software system, relationships naturally exist between them. Requirements influence design decisions, design decisions are realized in source code, and verification activities provide evidence that the resulting implementation satisfies its requirements. Traceability provides an explicit means of recording and following some of these relationships throughout the software lifecycle [29, 99]. Maintaining these relationships is important as the software evolves, since changes to one artifact may require corresponding changes to others.

Traceability and consistency are related, but distinct. Traceability concerns whether a relationship between elements of different softifacts can be followed, while consistency concerns whether those related elements agree with one another [26]. A trace link can help reveal a disagreement, but the presence of that link does not guarantee consistency. Conversely, two artifacts may agree even when the relationship between them has not been recorded. Maintaining software as it evolves requires attention to both properties.

The production, maintenance, and use of softifacts is largely dependent on the specific design process being used.

Software design can follow a number of different design processes, each with their own collection of softifacts. A common, traditional approach is the Waterfall model of software development [62] (Figure 2.1), which organizes development as a sequence of phases. After requirements have been baselined, resolving change requests caused by new requirements can increase effort and defects in Waterfall projects [15]. More

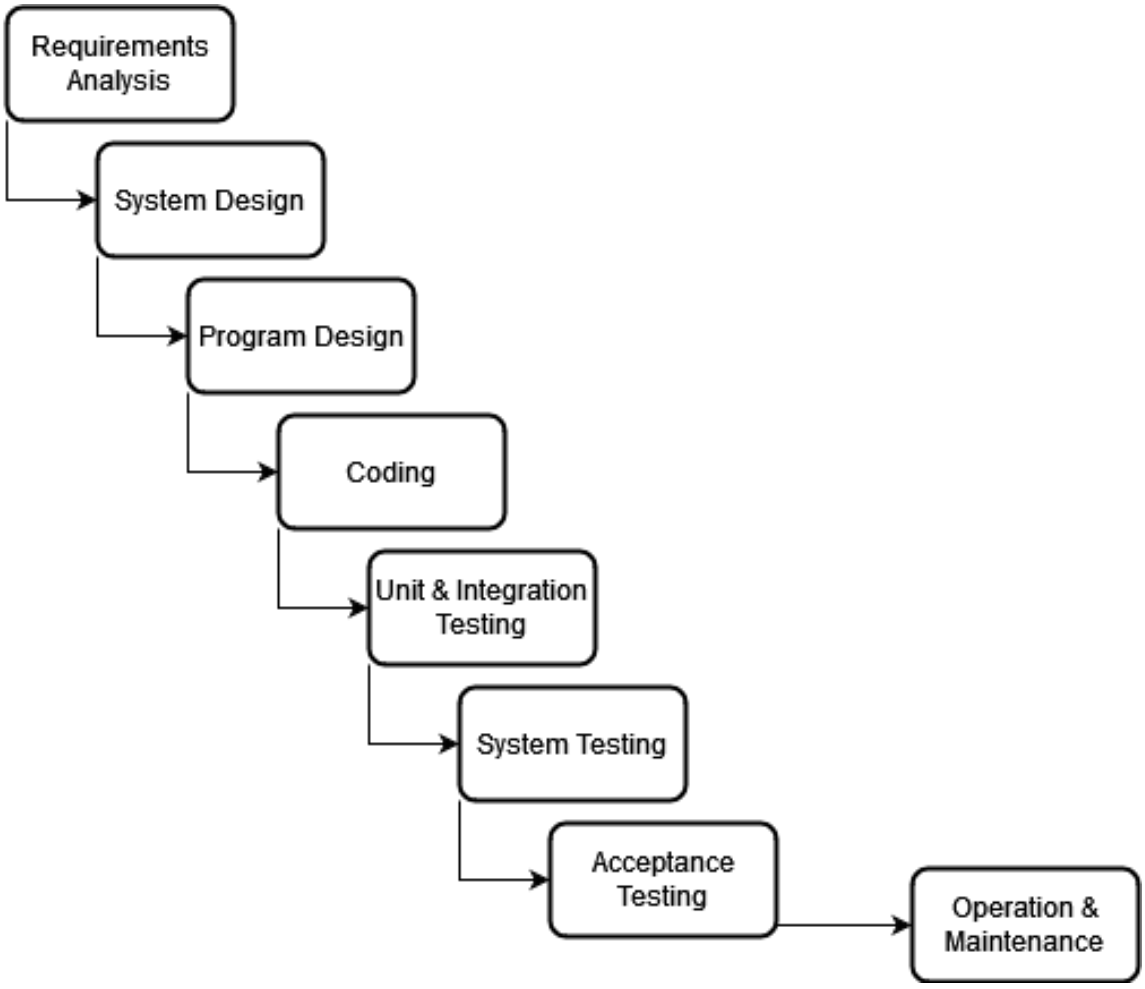


Figure 2.1: The Waterfall Model of Software Development

Table 2.1: Mapping of Rational Design Process to Waterfall Model and Common Software Artifacts

Rational Design Phase	**Waterfall Phase(s)**	**Common Artifacts**
Establish/Document Requirements	Requirements Analysis	System Requirements Specification; Verification and Validation Plan
Design and Document Module Structure	System Design	Design Document
Design and Document Module Interfaces	Program Design	Module Interface Specification
Design and Document Module Internal Structures	Program Design	Module Guide
Write Programs	Coding	Source Code; Build Instructions
Maintain	Testing (Unit, Integration, System, Acceptance) Operation & Maintenance	Test Cases; Verification and Validation Report

generally, requirements can continue to evolve throughout development as customer needs and organizational policies change and developers gain a better understanding of the product [53]. In contrast, Parnas and Clements [58] detailed what they dubbed a "rational" design process; an idealized version of software development that includes what needs to be documented in corresponding softifacts. The rational design process is not meant to be a linear process like the Waterfall model, but instead an iterative process using section stubs for information that is not yet available or not fully clear during the time of writing. Those stubs are then filled in over the development process, and existing documentation is updated so it appears to have been created in

a linear fashion for ease of review later in the software lifecycle.

The rational design process is particularly relevant to this thesis because it treats documentation as an integral part of software development rather than solely as a record produced after implementation. Parnas and Clements explicitly acknowledge that the information required to produce a rational design is not available all at once, and therefore distinguish the actual iterative development process from the organized documentation ultimately produced for review [58]. Later work by Parnas similarly emphasizes precise documentation as an important part of producing understandable and maintainable software [57, 59].

The rational design process involves the steps listed in the first column of Table 2.1.

Parnas provided a list of the most important documents [57] required for the rational design process, which Smith [82] expanded upon by including complementary artifacts such as source code, verification and validation plans, test cases, and build instructions. Smith's adaptation is particularly relevant in the context of scientific software, where the rational process is used to organize the development and maintenance of the collection of artifacts considered throughout this thesis [82]. While there have been many proposed artifacts, the following collection covers those most relevant to this thesis:

Name: **Software Requirements Specification (SRS)**
Description: Contains the functional and nonfunctional requirements detailing what the desired software system should do.

Name: **System Design Document**
Description: Explains how the system should be broken down and documents implementation-level decisions that have been made for the design of the system.

Name: **Module Guide**
Description: In-depth explanation of the modules outlined in the System Design Document.

Name: **Module Interface Specification**
Description: Interface specification for each of the modules outlined in the System Design Document/Module Guide.

Name: **Program Source Code**
Description: The source code of the implemented software system.

Name: **Verification and Validation Plan**
Description: Uses the system requirements to document acceptance criteria for each requirement that can be verified.

Name: **Test cases**
Description: Implementation of the Verification and Validation Plan in source code (where applicable) or as a step-by-step guide for testers.

Name: **Verification and Validation Report**
Description: Report outlining the results after undertaking all of the testing initiatives outlined in the Verification and Validation plan and test cases.

This collection is not exhaustive of all the types of softifacts, as there are many other design processes that use different types of softifacts. For example, looking at an agile approach using Scrum/Kanban, the softifacts tend to be distributed in different ways [35]. Requirements are documented in tickets under so-called "epics", "stories", and "tasks" as opposed to a singular requirements artifact, and the acceptance criteria listed on those tickets make up the validation and verification plan.

Regardless of the process, most attempt to document very similar information to that of the rational design process. Using the waterfall model as an example, we can see (Table 2.1) the rational design process and its artifacts map onto the model in a very straightforward manner.

Parnas [57] does an excellent job of defining the target audience for each of the most common softifacts, and we extend that perspective here. A summary can be found in Table 2.2.

Although these softifacts differ in purpose, audience, and level of abstraction, they describe different aspects of the same software system. Requirements constrain design and implementation, verification artifacts assess the resulting behaviour, and domain concepts recur wherever stakeholders need them. Because these relationships cross artifact boundaries, separately edited descriptions can diverge and weaken established traceability relationships [26, 99].

Some repetition remains unavoidable because artifacts serve different purposes.

Table 2.2: A summary of the Audience for each of the most common softifacts and what problem that softifact is solving

Softifact	**Who (Audience)**	**What (Problem/Solution)**
SRS	Software Architects Quality Assurance (QA) analysts	Defines exactly what specification the software system must adhere to.
Module Guide	All developers QA analysts	Outlines implementation decisions around the separation of functionality into modules and give an overview of each module.
Module Interface Specifications	Developers who implement the module Developers who use the module QA analysts	Details the exact interfaces of the modules from the Module Guide.
Source Code / Executable	All developers QA analysts End users	Implements the machine instructions designed to address the overall problem for which the software system has been specified
Verification and Validation Plan	Developers who implement the module QA analysts	Describes how the software should be verified using tests that can be validated. Includes module-specific and system-wide plans.

The distinction relevant to this thesis is not simply how often a piece of information appears, but how many independently editable versions of it exist. For example, an equation rendered in a document and implemented as source code may legitimately

appear twice; the maintenance risk comes from each version acting as its own authority. We use *unnecessary redundancy* to describe this latter situation.

This distinction leads naturally to software reuse. If common implementation, structure, or domain information can be identified and represented once, it may still be used in several places without requiring a corresponding number of authoritative versions, i.e. some of the repeated manual synchronization effort may be avoided.

2.2 Software Reuse and Software Families

We now look at ways in which others have attempted to avoid "reinventing the wheel" by providing means to reuse software, in part or whole, and reuse the analysis and design efforts of those that came before. We focus on software families (software product lines), component-based software engineering (CBSE), frameworks for reuse such as Draco, and the concept of reproducible research. These techniques were selected because they represent significant strategies for leveraging prior work, domain expertise, and automation in software development. Software families and CBSE illustrate structural and compositional reuse, Draco demonstrates early domain-specific reuse and code generation, while reproducible research highlights the importance of transparency and repeatability in computational work. Together, these approaches inform our emphasis on automating artifact generation and promoting knowledge reuse across the software lifecycle.

2.2.1 Software/Program Families

Software/program families, or software product lines, refer to a group of related systems sharing a common set of features, functionality, and design while differing at explicitly identified points of variation [34, 66]. These systems are typically designed to serve a specific domain or class of related problems, allowing development effort to be shared across multiple family members.

A central concept in software/program families is the distinction between *commonalities* and *variabilities* [9, 66]. *Commonalities* refer to the features, components, or design decisions shared by all members of the family, forming the stable core that defines the family's identity. In contrast, *variabilities* capture the aspects that can differ among family members, enabling customization or adaptation to specific requirements or contexts. These variabilities are typically managed through explicit *parameters of variation*, which may include configuration options, feature selections, or architectural choices [98]. Identifying and systematically managing these parameters is essential for effective reuse and automation in software product lines, as it allows developers to generate tailored systems while maintaining consistency and reducing duplication. Approaches such as Feature-Oriented Domain Analysis make these shared and variable properties explicit by analyzing a domain in terms of features that characterize members of the family [34].

Variability is also a correctness concern: feature-oriented correctness-by-construction is one approach for introducing variation points through refinement rules intended to preserve correctness across the product line [8].

A related approach is *commonality analysis*, which explicitly documents both the properties shared by members of a family and the parameters by which those members

may vary [98]. Rather than treating each program as an independent development effort, commonality analysis attempts to characterize the family itself. This supports reuse because the shared portions of the family can be identified once, while variation is captured explicitly rather than being rediscovered separately for each system.

These methods characterize a family, but do not prescribe how its members are implemented. Component-based and generative techniques provide two possible implementation strategies. For a component-based approach to support a family, the components or their composition must expose the parameters of variation so that family members can be constructed by selecting, configuring, or replacing components (see Section 2.2.2). Generative approaches instead use selected features or other higher-level specifications to generate or specialize family members (see Section 2.4) [19, 66].

This family-oriented view has also been applied specifically to scientific computing. Smith, McCutchan, and Cao examine program families in scientific computing as a means of identifying and exploiting common structure across related scientific applications [88]. Subsequent work applies commonality analysis to families of physical models, showing that the common information may include not only implementation structure, but also scientific concepts, assumptions, equations, and other model-level information [89]. Similar analyses have been carried out for mesh-generating software and other scientific software families [83, 84].

This broader notion of commonality is particularly important for this thesis. If two scientific software systems are members of the same family, their shared material may extend beyond reusable source-code components to the underlying scientific

and engineering information used to describe the problem. Requirements, mathematical models, assumptions, terminology, and documentation structure may therefore exhibit commonality alongside the implementation itself. Identifying these relationships provides an opportunity to reuse more than code: analysis and documentation effort may also be shared across family members [84, 89].

The Linux kernel has been studied as a large-scale example of explicitly managed variability. Its configuration space includes hardware support, device drivers, services, security options, and other features that can be selected to produce different system configurations. Studies of Linux as an evolving software product line, together with broader empirical work on feature lifecycles in highly-configurable systems, illustrate that such variability is not static, but must itself be managed as a family evolves over time [4, 46].

The main benefit of a family-oriented approach for this thesis is therefore systematic reuse. By identifying what is common across a family and representing variation explicitly, developers can avoid repeatedly constructing closely related systems from scratch [66, 98]. In scientific computing, this reuse can extend to physical models and requirements in addition to implementation structures [88, 89]. This motivates a broader discussion of software reuse: what portions of a software-development effort can be reused, and at what level should that reuse occur?

2.2.2 Software Reuse

Software reuse aims to reduce repeated development effort by making use of assets produced for previous systems. However, what constitutes a reusable asset can vary considerably. Reuse may occur at the level of complete applications, source-code

components, architectures, domain models, or the knowledge used to describe and construct a family of systems. Different approaches to reuse therefore differ not only in how reusable assets are combined, but in *what* they treat as reusable [19, 34, 66].

Component based software engineering (CBSE) is one such example. CBSE is an approach to software development that emphasizes using pre-built software components as the building blocks of larger systems. These reusable components can be developed independently and tested in isolation before being integrated into the larger system software [6, 100].

A key benefit of CBSE is that it can help to reduce the cost and time required for software development, since developers do not need to implement everything from scratch. Reusing components can also improve the quality and reliability of software systems when those components have already been tested and used successfully in other contexts [6].

One CBSE framework is the Common Object Request Broker Architecture (CORBA), which provides a standardized mechanism for components to communicate with each other across a network. CORBA defines a set of interfaces and protocols that allow components written in different programming languages to interact with each other in a distributed environment [54].

Another CBSE framework is the JavaBeans Component Architecture. It is a standard for creating reusable software components in Java. JavaBeans are self-contained software modules with a well-defined interface that can be easily integrated into a variety of development environments and combined to form larger applications [55].

A central challenge of CBSE is ensuring that components are compatible with one another and can be integrated into a larger system without conflicts or errors [6].

To help address this challenge, Wu et al. present a test model based on component interactions and dependencies [100]. A related challenge is that frameworks such as CORBA and JavaBeans preserve important architectural decisions into run time, where the abstractions that support reuse can also carry non-trivial performance costs. These limitations have motivated interest in forms of reuse that go beyond composition alone and instead emphasize the capture and reuse of domain knowledge, including approaches that shift more of the design burden to generation time rather than run time [19].

Others have attempted to provide frameworks for reuse as well. For example, Neighbors [49, 50, 51] proposed a method for engineering reusable software systems known as "Draco". Draco was among the earliest frameworks for software reuse, particularly in the context of domain-specific software generation, though its direct influence was more limited compared to later approaches. The core idea behind Draco is to construct software systems by composing reusable components organized by problem domain. Unlike CBSE, which focuses on assembling systems from independently developed, interchangeable components, Draco emphasizes the reuse of domain analysis, design information, and code through domain-specific languages, reusable transformations, and component refinements.

In Draco, a software engineer first performs a domain analysis, identifying the essential objects and operations relevant to a class of related systems. The resulting domain description provides a domain-specific language, reusable transformations, and software components whose refinements express implementation choices in terms of other domains. By expressing system requirements in a domain language and applying these transformations and refinements, developers can automatically generate

significant portions of a software system, reducing manual coding effort while reusing earlier analysis and design effort. While Draco itself is not widely used, its approach anticipated later developments in software product lines and generative programming by emphasizing the explicit capture and reuse of domain knowledge [19, 50].

Reuse is not limited to constructing new software; preserving the information associated with computational work can also allow earlier results to be revisited.

2.2.3 Reproducible Research

Being able to reproduce computational results is fundamental to good science [61]. For research software, however, reproducibility is often undermined by undocumented assumptions, missing dependencies, version drift in libraries and build tools, and changes in the computational environment [2, 11, 91]. These problems make previously published results difficult to recreate reliably and can threaten confidence in the software and the scientific conclusions derived from it, echoing the broader reproducibility crisis highlighted in the scientific literature [5].

It is also useful to distinguish reproducibility from replicability. Following the terminology used by Rougier et al. [70], reproducibility concerns rerunning the original computational workflow and obtaining the same results from the same code and data. Strict reproducibility assumes an identical computational environment, although scientific software should also remain robust under relevant changes to that environment. Replicability is a stronger goal: an independent researcher should be able to reconstruct the results from the documented theories, assumptions, and design decisions without relying on the original implementation. Both goals become difficult when essential knowledge remains tacit or scattered across papers, scripts, notebooks, and

build infrastructure [2, 70].

Existing work in reproducible research has often emphasized combining narrative, code, and data in a single package. For example, *compendia* were introduced as a way of encapsulating the full scope of a computational study so that readers can inspect details and rerun computations performed by the author [27]. Similarly, tools such as Sweave [41], SASweave [43], StatWeave [42], Scribble [24], Org-mode [72], and Jupyter notebooks [67] support literate workflows (see Section 2.3), in which executable code is presented alongside explanatory text.

These approaches improve transparency, but they do not guarantee reproducibility on their own. In practice, notebook-based workflows can still suffer from hidden dependencies, execution-order problems, and weak capture of environment details [65]. Containers help address some of the environment problem by packaging software stacks more explicitly [11]. However, preserving a snapshot of an environment is not the same as sustaining reproducibility as technologies evolve: host platforms, package sources, languages, and libraries can become incompatible or unavailable. Long-term reproducibility therefore also requires dependency and build information that is sufficiently complete, inspectable, and maintainable to reconstruct or adapt the workflow. Environment capture does not by itself preserve the domain knowledge, assumptions, and design rationale needed for independent reconstruction [2].

2.3 Literate Approaches to Software Development

There have been several approaches attempting to combine development of program code with documentation. Literate Programming and literate software are two such approaches that have influenced the direction of this thesis, which we now outline.

2.3.1 Literate Programming

Literate Programming (LP) is a method for writing software introduced by Knuth that focuses on explaining to a human what we want a computer to do rather than simply writing a set of instructions for the computer on how to perform the task [39].

Developing literate programs involves breaking algorithms down into *chunks* [32] or *sections* [39] which are small and easily understandable. The chunks are ordered to follow a "psychological order" [64] if you will, that promotes understanding. They do not have to be written in the same order that a computer would read them. It should also be noted that in a literate program, the code and documentation are kept together in one source. To extract runnable code, a process known as *tangle* must be performed on the source. A similar process known as *weave* is used to extract and typeset the documentation.

There are many advantages to LP beyond understandability. As a program is developed and updated, the documentation surrounding the source code is more likely to be updated simultaneously. It has been experimentally found that using LP ends up with more consistent documentation and code [75]. There are many downsides to having inconsistent documentation while developing or maintaining code [40, 95], while the benefits of consistent documentation are numerous, including improved maintainability and ease of future modifications, reduced risk of introducing errors during updates, enhanced onboarding and training for new team members, better communication among stakeholders, and increased confidence in overall software quality [31, 40, 57, 58, 59]. With the advantages and disadvantages of good documentation in mind we see that more effective, maintainable code can be produced if properly using LP [64].

Despite these proposed benefits, early accounts found that LP had not been widely adopted [75]. Some successful applications, however, demonstrate additional benefits of the approach. VNODE-LP was developed entirely using LP so that its mathematical theory, documentation, and source code could be reviewed together, making it easier for a human expert to verify that the implementation correctly translates the underlying theory [48]. Another example is "Physically Based Rendering: From Theory to Implementation", a literate program and textbook on the subject matter [63]. Shum and Cook [75] discuss the main issues behind LP's lack of popularity, which can be summed up as dependency on a particular output language or text processor, and the lack of flexibility in what should be presented or suppressed in the output.

There are several other factors which contribute to LP's lack of popularity and slow adoption thus far. While LP allows a developer to write their code and its documentation simultaneously, that documentation is comprised of a single artifact which may not cover the same material as standard artifacts software engineers expect (see Section 2.1 for more details). LP also does not simplify the development process: documentation and code are written as usual, and there is the additional effort of organizing and relating the chunks. It is also challenging to use LP when requirements and design choices change, as those changes may affect many different chunks. That is not to say LP does not have benefits, the LP development process allows developers to follow a more natural flow in development by writing chunks in whichever order they wish, keep the documentation and code updated simultaneously (in theory) because of their co-location, and automatically incorporate code chunks into the documentation to reduce some information duplication.

There have been many attempts to increase LP's popularity by focusing on changing the output language or removing the text processor dependency [64]. Several new tools such as CWEB (for the C language), noweb (programming language independent), and others were developed. Other tools such as DOC++ (for C++), javadoc (for Java), and Doxygen (for multiple languages) were also influenced by LP, but differ in that they are merely document extraction tools [22, 56, 96]. They do not contain the chunking features which allow for re-ordering algorithms.

With new tools came new features including, but not limited to, phantom abstracting [75], a "What You See Is What You Get" (WYSIWYG) editor [25], and even movement away from the "one source" idea [76].

Classical LP did not become a mainstream software-development methodology [33], but its core idea of combining code with explanatory text has been widely adopted through computational notebooks such as Jupyter, particularly in data science [7, 65, 67]. These more robust tools helped drive the understanding behind what exactly LP tools must do. Some languages, including Agda and Haskell, directly support literate source files [93, 94]. R instead has an ecosystem of tools for literate data analysis and dynamic documents, including R Markdown and the earlier Sweave [41, 101]. These tools, however, are primarily designed to dynamically create up-to-date reports and other documents by running embedded code, as opposed to being used as part of the software development process.

For our purposes, LP remains valuable for its emphasis on explanation and readability, but conventional LP uses a single source artifact to generate a program and its accompanying documentation [64]. Although the code and explanatory text are co-located, their semantic consistency must still be maintained manually. Any other,

related softifacts outside the literate source must likewise be synchronized separately, making it difficult to support broader traceability needs in software engineering practice.

2.3.2 Literate Software

A combination of LP and Box Structure [47] was proposed as a new method called "Literate Software Development" (LSD) [3]. Box structure can be summarized as the idea of different views that are abstractions that communicate the same information at different levels of detail, for different purposes. Box structures consist of black box, state box, and clear box structures. The black box gives an external (user) view of the system and consists of stimuli and responses; the state box makes the state data of the system visible (it defines the data stored between stimuli); and the clear box gives an internal (designer's) view describing how data are processed, typically referring to smaller black boxes [47]. These three structures can be nested as many times as necessary to describe a system.

LSD was developed with the intent to overcome the disadvantages of both LP and box structure [3]. It was intended to overcome LP's inability to specify interfaces between modules, the inability to decompose boxes and implement the design created by box structures, as well as the lack of tools to support box structure [3, 21].

The framework developed for LSD, "WebBox", expanded LP and box structures in a variety of ways. It included new chunk types, the ability to refine chunks, the ability to specify interfaces and communication between boxes, and the ability to decompose boxes at any level [3]. For our purposes, however, these extensions remain focused on code and its box-structure design, with little support for creating other

kinds of softifacts. That emphasis limits its usefulness in settings where teams rely on a broader collection of artifacts for communication, review, and maintenance.

2.4 Generative Programming and Metaprogramming

The ideas now associated with generative programming and metaprogramming are more general than early code-generation tools alone. At a high level, they concern the use of specifications, models, templates, and staged transformations to synthesize software from a more abstract representation [19, 92, 97]. This reflects a long-standing goal in software engineering: to capture recurring structure and variability explicitly so that families of related systems can be derived consistently rather than assembled one-by-one by hand. The broader literature on domain-specific languages (DSLs), software product lines (SPLs), and feature-oriented development frames this as a problem of structuring the generation space, not simply of emitting source code from a template [19, 97].

This broader view also explains why metaprogramming is often treated as a more general concept than generation alone. Template metaprogramming, macros, partial evaluation, DSLs, and compiler/interpreter generation all share the same underlying idea: the programmer expresses a problem at a higher level of abstraction, and the system constructs or specializes lower-level implementation artifacts from that specification [19, 74, 92]. In DSL-based settings, language-level self-extension can allow modelers to introduce new concepts and package them in reusable model libraries without modifying the language's metamodel [28]. Recent work on low-code platforms similarly uses a set of DSLs to represent platform models, variability, and product configurations in support of SPL engineering [10].

The main benefits of this broader perspective are familiar: increased productivity, systematic reuse, and reduced opportunity for inconsistency across generated artifacts. However, the modern literature also highlights a more subtle point: the true challenge is not simply generation, but the design of the generation space itself. This includes the specification language, the constraints that define valid combinations, the feature model or variability model, the transformation rules, and the mechanisms by which generated artifacts are assembled and validated. In product-line and DSL settings, these issues are often more important than the mechanics of template expansion alone. This additional layer can also make a generated system more difficult to understand and debug. Developers may need to reason across the input specification, transformation rules, and generated artifacts, while the output alone may not reveal why a particular structure or design decision was produced [19].

Examples from the wider ecosystem illustrate the same trend. Parser generators such as ANTLR generate recognizers from declarative grammar descriptions [60]; API tooling such as Swagger Codegen generates client libraries and server stubs from OpenAPI descriptions [77]. Together, these tools show how structured specifications can drive the generation of particular implementation artifacts.

For this thesis, the significance of generative programming lies in applying this idea across the multiple related artifacts that describe a software system. The important point is not simply that code can be generated automatically, but that engineering knowledge can be captured once, expressed declaratively, and then reused to produce consistent artifacts across a family of related software products or deliverables.

Model-Driven Engineering (MDE) provides a closely related perspective. It treats

models as primary development artifacts and uses transformations to derive or synchronize other representations, including implementation artifacts. Model-transformation approaches differ in their source and target representations, directionality, and degree of automation [20]. Making these transformations explicit can also provide opportunities to record and maintain traceability as the resulting artifacts evolve [99].

Although MDE shares the general goal of deriving software artifacts from higher-level representations, it was not used as a methodological starting point for this thesis. Our work instead developed from the bottom up through the analysis of concrete scientific-computing case studies and the practical requirements of the artifacts they contained. We therefore did not adopt MDE as an overarching methodology or commit to a particular modeling standard, development process, or tooling ecosystem. Doing so at the outset would have expanded the scope of the work and introduced the commitments of a broader modeling framework before their value had been demonstrated for our context.

MDE is included here as relevant related work and as a retrospective point of comparison, rather than as the theoretical basis from which our approach was derived. Any similarities should therefore not be read as indicating that this work adopted MDE terminology, methods, or frameworks.

Taken together, the approaches reviewed in this chapter address different aspects of representing, reusing, and transforming software knowledge. They frame the broader problem of maintaining consistent relationships among the artifacts that describe an evolving software system.

Chapter 3

A Look Under the Hood: Our Process

This chapter examines the process by which we developed the Drasil framework itself. Rather than describing the general software development process that Drasil is intended to support or automate, we turn our attention inward to examine the steps, decisions, and rationale that guided its creation. Our goal is to provide transparency into the reasoning that shaped Drasil, highlighting the challenges encountered and the strategies employed to address them. By doing so, we aim to give the reader a clear understanding of how the framework's design emerged from a careful analysis of redundancy and knowledge organization across software artifacts.

As one of the central motivations for developing Drasil was the observation that softifacts contain significant, unnecessary redundancy, we begin by motivating our approach through concrete examples of redundancy in softifacts. We then describe how we selected and analyzed a set of representative case studies, breaking down their artifacts to identify recurring patterns and organizational structures. This analysis leads to a general methodology for knowledge capture and projection, which ultimately shaped the core principles of Drasil. Because this methodology depends on

what we mean by *knowledge*, the chapter also introduces a working definition of that term that underlies the framework developed later in the thesis. Throughout, we emphasize the interplay between practical experience and theoretical insight, illustrating how each informed our process and contributed to the framework's evolution.

3.1 A Minimal Motivating Example: HGHC

Before delving into the full complexity of our case studies and artifact templates, it is helpful to illustrate the core problem of knowledge redundancy with a minimal, toy example. For this, we use the "HGHC" system, a simple heat transfer case study included in the Drasil repository.

The HGHC example's SRS specifies what problem the software is trying to solve. In this case, it should model heat transfer coefficients in nuclear fuel rods, specifically h_g (gap conductance) and h_c (effective heat transfer coefficient between clad and coolant). Even in this minimal system, the same core knowledge appears in multiple places within the SRS.

To illustrate, consider the following: The SRS defines variables such as h_g, h_c, τ_c (clad thickness), h_p (initial gap film conductance), h_b (initial coolant film conductance), and k_c (clad conductivity). The defining equations for h_g and h_c are also given as:

$$h_g = \frac{2k_c h_p}{2k_c + \tau_c h_p}$$

$$h_c = \frac{2k_c h_b}{2k_c + \tau_c h_b}$$

We can already see repeated knowledge (k_c and τ_c) in these two equations. The redundancy goes even deeper in the full SRS (which can be found in Appendix A).

In the given SRS we can see repeated descriptions and units for h_g and h_c in the table of symbols and the data definitions. We also observe something more troubling: missing symbols. None of k_c, h_p, h_b, or τ_c have units associated with them nor are they defined in the table of symbols, despite their presence in the equations and data definitions for h_g and h_c.

As this is a toy example, it may seem contrived to have missing information. However we ran into exactly these sorts of issues with our real-world case studies. In this example, while we can easily identify and rectify omissions, it highlights a key issue: even in a simple system, knowledge is repeated across different sections of a single softifact, and critical definitions can be overlooked. This kind of repetitive, reference-style information is also exactly the sort of tedious material that is easy to omit from real-world, manually produced documentation, which helps explain both why practitioners dislike writing it and why manual processes are unreliable for maintaining it consistently. Once we add more complexity, with multiple softifacts and larger systems, these issues compound.

Imagine now that we wish to validate the units of both h_g and h_c. The knowledge required for this validation is currently missing. Suppose we add the appropriate symbol definitions and units to the SRS; if we then discover an inconsistency in the units due to an error in the defining equations, we would need to update multiple sections of the SRS to correct it, increasing the risk of further mistakes. If additional softifacts, particularly code, have already been created based on these definitions, the challenge of maintaining consistency becomes even greater. Corrections would now be required across multiple input languages (for example, LaTeX and Python),

significantly increasing the verification burden. Ultimately, these softifacts are communicating much of the same knowledge (i.e. the definitions of h_g and h_c), merely formatted for different audiences. In the case of the SRS and the source code, the audiences are human stakeholders and computers, respectively.

This small example foreshadows the larger patterns of redundancy and risks of inconsistency that we observe in more complex systems. The rest of this chapter generalizes from such examples, showing how we systematically identified, categorized, and addressed these issues in the development of Drasil.

3.2 A (Very) Brief Introduction to Our Case Study Systems

To ground our analysis in reality and move beyond abstract principles, this thesis draws on a set of detailed case studies. These case studies serve not only to illustrate the prevalence and impact of redundant knowledge in softifacts, but also to provide a concrete basis for systematically identifying, comparing, and generalizing patterns of redundancy and knowledge organization. By examining systems developed using common artifact templates, we are able to motivate the need for the Drasil framework and inform its design.

To simplify the process of identifying redundancies and patterns, we have chosen several case studies developed using common artifact templates, specifically those used by Smith et al.[86, 87] Also, as previously mentioned in Section 1.3, our case study software systems follow the '*input*' $\rightarrow$ '*process*' $\rightarrow$ '*output*' pattern. These systems cover a variety of use cases, to help avoid over-specializing into one particular

Table 3.1: Summary of the case study systems

Case study	**Problem**
GlassBR	Predict whether a glass slab can withstand a blast under specified conditions.
SWHS	Simulate how phase change material (PCM) affects heating and energy storage in a solar water heating tank.
NoPCM	Simulate heating and energy storage in a solar water heating tank without phase change material.
SSP	Evaluate slope stability by identifying the critical slip surface and determining the associated factor of safety and interslice forces.
Projectile	Predict the landing position of a projectile.
GamePhysics	Provide a physics library that simulates objects under varied physical conditions efficiently enough for soft real-time use in games.

system type.

The majority of the aforementioned case studies were developed by other teams at McMaster University to address real engineering problems outside of our research group. Table 3.1 provides a high-level reference to each case study. For each system, the GitHub repository contains the artifacts relevant to our analysis, namely the documents and implementation materials that expose the system's requirements, design decisions, and implementation details. These include, where available, the Software Requirements Specification (SRS), Module Guide (MG), Module Interface Specification (MIS), source code, and other supporting materials.

The NoPCM case study was created as a software family member for the SWHS

case study. It was manually written, removing all references to PCM and thus remodeling the system.

The Projectile case study, was the first example of a system created solely in Drasil, i.e. we did not have a manually created version to compare and contrast with through development. As such, it will not be referenced often since it did not inform Drasil's design or development until much further in our process. The Projectile case study was created post-facto to provide a simple, understandable example for a general audience as it requires, at most, a high-school level understanding of physics.

After carefully selecting our case studies, we went about a practical approach to find and remove redundancies. The first step was to break down each artifact type and understand exactly what they are trying to convey.

3.3 Breaking Down Softifacts

To meaningfully address redundancy in software artifacts, we must understand the purpose, content, and audience of each artifact—not just observe repetition. Building on the foundation provided by our case studies, we use the term *core knowledge* to refer to the system information that recurs across artifacts, such as the quantities being modeled, the relations among them, the requirements on acceptable behavior, and the design decisions needed to realize those requirements. This section examines the structure and role of each major softifact[1] in our process. By breaking down these artifacts, we aim to reveal both their commonalities and their differences, setting the

[1]For brevity, we focus our analysis on the SRS, Module Guide, and Source Code, as these artifacts capture complementary aspects of a software system: requirements, design decisions, and implementation. Together, they provide a useful basis for identifying the patterns of knowledge organization and redundancy that informed the later design of Drasil.

stage for identifying patterns of knowledge organization and redundancy that inform the design of Drasil.

As noted earlier, for our approach to work we must understand exactly what each of our softifacts are trying to say and to whom.[2] By selecting our case studies from those developed using common artifact templates, we have given ourselves a head start on that process. The following subsections present a brief sampling of our process of breaking down softifacts, using the GlassBR case study as a running, motivating example to ground the analysis, acknowledging that a comprehensive overview would be excessively lengthy.

3.3.1 SRS

To start, we look at the SRS. The SRS (or some incarnation of it) is one of the most important artifacts for any software project, as it specifies what problem the software is trying to solve. There are many ways to state this problem, and the template from Smith et al. has given us a strong starting point. Figure 3.1 shows the table of contents for an SRS using the Smith et al. template.

With the structure of the document in mind, let us look at several of our case studies' SRS documents to get a deeper understanding of what each section truly represents. Figures 3.2, 3.3, and 3.4 show the reference sections of the SRS for GlassBR. Each of the case studies' SRS contains a similar section so for brevity we will omit the others here, but they can be found in the GitHub repository. We will try to ignore any superficial differences (spelling, grammar, phrasing, etc.) in each of the case studies while we look for commonality. We are also trying to determine

[2]Refer to Section 2.1 for a general summary of softifacts.

1. Reference Material
   1.1. Table of Units
   1.2. Table of Symbols
   1.3. Abbreviations and Acronyms
2. Introduction
   2.1. Purpose of Document
   2.2. Scope of Requirements
   2.3. Characteristics of Intended Reader
   2.4. Organization of Document
3. Stakeholders
   3.1. The Customer
   3.2. The Client
4. General System Description
   4.1. System Context
   4.2. User Characteristics
   4.3. System Constraints
5. Specific System Description
   5.1. Problem Description
      5.1.1. Physical System Description
      5.1.2. Goal Statements
   5.2. Solution Characteristics Specification
      5.2.1. Assumptions
      5.2.2. Theoretical Models
      5.2.3. General Definitions
      5.2.4. Data Definitions
      5.2.5. Instance Models
      5.2.6. Data Constraints
      5.2.7. Properties of a Correct Solution
6. Requirements
   6.1. Functional Requirements
   6.2. Non-Functional Requirements
7. Likely Changes
8. Unlikely Changes
9. Traceability Matrices and Graphs
10. Values of Auxiliary Constants
11. References
12. Appendix

Figure 3.1: The Table of Contents from the expanded Smith et al. template

how the non-superficial differences relate to the document template, general problem domain, and specific system information. In the annotated Table of Symbols figures used in this chapter, 'B' denotes repeated boilerplate structure and 'P' denotes a bundled set of problem-specific entries, including both domain-specific and system-specific content. We use that coarser label here because these figures make the shared structure clear, but do not visually support a reliable finer distinction. We return to that finer distinction later in the SWHS/NoPCM comparison, where the few system-specific differences can be marked directly.

1.1 Table of Units

The unit system used throughout is SI (Système International d'Unités). In addition to the basic units, several derived units are also used. For each unit, the Table of Units lists the symbol, a description and the SI name.

Symbol	Description	SI Name
kg	mass	kilogram
m	length	metre
N	force	newton
Pa	pressure	pascal
s	time	second

Figure 3.2: Table of Units section from GlassBR SRS

1.2 Table of Symbols

B

The symbols used in this document are summarized in the Table of Symbols along with their units. The symbols are listed in alphabetical order.

Symbol	Description	Units
a	Plate length (long dimension)	m
AR	Aspect ratio	–
$AR_{\max}$	Maximum aspect ratio	–
B	Risk of failure	–
b	Plate width (short dimension)	m
capacity	Capacity or load resistance	Pa
$d_{\max}$	Maximum value for one of the dimensions of the glass plate	m
$d_{\min}$	Minimum value for one of the dimensions of the glass plate	m
E	Modulus of elasticity of glass	Pa
⋮		

P

Figure 3.3: Table of Symbols (truncated) section from GlassBR SRS

1.3 Abbreviations and Acronyms

Abbreviation	Full Form
A	Assumption
AN	Annealed
AR	Aspect Ratio
DD	Data Definition
FT	Fully Tempered
GS	Goal Statement
GTF	Glass Type Factor
HS	Heat Strengthened
IG	Insulating Glass
IM	Instance Model
LC	Likely Change
LDF	Load Duration Factor
LG	Laminated Glass
⋮	

Figure 3.4: Table of Abbreviations and Acronyms (truncated) section from GlassBR SRS

Looking at the (truncated for space) Table of Symbols, Table of Units, and Table of Abbreviations and Acronyms sections (Figures 3.2, 3.3, and 3.4) across the case studies, we can see that, barring the table values themselves, they are almost identical between the case studies. More importantly, this repetition makes explicit one of the structural patterns that recurs throughout the artifacts: reference-style knowledge is often organized and projected in tabular form. In other words, the recurring feature is not only the specific entries in these sections, but the table structure itself as a reusable way of presenting knowledge. The Table of Symbols is simply a table of values, akin to a glossary, specific to the symbols that appear throughout the rest of the document. For each of those symbols, we see the symbol itself, a brief description of what that

symbol represents, and the units it is measured in, if applicable. Similarly, the Table of Units lists the Système International d'Unités (SI) Units used throughout the document, their descriptions, and the SI name. Finally, the table of Abbreviations and Acronyms lists the abbreviations and their full forms, which are essentially the symbols and their descriptions for each of the abbreviations.

While the reference material section should be fairly self-explanatory as to what it contains, other sections and subsections are not always so clear from their names alone. For example, a section labeled as a theoretical model, data definition, or instance model does not by itself reveal exactly what kind of knowledge is being presented, how detailed that knowledge is, or how it relates to other sections of the document. This matters for our analysis because, to identify patterns of redundancy and organization across artifacts, we first had to determine what each section was actually saying rather than rely only on the template labels. Refer to Section 2.1 for more details on the roles of these artifacts; a brief summary is available in Table 2.2.

Returning to our exercise of breaking down each section of the SRS to determine the subtleties of *what* is contained therein[3] it should be unsurprising that each section maps to the definition provided in the Smith et al. template. However, the kinds of information contained in those sections differ substantially. Some material is generic boilerplate, such as the introductory text describing the purpose and organization of each section of the SRS. Some is clearly system-specific, such as GlassBR's goal statements, its instance models, and its data definitions for quantities like demand, stress distribution factor, and probability of breakage. Between those extremes is material that belongs to the problem domain more than to the individual software system.

[3]The breakdown details are omitted for brevity and due to their monotonous nature, although the overall process is very much akin to the breakdown of the Reference Material section.

In GlassBR, for example, the theoretical background concerning glass mechanics, load, and resistance is not unique to that one program; rather, it is broader domain knowledge that the SRS uses to derive the system-specific models, constraints, and calculations presented later in the document.

Observing the contents of an SRS template adhere to said template may seem mundane, but it is a necessary step before we can move on to other softifacts. Without understanding what the SRS template intends to convey it is hard to assess whether or not the case study SRS conveys that information. With that in mind, we can move on to the MG and source code.

3.3.2 Module Guide

The MG is a softifact that details the architecture of a given software system. It holds a number of design decisions around sensibly grouping functionalities within the system into modules to fulfill the requirements laid out in the SRS. For example, one might have an input/output module for handling user input and giving the user feedback through the display (i.e. via print commands or some other output), or a calculations module that contains all of the calculation functions being performed in the normal operation of the given software system. The Smith et al. MG template also includes a traceability matrix for ease of verifying which requirements are fulfilled by which modules. Finally, the MG includes considerations for anticipated or unlikely changes that the system may undergo during its lifecycle.

Figure 3.5 shows the table of contents for the GlassBR case study's MG. Again, for the sake of brevity we will omit the other case studies here, even though we engaged in the same breakdown process for each of them. Just as with the SRS we are looking

Module Guide for GlassBR

Spencer Smith and Thulasi Jegatheesan

July 25, 2018

Contents

Figure 3.5: Table of Contents for GlassBR Module Guide

5.2.4 Calc Module (M5)

Secrets: The equations for predicting the probability of glass breakage, capacity, and demand, using the input parameters.

Services: Defines the equations for solving for the probability of glass breakage, demand, and capacity using the parameters in the input parameters module.

Implemented By: GlassBR

Figure 3.6: Calc Module from the GlassBR Module Guide

for commonality and understanding of what the document is trying to portray to the reader. As such we will ignore superficial differences between the MG sections. As the MG is a fairly short document we will look at each of the most relevant sections as part of this exercise.

Breaking down the MG by section, we can see that the introduction is itself completely generic boilerplate explaining the purpose of the MG, the audience, and some references to other works that explain why we would make certain choices over other (reasonable) ones given the opportunity. There is nothing system-specific, nor specific to the given problem domain of the case study.

Following through the table of contents into the "Anticipated and Unlikely Changes" section, we see that again the introductions to this section and its subsections are generic boilerplate, however the details of each section are not. Both subsections are written in the same way: as a list of labeled changes (AC# for anticipated change, UC# for unlikely change). This is the first place we see both problem-domain and system-specific information. Interestingly, the Module Hierarchy section follows the same general style: it is a list of modules that represent the leaves of the module hierarchy tree and each one is labeled (M#).

Skipping ahead to the module decomposition, we find a section heading for each Level 1 module in the hierarchy, followed by subsections describing the Level 2 modules. The Level 1 headings are almost entirely generic boilerplate, with common examples including Hardware-Hiding, Behaviour-Hiding, and Software Decision modules. The Level 2 modules, by contrast, are problem-domain or system-specific. An example of a system-specific module is shown in Figure 3.6.

Each module is described by its secrets, services, and what it will be implemented by. For example, a given module could be implemented by the operating system (OS), the system being described (ex. GlassBR), or a third party system/library that will inter-operate with the given system.

Finally, the MG includes a traceability matrix and use hierarchy diagram. These provide visual summaries of how modules relate to the system's requirements and to one another, respectively. The traceability matrix makes the connection between the SRS and MG especially explicit, whereas many of the other links between these two softifacts have been implicit until this point. We do not analyze these representations in detail here, however, because for our purposes they largely restate relationships already visible in the module decomposition. Generally, the next softifact would be the MIS, but as it is structured so similarly to the MG (one section per module, each section organized in a very similar way, a repeated use hierarchy, etc.) we will skip it for brevity. The MIS includes novel system-specific, implementation-level information denoting the interfaces between modules, but for our current exercise does not provide any revelations beyond that of the MG.

The MG gives us a very clear picture of *decisions* made by the system designers,

as opposed to the knowledge of the system domain, problem being solved, and requirements of an acceptable solution provided in the SRS. The MG provides platform and implementation-specific decisions, which will eventually be translated into implementation details in the source code. With that in mind, let us move on to dissecting the source code.

3.3.3 Source Code

The source code is arguably the most important softifact in any given software system since it serves as the set of instructions that a computer executes to solve the given problem. With only the other softifacts and without the source code, we would have a very well defined problem and acceptance criteria for a possible solution, but would never actually solve the problem.

As the source code is the executable set of instructions, one would expect much of it to be system- and problem-domain specific. Looking into the source of our case studies, we find this to be largely true for the computational core, while also observing recurring implementation-level boilerplate across projects. For example, tasks such as reading inputs, writing outputs, and managing files are often structured very similarly between systems, and some later-added numerical components, such as ordinary differential equation (ODE) solvers, are generic rather than domain-specific.

Returning to our example of the MG from GlassBR (Figure 3.5) and comparing it to the python source code structure shown in Figure 3.7, we can see that in this manually maintained case study the source code follows almost identically in structure to the module decomposition. The only difference is the existence of an exceptions module defining the different types of exceptions that may be thrown by the other

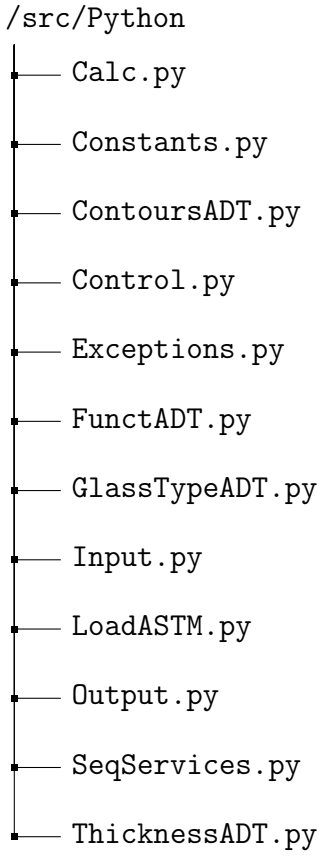


Figure 3.7: Python source code directory structure for GlassBR

modules. While this can be considered a fairly trivial difference, likely made for ease of maintenance, readability, and extensibility, it highlights that **the two softifacts can drift out of sync when maintained separately**. We speculate this difference was caused by a change made during the implementation phase, wherein the MG was not updated to reflect the addition of an exceptions module.

Let us look deeper into the code for one specific module, for example the Calc module introduced in the MG (Figure 3.6). The source code for said module can be found in Figure 3.8. In the source code we see a number of calculation functions, including those that calculate the probability of glass breakage, demand (also known as *load* or q), and capacity (also known as *load resistance* or LR) as outlined in the *secrets* section of the Calc module definition in the MG. We also see a number of intermediary calculation functions required to calculate these values (for example, `calc_NFL`, which calculates the non-factored load (NFL), and its dependencies).

The source code provides clear instructions to the machine on how to calculate each of these values and their intermediaries; it provides the actionable steps to solve the given problem. When we compare the code with relevant sections of the SRS, specifically the Data Definitions (DDs) for each term, we can see a very obvious transformation from one form to the other. For a concrete example, compare the data definition for $\hat{q}$ shown later in Figure 3.10 with its corresponding implementation (`calc_q_hat`) in Figure 3.8: the symbol used by the DD is reflected in the (partial) name of the function in the source code, and the equation from the DD is calculated within the source code. This is one of many patterns we see across our softifacts within each case study.

```
5  from Input import *
6  from ContoursADT import *
7  from math import log, exp
8
9  ## @brief Calculates the Dimensionless load
10 #  @return the unitless load
11 def calc_q_hat( q, params ):
12     upper = q * (params.a * params.b) ** 2
13     lower = params.E * (params.h ** 4) * params.GTF
14     return ( upper / lower )
15 ## @brief Calculates the Stress distribution factor based on Pbtol
16 #  @return the unitless stress distribution factor
17 def calc_J_tol( params ):
18     upper1 = 1
19     lower1 = 1 - params.Pbtol
20
21     upper2 = (params.a * params.b) ** (params.m - 1)
22     lower2 = params.k * ((params.E * (params.h ** 2)) ** (params.m)) * params.LDF
23
24     return (log( (log(upper1 / lower1)) * (upper2 / lower2) ))
25 ## @brief Calculates the Probability of glass breakage
26 #  @return unitless probability of breakage
27 def calc_Pb( B ):
28     output = 1 - (exp(-B))
29     if not (0 < output < 1):
30         raise InvalidOutput("Invalid output!")
31     return (output)
32 ## @brief Calculates the Risk of failure
33 #  @return unitless risk of failure
34 def calc_B( J, params ):
35     upper = params.k * ((params.E * (params.h) ** 2) ** params.m) * params.LDF * exp(
       J)
36     lower = ((params.a * params.b) ** (params.m - 1))
37     return ( upper / lower )
38 ## @brief Calculates the Non-factores load
39 #  @return unitless non-factored load
40 def calc_NFL( qÌĆtol, params ):
41     upper = qÌĆtol * params.E * (params.h ** 4)
42     lower = (params.a * params.b) ** 2
43     return ( upper / lower )
44 ## @brief Calculates the Load resistance
45 #  @return unitless load resistance
46 def calc_LR( NFL, params ):
47     return ( NFL * params.GTF * params.LSF )
48 ## @brief Calculates Safetey constraint 1
49 #  @return true if the calculated probability is less than the tolerable probability
50 def calc_is_safePb( Pb, params ):
51     if (Pb < params.Pbtol):
52         return True
53     else:
54         return False
55 ## @brief Calculates Safetey constraint 2
56 #  @return true if the load resistance is greater than the load
57 def calc_is_safeLR( LR, q ):
58     if (LR > q):
59         return True
60     else:
61         return False
```

Figure 3.8: Source code of the Calc.py module for GlassBR

3.3.4 Wrapping Up

A close examination of each major software artifact reveals that, while their purposes and audiences differ, they are all fundamentally built upon the same core knowledge: the system quantities and definitions, the relations among them, the requirements on acceptable behavior, and the decisions needed to realize those requirements. Redundancy is introduced both within and across these artifacts, often as a result of tailoring the same information for different contexts or audiences. Structural similarities emerge across artifacts, highlighting opportunities for systematic knowledge capture and reuse. At the same time, inconsistencies and omissions can easily arise when artifacts are maintained separately, underscoring the risks inherent in manual, disconnected processes. Understanding the intent, content, and audience of each artifact is therefore essential for identifying patterns of redundancy and for informing the design of tools that aim to improve consistency and reduce unnecessary repetition.

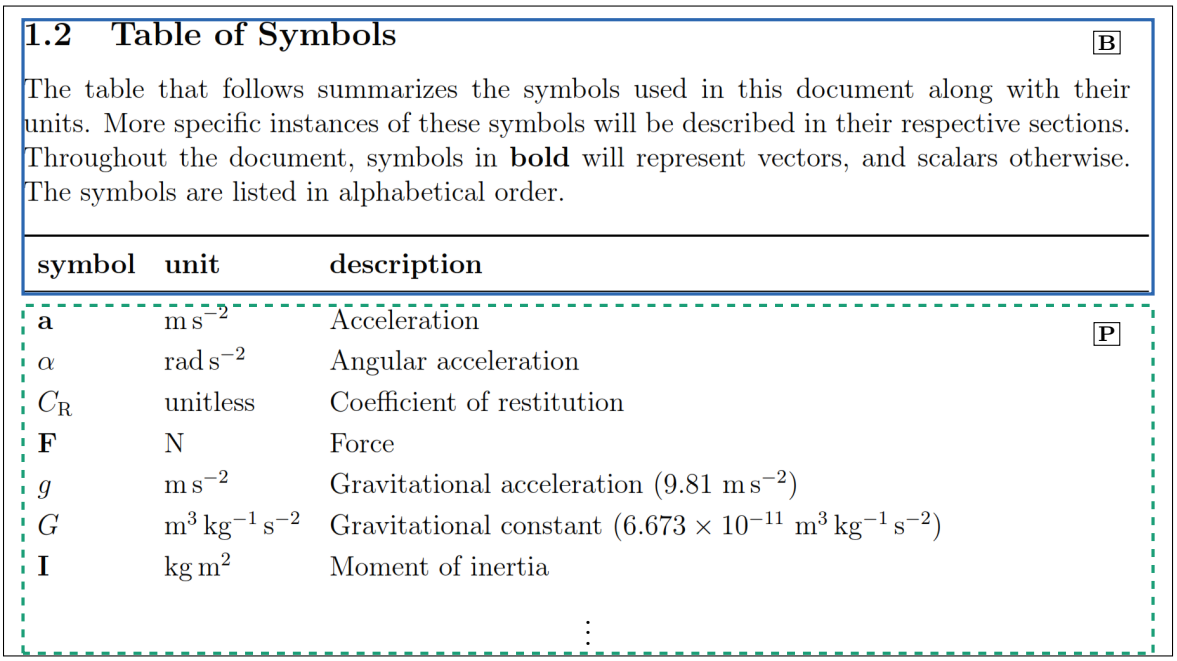
By systematically dissecting each artifact type, we clarify what information each artifact carries and how overlapping knowledge is represented across them. That decomposition sets up the more explicit discussion of recurring patterns of redundancy, and of why those patterns arise, in the following sections, while also clarifying the requirements for any framework (Drasil) that aims to automate or improve the generation of these artifacts.

3.4 Identifying Repetitive Redundancy

Having examined how the artifacts are structured, we can now turn to the recurring patterns of redundancy and information organization that appear within and across

the case studies. Some of these patterns follow directly from the templates used to construct particular softifacts, while others reflect repeated ways of organizing and reusing information within a single artifact. In the examples that follow, we make these patterns more concrete by examining recurring tabular structures, repeated boilerplate text, and repeated references to the same underlying system knowledge across sections and artifacts.

Returning to our example from Section 3.3.1, and in particular to Figures 3.2, 3.3, and 3.4, the reference section of the SRS template already shows one clear recurring pattern: its three subsections are organized in nearly the same way, each presenting a table of terms with short descriptive information. Figures 3.2 and 3.3 further show that the Table of Units and Table of Symbols include a short introductory blurb before the table itself, whereas Figure 3.4 shows that the Table of Abbreviations and Acronyms does not. Across case studies, the introduction to the Table of Units is pure boilerplate, repeated verbatim and applicable to *any* software system using SI units. The introduction to the Table of Symbols is similarly boilerplate, although it admits minor variations. This is easy to see by comparing the earlier GlassBR example in Figure 3.3 with Figure 3.9 (GamePhysics): GamePhysics explicitly states its vector notation convention, whereas GlassBR does not, because vectors do not arise in that case study. In both figures, 'B' highlights repeated boilerplate structure while 'P' highlights the problem-specific entries. Here, "problem-specific" intentionally bundles together domain-specific and system-specific content, since these figures support that coarser distinction much more clearly than a finer split. The later SWHS/NoPCM example will make that finer split more visible by marking the small number of explicitly system-specific additions. Even at this coarser level, the variability between



1.2 Table of Symbols

B

The table that follows summarizes the symbols used in this document along with their units. More specific instances of these symbols will be described in their respective sections. Throughout the document, symbols in **bold** will represent vectors, and scalars otherwise. The symbols are listed in alphabetical order.

symbol	unit	description
a	$\mathrm{m\,s^{-2}}$	Acceleration
α	$\mathrm{rad\,s^{-2}}$	Angular acceleration
C_{R}	unitless	Coefficient of restitution
F	N	Force
g	$\mathrm{m\,s^{-2}}$	Gravitational acceleration (9.81 $\mathrm{m\,s^{-2}}$)
G	$\mathrm{m^3\,kg^{-1}\,s^{-2}}$	Gravitational constant (6.673×10^{-11} $\mathrm{m^3\,kg^{-1}\,s^{-2}}$)
I	$\mathrm{kg\,m^2}$	Moment of inertia
⋮		

P

Figure 3.9: Table of Symbols (truncated) section from GamePhysics SRS

systems is greater than a mere change in symbol choice, confirming that genuinely problem-specific knowledge is being encoded within this repeated structure.

The reference section of the SRS conveys a substantial amount of knowledge in a very straightforward and organized manner. The basic units provided in the table of units give a prime example of fundamental, global knowledge shared across domains. Nearly any system involving physical quantities will use one or more of these units. On the other hand, the table of symbols provides problem-specific knowledge that will not be useful across unrelated domains. For example, the stress distribution factor J from GlassBR may appear in several related problems, but would be unlikely to be seen in something like SWHS, NoPCM, or Projectile. Finally, acronyms are very context-dependent. They are often specific to a given domain and, without a coinciding definition, it can be very difficult for even the target audience to understand what

Number	DD7
Label	**Dimensionless Load** ($\hat{q}$) ①
Equation	$\hat{q} = \frac{q(ab)^2}{Eh^4 \text{GTF}}$
Description	q is the 3 second equivalent pressure, as given in IM3 ② a, b are dimensions of the plate, where $(a > b)$ E is the modulus of elasticity h is the true thickness, which is based on the nominal thickness as shown in DD2 ③ GTF is the Glass Type Factor, as given by DD6 ④
Source	[2], [8, Eq. 7]
Ref. By	DD4 ⑤

Figure 3.10: Data Definition for Dimensionless Load ($\hat{q}$) from GlassBR SRS

they refer to. Within one domain, there may be several acronyms that look identical, but mean different things, for example: PM can refer to a Product Manager, Project Manager, Program Manager, Portfolio Manager, etc.

By continuing to breakdown the SRS and other softifacts, we are able to find many more patterns of knowledge repetition. For example, we see the same concept being introduced in multiple areas within a single artifact and across artifacts in a project. Figure 3.10 shows the data definition for $\hat{q}$ in GlassBR. The same term was previously defined with fewer details in the table of symbols, as well as showing up implicitly or in passing in the MG (Figure 3.6 and the *loadASTM* module respectively), and implemented in the Source Code (Figure 3.8 lines 11-14). A non-exhaustive list of places where $\hat{q}$ appears helps make the repetition more explicit. The numbered call-outs in Figure 3.10 highlight several of the direct references within the data definition itself:

- Callout 1 marks the definition of $\hat{q}$ itself within the data definition.
- Callout 2 marks the reference to IM3 in the description.
- Callout 3 marks the reference to DD2 in the description.
- Callout 4 marks the reference to DD6 in the description.
- Callout 5 marks the reverse reference from DD4 in the "Ref. By" row.
- Beyond this figure, $\hat{q}$ also appears in the SRS table of symbols, in other SRS data definitions such as the Stress Distribution Factor (J) and Non-Factored Load (NFL), in the SRS calculation of Capacity (LR) through intermediary quantities such as NFL, in the MG, and in the source code (Figure 3.8, lines 11-14).

Although the full definition of $\hat{q}$ is initially provided for a human audience only once, it is necessary to reference it in different ways for different audiences. Each audience is expected to grasp the symbol's meaning within their given context or consult other softifacts for more comprehensive understanding. When reading the SRS, the data definitions and other reference materials play a crucial role in swiftly comprehending the complete definition of $\hat{q}$ in relation to the system's inputs, outputs, functional requirements, and properties of a correct solution.

The MG, on the other hand, briefly mentions $\hat{q}$ when defining the responsibilities of both the *loadASTM* and *Calc* modules (the former being responsible for loading values from a file, and the latter utilizing those values for calculations), whereas the source code provides a highly detailed definition to ensure accurate execution of the relevant calculation(s).

The varying level of detail across the softifacts should not come as a surprise since each softifact targets a different audience and their specific needs at various stages of the software development process. Although the level of verbosity may differ, the core information remains consistent: the authors are consistently referring to the definition of $\hat{q}$ via its symbolic representation, regardless of the level of detail incorporated. The goal is to convey relevant aspects of knowledge of a given term, while eliding that which is deemed superfluous, based on the context and the specific requirements of our audience. In other words, the authors only *project* some portion of their knowledge of given terms at a given time, depending on their needs (precision, brevity, clarity, etc.), the expectations of the audience, and contextual relevance.[4] The audience, on the other hand, engages in *knowledge transformation*, whereby they consume the representation and transform it into their own internal representation, based on their personal knowledge-base.

Returning to the context of software systems, if we broaden our view from a single system, to a software family, we can also find patterns of commonality and repeated knowledge across the various softifacts of the family members (For example our SWHS and NoPCM case studies) as they have been developed to solve similar, or in our case nearly identical, problems. Software family members are good examples to help determine what types of information or knowledge provided in the softifacts belong to the system-domain, problem-domain, or are simply general (boilerplate).

Looking at SWHS and NoPCM, we can readily identify identical theoretical models (TMs), because the underlying theory is determined by the shared problem domain. For example, both systems use the principle of conservation of thermal energy,

[4]We have only referred to the term as $\hat{q}$ in this section to emphasize our argument and make a meta-argument that the definition is irrelevant to our audience in this example. What matters is the symbolic reference, which we share a common understanding of.

Number	T1
Label	**Conservation of thermal energy**
Equation	$-\nabla \cdot \mathbf{q} + g = \rho C \frac{\partial T}{\partial t}$
Description	The above equation gives the conservation of energy for transient heat transfer in a material of specific heat capacity C ($\mathrm{J\,kg^{-1}\,°C^{-1}}$) and density ρ ($\mathrm{kg\,m^{-3}}$), where $\mathbf{q}$ is the thermal flux vector ($\mathrm{W\,m^{-2}}$), g is the volumetric heat generation ($\mathrm{W\,m^{-3}}$), T is the temperature (°C), t is time (s), and ∇ is the gradient operator. For this equation to apply, other forms of energy, such as mechanical energy, are assumed to be negligible in the system (A1). In general, the material properties (ρ and C) depend on temperature.
Source	http://www.efunda.com/formulae/heat_transfer/conduction/overview_cond.cfm
Ref. By	GD2

Figure 3.11: Theoretical Model of conservation of thermal energy found in both the SWHS and NoPCM SRS

shown in Figure 3.11, which appears unchanged in both SRS documents. However, when we move from that shared theoretical model to the corresponding instance models (IMs), the equations change to reflect the system context. In this case, the absence of PCM changes the equations used to calculate the energy balance on water in the tank, as shown in Figure 3.12. At the same time, the strong visual similarity between the two IMs shows that most of the content is still shared. The only annotations we need are the 'S' markers on the SWHS model, which identify the few PCM-specific inputs and terms that disappear in NoPCM.

While these examples are fairly small and specific, they are indicative of a larger, more generalizable, set of patterns of knowledge organization and repetition. Across the case studies, we repeatedly see the same core knowledge organized in recurring tabular structures, introduced by reusable boilerplate, referenced symbolically across multiple sections, and re-expressed across artifacts such as the SRS, MG, and source

Number	IM1
Label	**Energy balance on water to find T_W**
Input	m_W, C_W, h_C, A_C, h_P, A_P [S], t_{final}, T_C, T_{init}, $T_P(t)$ from IM2 [S] The input is constrained so that $T_{\text{init}} \leq T_C$ (A11)
Output	$T_W(t)$, $0 \leq t \leq t_{\text{final}}$, such that $\frac{dT_W}{dt} = \frac{1}{\tau_W}[(T_C - T_W(t)) + \eta(T_P(t) - T_W(t))]$, [S] $T_W(0) = T_P(0) = T_{\text{init}}$ (A12) and $T_P(t)$ from IM2 [S]
Description	T_W is the water temperature (°C). T_P is the PCM temperature (°C). [S] T_C is the coil temperature (°C). $\tau_W = \frac{m_W C_W}{h_C A_C}$ is a constant (s). $\eta = \frac{h_P A_P}{h_C A_C}$ is a constant (dimensionless). [S] The above equation applies as long as the water is in liquid form, $0 < T_W < 100$°C, where 0°C and 100°C are the melting and boiling points of water, respectively (A14, A19).
Sources	[4]
Ref. By	IM2

(a) SWHS Instance Model for Energy Balance on Water

Number	IM1
Label	**Energy balance on water to find T_W**
Input	m_W, C_W, h_C, A_C, t_{final}, T_C, T_{init} The input is constrained so that $T_{\text{init}} \leq T_C$ (A9)
Output	$T_W(t)$, $0 \leq t \leq t_{\text{final}}$, such that $\frac{dT_W}{dt} = \frac{1}{\tau_W}(T_C - T_W(t))$, $T_W(0) = T_{\text{init}}$
Description	T_W is the water temperature (°C). T_C is the coil temperature (°C). $\tau_W = \frac{m_W C_W}{h_C A_C}$ is a constant (s). The above equation applies as long as the water is in liquid form, $0 < T_W < 100$°C, where 0°C and 100°C are the melting and boiling points of water, respectively (A10).
Sources	Original SRS with PCM removed
Ref. By	

(b) NoPCM Instance Model for Energy Balance on Water

Figure 3.12: Instance Model difference between SWHS and NoPCM

code. Across related systems, we also see shared domain knowledge retained while a smaller set of system-specific details varies, as in the SWHS/NoPCM comparison. In these examples, the shared knowledge may be common to softifacts as a whole, common to a problem domain, or specific to an individual system, and it may be projected either in full or in abbreviated form depending on the audience and purpose of the given softifact. Unnecessary redundancy arises when those repeated projections must be maintained separately: the same definition, equation, table entry, or boilerplate explanation must be updated in multiple places, increasing the risk of omissions, inconsistencies, and drift between artifacts. Improving maintainability therefore means reducing the number of independent edits required to keep these related projections consistent.

These observations are not peculiar to software artifacts alone. More broadly, they reflect a general feature of human communication: we routinely rely on common representations (shared symbols) and elided definitions rather than restating the full context every time. In that sense, what we have called knowledge projection and knowledge transformation are not ad hoc mechanisms invented to describe softifacts; they are a domain-specific instance of how people communicate more generally. This matters for Drasil because it suggests that the redundancy we observe across softifacts is not simply the result of poor writing or careless process. Rather, it arises in part because the same underlying knowledge must be projected differently for different audiences, contexts, and purposes, even in carefully structured artifacts.

If this is the case, then the central challenge is not merely to eliminate repetition, but to determine how that underlying knowledge should be captured and organized so that appropriate projections can be generated consistently. This is the problem to

which we now turn.

These patterns directly inform the requirements and architecture of Drasil. By understanding them, we reveal specific challenges and opportunities for automation and consistency in software artifact generation. Drasil must be able to capture shared definitions, relations, and explanations once, distinguish reusable boilerplate from domain-specific and system-specific knowledge, and project that knowledge consistently into different artifacts and sections. This directly targets the maintenance problems identified above: repeated manual updates, cross-artifact drift, and inconsistencies between related representations of the same knowledge. In this sense, Drasil is intended to operationalize these patterns by reducing unnecessary duplication while preserving the context-specific projections that different audiences require. The following sections build on this foundation.

3.5 Organizing Knowledge: A Fluid Approach

Recognizing patterns of redundancy and knowledge projection, we are left with a crucial question: how should knowledge itself be captured and organized to support consistent, flexible reuse across artifacts? While our analysis has revealed the importance of knowledge projections and the risks of redundancy, it remains unclear what form the underlying knowledge base should take, or how it should be structured to enable effective projection. In the following section, we turn to this open problem, exploring conceptual approaches to knowledge capture and organization that can lay the groundwork for Drasil's design.

To this point we have been referring to an intuitive, albeit nebulous concept of

knowledge without explicitly defining it. Because this concept is central to the remainder of this chapter and to the design of Drasil, we make our intended meaning explicit here. In this thesis, we propose the following working definition of knowledge: knowledge is information that has been structured and contextualized to be meaningful and useful. On this view, knowledge encompasses facts, concepts, definitions, theories, and relationships relevant to a particular domain or problem. It may be represented in various forms, such as mathematical equations, natural language descriptions, code snippets, diagrams, and more, while remaining the same underlying knowledge. This definition underlies our later discussion of knowledge chunks and projections in Chapter 4, especially Section 4.2.1: Capturing knowledge via chunks.

For clarity in the following examples, we will use set notation to describe relationships between knowledge and projections. This is not intended as a formalization, but rather as a familiar mathematical juxtaposition to help illustrate the concepts. Here, we use K to denote a set, or in our terminology a *chunk*, of one or more related pieces of information. Under the working definition above, each K may represent a single fact, a definition, or a tightly connected set of information relevant to a particular concept or artifact.

Given the patterns across softifacts we observed in the previous section, we can generalize knowledge projections to their projection functions. Identity projection functions directly repeat knowledge verbatim i.e. $p(K) \equiv K$ for some set of knowledge K. It should be noted in our case that as long as the projection used contains the full definition and context of a piece of knowledge, that projection is considered an identity projection **regardless of changes to notation** (ex. "$x = y$" vs "$y = x$") or language (ex. "$x = y$" vs "x is equal to y" vs "x est égal à y"). An identity

projection is providing the knowledge chunk in full. Non-identity projections require using representations that elide some details of the knowledge at hand to make it more palatable for the audience of the given softifact i.e. $p(K) \subset K$. Similarly, we can have multivariant projection functions (whether identity or not) which project knowledge from several places into one form, i.e. $p(K, L, M) \subseteq K \cup L \cup M$.

In all cases, we have some fully defined knowledge and then we apply a projection function to retrieve the necessary information for our softifact. We can postulate that organizing and grouping knowledge into some type of structure, with an assortment of projection functions (For example, to look up the full definition in different representations, pull out relevant portions for a given audience, or provide useful abstractions) will allow us to reduce the need for manual duplication and remove unnecessary redundancies, as anything we need to include in our softifacts can be retrieved from a given source with a given projection function.

Keeping in mind that our core knowledge is used across all softifacts via projections, we may naively choose to consolidate all knowledge into a single database. This naive approach works well enough for a limited set of examples, but it quickly becomes apparent (refer to the PM example from Section 3.4) that context is highly important and the sheer scope of knowledge to be organized may become unwieldy. After breaking down multiple case studies, we believe collecting knowledge chunks into categories based on their domain(s) is a more easily navigable and maintainable approach. This also allows us to keep some context information at a meta-level (ex. Physics knowledge would be categorized into a Physics knowledge-base). Then for any given system, we would likely only need to reference across a handful of contexts (knowledge-bases) relevant to the domain.

There is also knowledge fundamental to all softifacts—it is contextual to the domain of softifact writing itself. This kind of meta-knowledge would be useful to have readily available in its own knowledge-base. The same could be said for things like SI Unit definitions, while they only apply to measuring physical properties, we see some domains built off of physics that operate at a higher level of assumed understanding (ex. Chemistry abstracts some of the physics details, while being directly reliant on them).

Some of the knowledge used in our softifacts is derived from other, more fundamental, knowledge chunks. For example, when using SI units, we may choose to use a derived unit (newton, joule, radian, etc.) that is a better fit for the application domain of the system being documented. While we want to avoid unnecessary redundancy, we can argue that derived units are good candidates for acceptable redundancy. For example, if anywhere we use *Joules* we replace that with the definition ($J = \frac{kg \cdot m^2}{s^2}$) we then run into a problem of context and complexity. Generally, the audience for a given softifact will have an internal representation of context-specific knowledge, so even something as straightforward as changing the units from J to $\frac{kg \cdot m^2}{s^2}$ will put unnecessary load on said audience and force them to engage in more intensive knowledge transformations, while also potentially making the softifacts harder to parse for experts in the domain. In these cases, we want to use the derived knowledge in place of the core knowledge.

Being able to specify our level of abstraction through the progressive application of projection functions eludes to another necessary piece of knowledge organization: the projection functions themselves. As we project out core knowledge that we know of as otherwise commonly derived concepts (like the Joule example above), we should also

like to store them in some form. For example, derived units may end up in the same context as the SI Units, defined by specific projection functions applied to SI Units. Continuing the Joule example, we would be applying a projection function across core knowledge related to energy and specific SI Units, then calling that projection *Joule* and giving it a symbolic representation J that we can refer to later.

We want to take a fluid and practical approach to organizing knowledge, such that we can keep domain-related knowledge chunks together with useful derivations. We want to separate knowledge in unrelated domains, such that it is straightforward to look up whatever we need with relative ease. The specific implementation for organization will be detailed later (See Chapter 4.2).

In summary, this approach to organizing knowledge (grouping related information into domain-specific knowledge-bases and supporting flexible projection) lays the conceptual foundation for Drasil's framework. By making explicit how knowledge is structured and accessed, we address the root causes of redundancy and inconsistency identified earlier in this chapter. This organizational strategy not only clarifies the requirements for knowledge capture and reuse, but also directly informs the design principles and implementation choices discussed later in the thesis.

3.6 Summary: The Seeds of Drasil

This final section synthesizes the chapter's findings, highlighting the key insights that inform the design of Drasil. By summarizing our process and rationale, we prepare the reader for the transition from analysis to implementation in the following chapter.

Through this chapter, as part of our effort to reduce unnecessary redundancy across the software development process, we have taken an approach to breaking

down softifacts to the core knowledge they present and looked for commonalities in that knowledge between them. We use several case study systems that fit our scope (input → process → output) as examples to give a concrete, applied base to the work.

Taken together, the case studies show that the same information recurs in a small number of predictable ways: artifact templates reuse familiar tabular and boilerplate structures, key quantities are referenced repeatedly across sections and artifacts, and software family members preserve shared domain content while varying only limited system-specific details. The recurring material is the substance of the system itself—its quantities, relationships, behavioral expectations, and design decisions—but each softifact presents that material differently to suit a particular audience and purpose.

We delved into the idea of knowledge chunks and projection functions for producing context-relevant pieces of knowledge that are consumable by a given audience. We have also explored strategies for organizing that knowledge in a practical manner.

We have determined the three main components necessary for any useful softifact: knowledge, context (i.e. audience), and organizational structure. From here we can operationalize each component in a reusable and (relatively) redundancy-free manner. This operationalization informs the initial design for our framework Drasil, which will be covered in depth in Chapter 4.

Effectively, we want to automate the generation of softifacts through applying projections to knowledge and presenting it in a given structure. The structure of softifacts is relatively straightforward to deal with, we can use templates, blueprints, or deterministic generation which rely on relatively common technologies. The knowledge-capture and projection is much more interesting as it relies on some yet-to-be-determined

knowledge-capture mechanism that can provide us with chunks of knowledge that can then be fed to projection functions in some context-aware manner.

Chapter 4

Drasil

Drasil is a framework designed to address the challenges of generating consistent, traceable, and maintainable scientific software artifacts from a single source of domain knowledge. The name “Drasil" is derived from Norse Mythology’s world tree Yggdrasil whose branches spread across the many realms and reflects how our framework spans the many domains and contexts relevant to software generation. In this chapter, we provide a guided tour of Drasil’s architecture and implementation, focusing on how knowledge is captured, organized, and operationalized to produce both documentation and executable code.

We begin by motivating the need for Drasil and outlining its core principles. Next, we delve into the mechanisms for capturing and classifying knowledge, introducing the concept of "chunks" and their role in building a reusable knowledge base. We then describe how this knowledge is structured into recipes that specify the content and organization of generated artifacts. The chapter continues with a detailed look at Drasil’s generation and rendering pipeline, illustrating how abstract representations are transformed into concrete outputs. Finally, we discuss the iterative refinement

process that underpins Drasil's development and summarize the key takeaways for our approach to software generation.

Throughout, our aim is to show not just what Drasil does, but why each component is necessary and how the pieces fit together to support reliable, maintainable software.

4.1 Drasil in a Nutshell

This section provides a high-level overview of Drasil's motivation and approach, setting the stage for the technical details that follow. To understand why Drasil is needed, consider the challenges of manually writing and maintaining a full range of softifacts: the process is redundant, tedious, and often leads to divergent softifacts. Drasil is a purpose-built framework created to tackle these problems. It is designed to generate a wide range of softifacts from a single, unified source of domain knowledge. By capturing the essential concepts, relationships, and requirements of a software system as reusable, well-typed "chunks," Drasil enables the automatic production of consistent, traceable, and maintainable artifacts tailored to different stakeholders. The framework emphasizes reducing manual duplication, ensuring traceability, and supporting long-lived, well-understood domains where frequent maintenance or evolution is expected.

Unlike documentation generators like Doxygen, Javadoc, and Sphinx that take a code-centric view of the problem and rely on manual redundancy – i.e. natural-language explanations written as specially delimited comments that can then be weaved into API documentation alongside code – Drasil takes a knowledge-centric, redundancy-limiting, fully traceable, single-source approach to generating softifacts.

The framework builds up knowledge *as needed* to solve concrete problems rather than attempting to encode all possible information from the start, allowing it to be flexible and adaptable to different domains and contexts early in its development.

However, Drasil is not, nor is it intended to be, a panacea for all the woes of software development. Even the seemingly well-defined problems of unnecessary redundancy and manual duplication turn out to be large, many-headed beasts, which exist across a multitude of software domains; each with their own benefits, drawbacks, and challenges.

To reiterate: Drasil has not been designed as a silver-bullet. It is a specialized tool meant to be used in well-understood domains for software that will undergo frequent maintenance and/or changes. In deciding whether Drasil would be useful for developing software to tackle a given problem, we recommend identifying those projects that are long-lived (10+ years) with softifacts relevant to multiple stakeholders. For our purposes, as mentioned earlier, we have focused on SC software that follows the input $\rightarrow$ process $\rightarrow$ output pattern. Scientific computing often appears both stable and change-prone, and that distinction is important for understanding Drasil's intended scope. The underlying scientific knowledge is often well-understood[13] enough to be captured explicitly, even when the software built around it still evolves through maintenance, changing assumptions, revised constraints, new outputs, or iterative model refinement. In other words, Drasil is aimed at domains where the foundational theory is stable enough to encode, while the artifacts derived from that theory still benefit from frequent regeneration as the software changes. SC software has the benefit of being grounded in relatively stable theories, so even when models are revised or refined, older formulations are often still applicable under a set of assumptions or

within acceptable margins for error.

With Drasil being built around long-lived, well-understood scientific computing problems that follow an input → process → output structure and involve softifacts for multiple stakeholders, we remain aware that there are likely many in-built assumptions in its current state that could affect its applicability to other domains. As such, we consider expanding Drasil's reach as an avenue for future work.

The Drasil framework relies on a knowledge-centric approach to software specification and development. We attempt to codify the foundational theory behind the problems we are attempting to solve and operationalize it through the use of generative technologies. By doing so, we can reuse common knowledge across projects and maintain a single source of truth in our knowledge database.

Given how important knowledge is to Drasil, one might think we are building ontologies or ontology generators. We must make it clear this is *not* the case. We are not attempting to create a source for all knowledge and relationships inside a given field. We are merely using the information we have available to build up knowledge as needed to solve problems. Over time, this may take on the appearance of an ontology, but Drasil does not currently enforce any strict rules on how knowledge should be captured, outside of its type system and some best practice recommendations. We will explore knowledge capture in more depth in Section 4.2.

4.2 Our Ingredients: Organizing and Capturing Knowledge

Before we can generate useful software artifacts, we must first capture and organize the underlying domain knowledge in a form that is both reusable and operational. To illustrate what this entails, consider a concrete example from the domain of scientific computing: Newton's Second Law of motion [52]. To use this law in both documentation and code, we need to capture several distinct pieces of knowledge:

- The name (or label) for the law ("Newton's Second Law").
- A natural language definition ("the net force on a body at any instant of time is equal to the body's acceleration multiplied by its mass").
- The mathematical relationship ($F = ma$).
- The symbolic representation (in this case is identical to the equation).
- The definitions and symbols for each variable (F, m, a).
- The units associated with each quantity (newtons, kilograms, metres per second squared).

Capturing all of this information in a way that is reusable, extensible, and traceable is non-trivial. We need a knowledge capture method robust enough to represent not only the specific information needed for the software systems we are building, but also common knowledge relevant across the entire target domain of SC software.

To address this, Drasil introduces a set of mechanisms for representing and structuring knowledge, centered around the concept of "chunks" and the progressive refinement of knowledge structures. By understanding these building blocks, the reader will be equipped to follow how Drasil supports traceability and reuse throughout the software generation process.

Before we can design a knowledge capture mechanism, we must first define what exactly we believe we need to capture. It is nearly impossible to consider every case of knowledge that could be used in the domain of SC software, not to mention an extremely large undertaking to begin with "all the things". As such we have decided to take an iterative, progressive refinement approach to our knowledge capture mechanisms. This should come as no surprise and follows from our general process for developing the Drasil framework.

4.2.1 Capturing Knowledge via Chunks

Having identified some of the kinds of information we need to capture—names/labels, definitions, symbols, units, and equations (as illustrated by our Newton's Second Law example)—we now turn to the mechanism Drasil uses to represent this knowledge in a reusable and extensible way. To address these requirements, we borrow and re-purpose the *chunk* term from Literate Programming (LP), creating a simplified, extensible, and ever-expanding hierarchy of chunks based on the needs for capturing individual pieces of knowledge. A chunk hierarchy is the organization of chunk types from simple, general building blocks to more specialized ones that add additional structure or meaning. In Drasil, this means later chunk types are constructed by extending earlier ones and reusing the information they already encode. We use a

```
33 -- === DATA TYPES/INSTANCES === --
34 -- | Used for anything worth naming. Note that a 'NamedChunk' does not have an
       acronym/abbreviation
35 -- as that's a 'CommonIdea', which has its own representation. Contains
36 -- a 'UID' and a term that we can capitalize or pluralize ('NP').
37 --
38 -- Ex. Anything worth naming must start out somewhere. Before we can assign equations
39 -- and values and symbols to something like the arm of a pendulum, we must first give
       it a name.
40 data NamedChunk = NC {
41   _uu :: UID,
42   _np :: NP
43 }
```

Figure 4.1: NamedChunk Definition

hierarchy because many pieces of scientific knowledge share common properties, and factoring those shared properties into simpler chunk types reduces duplication while making it easier to build richer, more specialized representations as needed.

Our base chunk, `NamedChunk` is defined in Figure 4.1 and can be thought of as any uniquely identifiable term. It is composed of a `UID`, or unique identifier, and a `NP`, or noun phrase, representing the term.

A single node does not a hierarchy make, but with the root `NamedChunk` defined, we can begin to progressively extend it to cover any new chunk types we may need for our knowledge capture requirements. For example, if we want to capture a simple term with its definition, we need more than just a `NamedChunk`. We need to extend `NamedChunk` to include a definition, and we refer to this particular variant chunk as a `ConceptChunk`. For ease of creation, we define a number of semantic, so-called *smart constructors* for each chunk type which allow us to more simply define our chunks and their intermediary datatypes (ie: `UID` and `NP` from the `NamedChunk` example). An example of such smart constructors being used to create simple instances of `ConceptChunk` can be seen in Figure 4.2. Note there are smart constructors for each datatype when multiple variants exist and could be used in a given place. Looking

```
76 probability = dcc "probability" (cnIES "probability") "The likelihood of an event to
       occur"
77 rate = dcc "rate" (cn' "rate") "Ratio that compares two quantities having different
       units of measure"
78 rightHand = dcc "rightHand" (cn' "right-handed coordinate system")  "A coordinate
       system where the positive z-axis comes out of the screen."
79 shape = dcc "shape" (cn' "shape") "The outline of an area or figure"
80 surface = dcc "surface" (cn' "surface") "The outer or topmost boundary of an object"
81 unit_ = dcc "unit" (cn' "unit") "Identity element"
82 vector = dcc "vector" (cn' "vector") "Object with magnitude and direction"
```

Figure 4.2: Some example instances of `ConceptChunk` using the `dcc` smart constructor

at the `ConceptChunk` example, we can see two different smart constructors (`cnIES` and `cn'`) being used to create the `NP` instance. Both smart constructors create an instance of `NP`, but in this case the smart constructor used defines given properties (i.e. pluralization rules of the noun phrase used as a term) for the instance to simplify construction.

These simple chunks are fine as a starting point, however, knowing the SC domain we can already foresee the need for a variety of other chunks. For brevity, we will only expand upon the chunks needed for the examples used in this thesis. More information, including detailed definitions of all the types and current state of Drasil can be found in the (wiki) or the Haddock documentation which can be generated from the Drasil source code.

4.2.2 Chunk Combinatorics

The flexibility and extensibility of Drasil's knowledge capture comes not just from the variety of chunk types, but from the ways in which chunks can be constructed, referenced, and composed. In this section, we explore how extending our chunk hierarchy multiplies the possible combinations and relationships between chunks, and

how this enables us to build richer knowledge structures. This approach is essential for reducing duplication and supporting traceability, as it allows definitions and relationships to be constructed from reusable components. To facilitate this, we introduce a domain-specific language (DSL) for constructing and combining English-language sentences (`Sentence`), which lets us define terms and their relationships in a flexible, compositional way.

Even in the small progression we have already seen, the need for multiple chunk types is apparent. A `NamedChunk` is sufficient for naming a concept, but once we also want to capture its definition we must move to a more specialized `ConceptChunk`. The more we extend our chunk hierarchy to capture additional information such as symbols, units, and equations, the larger the number of possible combinations we will need to account for. This is not only relevant to chunks themselves, but also to the information they encode. For instance, rather than defining acceleration in isolation, we may want to define it as the rate of change of velocity, where velocity is itself already a captured term. In our effort to reduce duplication and maintain a single source of information, we therefore intend to create chunks with references to other chunks such that a definition can reuse already-captured terminology.

To define terms (such as acceleration) in relation to previously captured terms (such as velocity), we need a way to project existing knowledge into new definitions while retaining the flexibility to extend that knowledge. With that in mind, we created the `Sentence` DSL for creating and combining English- language sentences. In its most basic form, `Sentence` will simply wrap a string. However, by defining a number of helper functions and other useful utilities, our `Sentence` DSL can be used to combine, change case, pluralize, and more. A simple example of a series

```
acceleration = dccWDS "acceleration" (cn' "acceleration")
  (S "the rate of change of a body's" +:+ phrase velocity)
position = dcc "position" (cn' "position")
  "an object's location relative to a reference point"
velocity = dccWDS "velocity" (cnIES "velocity")
  (S "the rate of change of a body's" +:+ phrase position)
```

Figure 4.3: Projecting knowledge into a chunk's definition

of chunks that utilize the `Sentence` DSL to derive their definitions can be seen in Figure 4.3. We use the `phrase` helper function to pull the appropriate sentence out of the `velocity` chunk and the combinator (`+:+`) to concatenate the sentences. Note that `position` uses a different smart constructor (for non-derived definitions as it is not derived from existing chunks) and so doesn't require the sentence constructor (`S`) around its definition.

We also make opportunistic use of composable accessors (commonly known in the Haskell ecosystem as "lenses") to help manage complexity when combining and projecting chunks. Conceptually, these are lightweight, composable getters and setters that let us focus on the logical structure of a chunk (its name, symbol, sentence, expression, units, etc.) without repeatedly writing boilerplate navigation code for nested fields[1]. In practice they let us build derived chunks by projecting the subparts we need, update specific components (for instance a symbol or an attached unit) in a principled, immutable way, and compose transformations cleanly when constructing larger chunks from smaller ones. For example, starting from an existing concept for force, we can reuse its name and definition while attaching a symbol and unit to construct a more specialized chunk without rewriting the underlying knowledge.

As seen in our earlier case studies (see, for example, Figure 3.10), the types of

[1]Concrete lens usage and examples are available in the code found in the project repository for readers who want the implementation details.

```
91 force = dcc "force" (cn' "force")
92   "an interaction that tends to produce change in the motion of an object"
```

Figure 4.4: The force `ConceptChunk`

```
acceleration = uc CP.acceleration (vec lA) accelU
force = uc CP.force (vec cF) newton
```

Figure 4.5: The force and acceleration `UnitalChunk`s

knowledge we need to capture often go well beyond simple names and definitions. Those examples highlighted the need for symbolic representations, equations, units, and explicit relationships between concepts. While our motivating example of Newton's Second Law ($F = ma$) covers some of these aspects, it also reveals gaps in our current chunk definitions particularly in our ability to capture symbols, units, equations, and relationships between related quantities in a reusable, traceable way.

Building on those gaps, let us return to NewtonâĂŹs Second Law and examine what is required to achieve an operational definition of *Force* as a chunk in Drasil. To fully represent force, we need not only a name and definition, but also a symbolic representation (F), a mathematical equation ($F = ma$), and associated units (N). Likewise, mass and acceleration require their own symbols (m, a), definitions, and units (kg, m/s^2). This illustrates the necessity for chunk types that can encode all these facets in a reusable and traceable manner.

We start by building a `ConceptChunk` for the Force concept as seen in Figure 4.4. It is built from a common noun ("force") and a definition. Next we need to capture the units of measurement for the force concept by defining what we have named a `UnitalChunk`. This definition for force can be seen in Figure 4.5. Note that we are not re-defining force in this instance. We are instead using a smart constructor to

```
21 -- * Fundamental SI Units
22
23 metre, kilogram, second, kelvin, mole, ampere, candela :: UnitDefn
24 metre    = fund "metre"    "length"               "m"
25 kilogram = fund "kilogram" "mass"                 "kg"
26 second   = fund "second"   "time"                 "s"
27 kelvin   = fund "kelvin"   "temperature"          "K"
28 mole     = fund "mole"     "amount of substance"  "mol"
29 ampere   = fund "ampere"   "electric current"     "A"
30 candela  = fund "candela"  "luminous intensity"   "cd"
```

Figure 4.6: The fundamental SI Units as Captured in Drasil

```
11 accelU          = newUnit "acceleration"          $ metre /: s_2
12 angVelU         = newUnit "angular velocity"      $ radian /: second
13 angAccelU       = newUnit "angular acceleration" $ radian /: s_2
14 forcePerMeterU  = newUnit "force per meter"       $ newton /: metre
15 impulseU        = newUnit "impulse"               $ newton *: second
16 momtInertU      = newUnit "moment of inertia"     $ kilogram *: m_2
17 momentOfForceU  = newUnit "moment of force"       $ newton *: metre
18 springConstU    = newUnit "spring constant"       $ newton /: metre
19 torqueU         = newUnit "torque"                $ newton *: metre
20 velU            = newUnit "velocity"              $ metre /: second
```

Figure 4.7: Examples of chunks for units derived from other SI Unit chunks

build atop the existing force chunk (`CP.force`) and adding a symbolic representation (`vec cF`, which is shorthand for the selected vector representation[2] of the capital letter F). We also must capture the units used to measure force (`newton`) which are defined in `Data.Drasil.SI_Units` and can be seen in Figure 4.6. The SI Units are captured using another type of chunk known as a `UnitDefn` which are built atop a `ConceptChunk` and `UnitSymbol`. The `UnitSymbol` is also a chunk that can be one of several other chunk types.[3]. Returning to the force `UnitalChunk`, the smart constructor `uc` we are using will assume we are working in the real number space. There are other smart constructors for other spaces, which are also captured using other chunks elsewhere in Drasil (see: `Language.Drasil.Space`).

[2]More on this in Section 4.3

[3]For brevity we are glossing over many chunk definitions and the differences between them as there are a large variety in use for even simple examples. They are covered in more depth in the full technical documentation found on our GitHub repository.

Now we approach the terms mass and acceleration in a similar manner. To clarify, for each physical quantity we wish to represent in Drasil (such as mass or acceleration), we follow a two-step process. First, we define a `ConceptChunk` for the term, which captures its name and a natural language definition. For example, the concept of acceleration is defined in relation to velocity and position, as shown earlier in Figure 4.3. Second, we create a `UnitalChunk` for each quantity, which extends the concept by associating it with a specific symbol (such as a for acceleration or m for mass) and its units of measurement (for example, metres per second squared for acceleration, or kilograms for mass).

This approach ensures that each quantity is not only described in words but is also fully specified for use in both documentation and code generation. For instance, acceleration's units are derived from other SI units, and this relationship is captured explicitly in Drasil (see Figure 4.7). By systematically defining both the conceptual meaning and the quantitative properties of each term, Drasil enables consistent reuse and traceability throughout the generated artifacts.

Now we have almost everything we need to finish our definition of Newton's second law. Everything thus far has been defined in natural language using our `Sentence` DSL, but it is not well-suited to the one piece we are currently missing: a way to define expressions relating chunks in a language agnostic (mathematics) representation.

4.2.3 Relating Chunks via Expressions

Continuing our Newton's Second Law example from Section 4.2.2, we need a means to capture how force is calculated. We know it is calculated relative to mass and acceleration, so we need a way to encode that, preferably in an operational manner.

```
194  -- | Multiply two expressions (Real numbers).
195  mulRe l (Lit (Dbl 1))      = l
196  mulRe (Lit (Dbl 1)) r      = r
197  mulRe l (Lit (ExactDbl 1)) = l
198  mulRe (Lit (ExactDbl 1)) r = r
199  mulRe (AssocA MulRe l) (AssocA MulRe r) = AssocA MulRe (l ++ r)
200  mulRe (AssocA MulRe l) r = AssocA MulRe (l ++ [r])
201  mulRe l (AssocA MulRe r) = AssocA MulRe (l : r)
202  mulRe l r = AssocA MulRe [l, r]
```

Figure 4.8: Defining real number multiplication in `Expr`

We also know acceleration is itself derived from time and velocity, which is derived from time and position.

When thinking about the types of information we would like to encode with expressions, we have some obvious candidates. We should be able to encode common arithmetic operations (addition, subtraction, multiplication, division, exponentiation, etc.), boolean operations (and, or, not, etc.), comparisons (equal, not equal, less than, greater than), trigonometric functions (sin, tan, cos, arctan, etc.), calculus (derivation, integration, etc.), and vector/matrix operations (dot product, cross product, etc.). For the sake of our example we'll focus on only those we need to define Newton's Second Law of motion: multiplication and derivatives.

As an aside, we will elide the full implementation history of the expression DSL known as `Expr` and instead focus on the design pressures that shaped it. We needed a representation that was language-agnostic, compositional, and expressive enough to capture the recurring mathematical relationships seen across our case studies, while remaining simple enough to support projection into multiple generated artifacts. As such, `Expr` was designed by identifying the common operations required in practice and factoring them into a reusable core rather than attempting to encode all of mathematics at once. With that in mind, we have encoded multiplication (for real numbers)

```
newtonSLEqn                = sy QPP.mass `mulRe` sy QP.acceleration

velocityEqn, accelerationEqn :: ModelExpr
velocityEqn                = deriv (sy QP.position) QP.time
accelerationEqn            = deriv (sy QP.velocity) QP.time
```

Figure 4.9: Encoding Newton's second law of motion

as seen in Figure 4.8. `AssocA` refers to an associative arithmetic operator which can be applied across a list of expressions (in this case, `mulRe`, though addition and other associative operations look very similar). Derivation is much simpler to encode, as it is simply a relationship between two existing chunks, i.e. `deriv a b` represents taking the derivative of `a` with respect to `b`. This allows us to display derivatives correctly as mathematical notation, but it does not mean `Expr` itself captures the semantics of differentiation beyond that structural relationship.

Continuing our example, we see (Figure 4.9) it is fairly trivial to encode the expression for Newton's Second Law using the `Expr` DSL. We are still not done building our Newton's Second Law chunk however, as this specific law should be referable by its own identifier. It is not simply "force", nor the relationship between mass and acceleration, but a combination of both with its own natural language semantics that allow us to refer to it specifically. The full definition for this particular chunk can then be found in Figure 4.10. The chunk is defined relative to force and the expression captured in Figure 4.9, alongside a new identifier and description. Notice we are building off of other chunks throughout each piece of the chunk's definition, thus giving us perfect traceability from beginning to end, regardless of whether we are looking at a defining expression or natural language description.

The `Expr` DSL grants us flexibility in defining relationships without imposing

```
newtonSLQD :: ModelQDef
newtonSLQD = fromEqn' "force" (nounPhraseSP "Newton's second law of motion")
  newtonSLDesc (eqSymb QP.force) Real newtonSLEqn

newtonSLDesc :: Sentence
newtonSLDesc = foldlSent [S "The net", getTandS QP.force, S "on a",
  phrase body `S.is` S "proportional to", getTandS QP.acceleration `S.the_ofThe`
  phrase body `sC` S "where", ch QPP.mass, S "denotes", phrase QPP.mass `S.the_ofThe`
  phrase body, S "as the", phrase constant `S.of_` S "proportionality"]
```

Figure 4.10: Newton's second law of motion as a *Chunk*

any particular structure on a chunk other than "contains an `Expr`", or more specifically "can be modeled using an `Expr`." Our example also shows us that a few simple chunk types can be combined repeatedly to build much more robust, complex, and information-dense chunks. On the other hand, encoding even a relatively simple theory like Newton's Second Law of motion requires us to define chunks down to the fundamentals. Luckily, that is another area where chunks shine as they are infinitely reusable. Once a chunk has been defined and added to our knowledge-base, we never need to redefine it. We simply use it wherever it is needed.

On the topic of using chunks, we now need a means for taking our chunks and structuring them such that we can actually do something useful with them. For example, how would we go from chunks to softifacts? Look forward to that in Section 4.3.

4.2.4 Chunk Classification

In the previous sections, we used the idea of "chunks" of knowledge as our building blocks. They range in complexity from simple single terms, to complex relationships between expressions and overarching conceptual frameworks. Each chunk is defined by a collection of structural properties, or fields, such as a unique identifier,

name, symbol, unit, or space[4]. This “atomic" information is generally encoded in simpler, more fundamental chunks. Complex chunks are constructed by combining these simpler chunks into more elaborate structures, which we can think of in terms of “molecules” compared to the “atoms" of basic chunks. With the increasing complexity and variety of chunk types, it becomes important to classify chunks based on their specific contents and the kinds of knowledge they encode and can project. This classification system not only helps avoid repetition and ensures each chunk is uniquely suited to its purpose, but also clarifies which chunks are best suited to capture particular types of domain knowledge.

We have developed a classification system for chunks grouping them based on the properties we can project. For example, all chunks encoding things that are measured with units would be instances of the `Unitary` type[5]. Chunks that capture quantities, whether Unitary or not, would be instances of the `Quantity` type. As we simplify, we end up determining what qualifies something as a chunk in the first place. That is, what is the root property that *all* chunks must have to be considered a chunk? That is the `HasUID` classification, which essentially states that a given chunk of knowledge can be referred to by some unique identifier. For a more intuitive example, the `NamedIdea` classification is used for chunks that encode knowledge represented by given terms such as “force", “computer", “priority", and so on.

Figure 4.11 and Figure 4.12 show a representative subset of the different ways we classify chunks and some example chunk types and how they would be classified,

[4]Here, a space refers to the kind of values a quantity may inhabit, such as the real numbers, integers, booleans, or vectors, and it constrains the operations and representations that make sense for that quantity.

[5] It is important to note that these classifications are based on the ability to project certain information. In the codebase we often use lenses to access or manipulate these fields, as they may be projected from component chunks rather than being directly stored in the chunk itself.

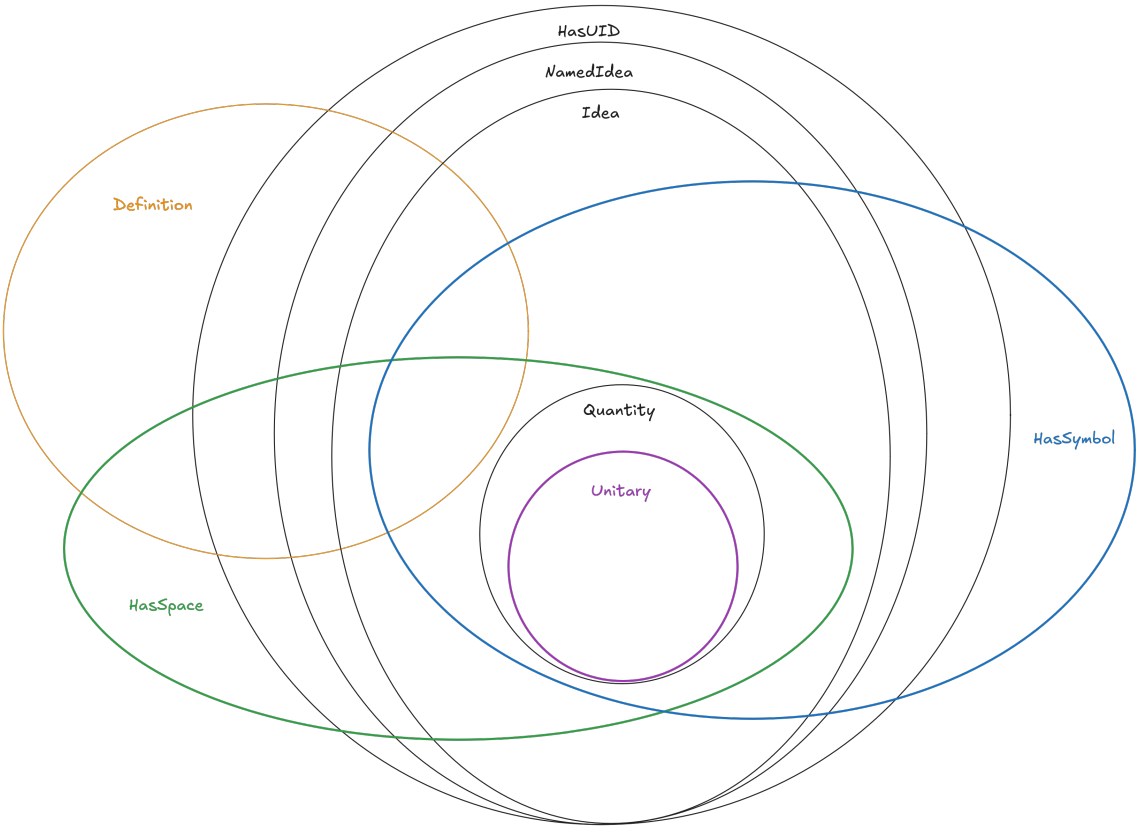


Figure 4.11: A representative subset of Drasil's Chunk Classification system showing some of the encoded property relationships

respectively. Note there is a one to many relationship between a given chunk and its classifications. As you can see, the `UnitalChunk` and `DefinedQuantityDict` are very closely related in that they both project their own `NamedIdea`, related `Space`, `Symbol`, and `Quantity`, however, the `UnitalChunk` is also classified as `Unitary` as it *must* have a unit associated with it.

We now have a compact, well-typed vocabulary for representing domain knowledge: chunks and classifications give us named concepts, quantities, units, and relationships that are easy to reason about. That vocabulary is only useful if we can turn

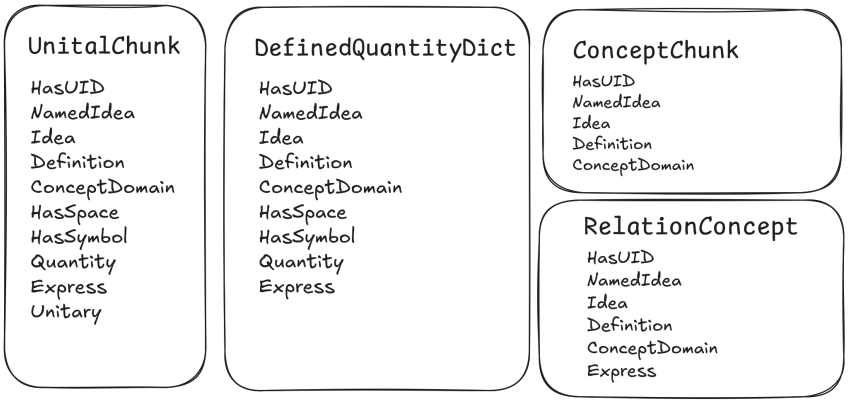


Figure 4.12: A representative subset of Drasil's chunks showing how chunks may belong to multiple classifications based on the underlying knowledge they project.

it into artifacts people consume. To convert that vocabulary into artifacts people actually use, we need a way to operationalize those chunks. In Drasil that means using recipes.

4.3 Recipes: Codifying Structure

Given our chunk classes and a compositional Expr/Sentence model in place, we have a knowledge core that can be selectively projected for any audience. The next step is thus operational: we must describe how those chunks are composed into artifact-specific recipes and how the recipes are rendered into concrete softifacts (documents, code, and supporting files).

Our knowledge base is essentially just a collection of chunks, which are themselves a collection of definitions, encoded in our Expr and Sentence DSLs. To generate softifacts we need a means to define the overall structure of the softifact as well as the knowledge projections we require to expose the appropriate knowledge from our

chunks in a meaningful way for the target audience of that particular softifact. This is where our *Recipe* DSL comes into play.

A recipe is, in it's simplest form, a specification for one or more softifacts. It is an intermediate representation of a given softifact and can be thought of as a "little program". With it, we select which knowledge is relevant and how it should be organized before passing the recipe to the generator/rendering engine to be consumed and produce the final softifact. A recipe not only allows us to structure where knowledge should be presented, but also gives us tools for automatically generating non-trivial sections of our softifacts. For example, a traceability matrix can be automatically generated by traversing the tree of relationships between chunks within a recipe. We'll discuss that in more depth in Section 4.4.

As we saw in Section 3.4, there are patterns of repetition for the way knowledge is projected and organized within and between our softifacts. The recipe DSL has been designed to take advantage of our knowledge capture mechanisms to reduce unnecessary repetition and duplication. We consider the softifacts as different views of the same knowledge, and project only that which is relevant to the audience of the specific softifact our recipe applies to. For example, a human readable document may use symbolic mathematics notation for representing formulae whereas the source code would represent the same formulae using syntax specific to the programming language in use (or, taking an extreme example the knowledge could be represented on punched cards). Regardless, we would define the underlying knowledge once in our chunks, and use the recipe DSL to select the representation.

As our case studies are following templates for our softifacts, they give us a solid foundation with which to write our recipes. We already know how the knowledge

```
25 -- | A Software Requirements Specification Declaration is made up of all necessary
       sections ('DocSection's).
26 type SRSDecl = [DocSection]
27
28 -- | Contains all the different sections needed for a full SRS ('SRSDecl').
29 data DocSection = TableOfContents        -- ^ Table of Contents
30   | RefSec DL.RefSec                     -- ^ Reference
31   | IntroSec DL.IntroSec                 -- ^ Introduction
32   | StkhldrSec DL.StkhldrSec             -- ^ Stakeholders
33   | GSDSec DL.GSDSec                     -- ^ General System  Description
34   | SSDSec SSDSec                        -- ^ Specific System Description
35   | ReqrmntSec ReqrmntSec                -- ^ Requirements
36   | LCsSec                               -- ^ Likely Changes
37   | UCsSec                               -- ^ Unlikely Changes
38   | TraceabilitySec DL.TraceabilitySec   -- ^ Traceability
39   | AuxConstntSec DL.AuxConstntSec       -- ^ Auxiliary Constants
40   | Bibliography                         -- ^ Bibliography
41   | AppndxSec DL.AppndxSec               -- ^ Appendix
42   | OffShelfSolnsSec DL.OffShelfSolnsSec -- ^ Off the Shelf Solutions
```

Figure 4.13: Recipe Language for SRS

should be organized, greatly simplifying how we can begin structuring our recipe DSL. The most straightforward, practical approach would be to start by defining a recipe for a given softifact type (SRS, MG, Code, etc.). Then flesh that out section by section, following the template as an organizational guide, using the patterns we discovered earlier (Section 3.4) as a means of avoiding unnecessary duplication. This is the approach we have chosen to follow, and it has lead to some interesting results.

As an example, let us take a look at the recipe language we have defined for an SRS using the Smith et al. template. Looking at the `DocSection` definition from Figure 4.13, we can see a striking resemblance to (a subset of) the template's Table of Contents (Figure 3.1). Each section is defined by a datatype, which is, through a similar mechanism, decomposed into multiple subsections (for example, Figure 4.14 shows the decomposition of the requirements section) which then defines the types of organizational structures we use for specifying the contents of a given subsection. When populated, the recipe specifies both the structure of our expected softifact and the knowledge (chunks) contained therein. While it may seem like it, this does not

```
88 -- | Requirements section (wraps 'ReqsSub' subsections).
89 newtype ReqrmntSec = ReqsProg [ReqsSub]
90
91 -- | Requirements subsections.
92 data ReqsSub where
93   -- | Functional requirements. 'LabelledContent' for tables (includes input values).
94   FReqsSub    :: Sentence -> [LabelledContent] -> ReqsSub
95   -- | Functional requirements. 'LabelledContent' for tables (no input values).
96   FReqsSub'   :: [LabelledContent] -> ReqsSub
97   -- | Non-Functional requirements.
98   NonFReqsSub :: ReqsSub
```

Figure 4.14: `ReqrmntSec` Decomposition and Definitions

collapse knowledge capture and presentation into a single concern: the knowledge remains encoded independently in reusable chunks, while the recipe specifies how that knowledge should be selected and organized for a particular artifact. For an SRS in particular, this is a natural fit, since the template is itself a systematic way of presenting scientific knowledge to stakeholders.

As the given recipe so far has provided nothing but structure, you may be wondering how we populate it with the appropriate knowledge chunks to produce a softifact? For that, we should look at a specific example from one of our case studies (GlassBR).

4.3.1 Example: SRS Recipe for GlassBR

The recipe for the GlassBR SRS can be seen in Figure 4.15. This should look familiar, as it follows a similar structure to the Smith et al. template table of contents (Figure 3.1). We use a number of helper functions and data types to condense and collect the chunks we require and avoid writing any generic boilerplate text. The structure of the document can be rearranged to an extent, in that we can add, remove, or reorder sections at will by simply moving them around within the `SRSDecl`. We can also make choices regarding the display of certain types of information in the final,

```
srs :: Document
srs = mkDoc mkSRS  (S.forGen titleize phrase) si
```

⋮

```
mkSRS = [TableOfContents,
  RefSec $ RefProg intro [TUnits, tsymb [TSPurpose, SymbOrder], TAandA],
  IntroSec $
    IntroProg (startIntro software blstRskInvWGlassSlab glassBR)
      (short glassBR)
    [IPurpose $ purpDoc glassBR Verbose,
     IScope scope,
     IChar [] (undIR ++ appStanddIR) [],
     IOrgSec orgOfDocIntro Doc.dataDefn (SRS.inModel [] []) orgOfDocIntroEnd],
  StkhldrSec $
    StkhldrProg
      [Client glassBR $ phraseNP (a_ company)
        +:+. S "named Entuitive" +:+ S "It is developed by Dr." +:+ S (name
    mCampidelli),
      Cstmr glassBR],
  GSDSec $ GSDProg [SysCntxt [sysCtxIntro, LlC sysCtxFig, sysCtxDesc, sysCtxList],
    UsrChars [userCharacteristicsIntro], SystCons [] [] ],
  SSDSec $
    SSDProg
      [SSDProblem $ PDProg prob [termsAndDesc]
        [ PhySysDesc glassBR physSystParts physSystFig []
        , Goals goalInputs],
       SSDSolChSpec $ SCSProg
        [ Assumptions
        , TMs [] (Label : stdFields)
        , GDs [] [] HideDerivation -- No Gen Defs for GlassBR
        , DDs [] ([Label, Symbol, Units] ++ stdFields) ShowDerivation
        , IMs [instModIntro] ([Label, Input, Output, InConstraints, OutConstraints]
    ++ stdFields) HideDerivation
        , Constraints auxSpecSent inputDataConstraints
        , CorrSolnPpties [probBr, stressDistFac] []
        ]
      ],
  ReqrmntSec $ ReqsProg [
    FReqsSub inReqDesc funcReqsTables,
    NonFReqsSub
  ],
  LCsSec,
  UCsSec,
  TraceabilitySec $ TraceabilityProg $ traceMatStandard si,
  AuxConstntSec $ AuxConsProg glassBR auxiliaryConstants,
  Bibliography,
  AppndxSec $ AppndxProg [appdxIntro, LlC demandVsSDFig, LlC dimlessloadVsARFig]]
```

Figure 4.15: SRS Recipe for GlassBR

```
si :: SystemInformation
si = SI {
  _sys         = glassBR,
  _kind        = Doc.srs,
  _authors     = [nikitha, spencerSmith],
  _purpose     = purpDoc glassBR Verbose,
  _quants      = symbolsForTable,
  _concepts    = [] :: [DefinedQuantityDict],
  _instModels  = iMods,
  _datadefs    = GB.dataDefs,
  _configFiles = configFp,
  _inputs      = inputs,
  _outputs     = outputs,
  _constraints = constrained,
  _constants   = constants,
  _sysinfodb   = symbMap,
  _usedinfodb  = usedDB,
   refdb       = refDB
}
```

Figure 4.16: System Information for GlassBR

generated softifact. For example, we alluded to vector representations earlier (Section 4.2.2) which we currently can display using either bold typeface or italics. This choice must be made using the `TypogConvention` datatype which lists the choices made (ex. `TypogConvention[Vector Bold]`) and is used by a relevant portion of the SRS recipe if vectors are included in the softifact. Our GlassBR example does not make use of this convention, however the GamePhysics case study does.

As for the chunks necessary for generating the softifact, each section includes references to them through supplied lists in something we call the `System Information` object (Figure 4.16).

The system information object uses helper functions to keep track of the lists of chunks necessary for filling in the sections of our softifact, the SRS. Each piece of the object consists of one or more chunks to be used in filling in the relevant sections. A quick description of each property can be found in Table 4.1, but it should be noted that some of the defined properties are only there for convenience (and/or remain only until the next refinement pass) as they are derived from other chunks with the

Table 4.1: System information object breakdown - every property is represented by chunks encoding given information

Property	**Chunk(s) Referenced**
_sys	System name (ex. "GlassBR").
_kind	Softifact type we are defining (ex. SRS).
_authors	List of authors of the softifact.
_purpose	Purpose of the softifact.
_quants	List of required quantities.
_concepts	List of required concepts not otherwise encoded.
_instModels	List of required instance models.
_datadefs	List of required data definitions.
_configFiles	List of required configuration files.
_inputs	List of required inputs to the system.
_outputs	List of required outputs from the system.
_constraints	List of constraints for the system (physical and software specific).
_constants	List of required constants.
_sysinfodb	Database of all required chunks.
_usedinfodb	Database of all used acronyms and symbols.
refdb	Database of all relevant external references/citations.

use of helper functions and could be inferred.

As should be obvious, the system information object is referencing other chunks which have been defined elsewhere. Most definitions can be found in the knowledge-base, with system-specific aggregations (i.e. lists of chunks) being defined within the scope of the specific example project. For example, `iMods` referenced as the `_instModels` property is defined as shown in Figure 4.17, alongside the two system-specific instance models listed within it. The two specific instance models can be seen to be derived from other existing chunks, through the use of a series of helper functions, thus ensuring their traceability to the knowledge-base.

Given that the system information object refers to all of the requisite chunks for the knowledge contained in our softifacts and the SRS recipe defines how that

```
iMods :: [InstanceModel]
iMods = [pbIsSafe, lrIsSafe]

⋮

pbIsSafe :: InstanceModel
pbIsSafe = imNoDeriv (equationalModelN (nounPhraseSP "Safety Req-Pb") pbIsSafeQD)
  [qwC probBr $ UpFrom (Exc, exactDbl 0), qwC pbTol $ UpFrom (Exc, exactDbl 0)]
  (qw isSafePb) []
  [dRef astm2009] "isSafePb"
  [pbIsSafeDesc, probBRRef, pbTolUsr]

⋮

lrIsSafe :: InstanceModel
lrIsSafe = imNoDeriv (equationalModelN (nounPhraseSP "Safety Req-LR") lrIsSafeQD)
  [qwC lRe $ UpFrom (Exc, exactDbl 0), qwC demand $ UpFrom (Exc, exactDbl 0)]
  (qw isSafeLR) []
  [dRef astm2009] "isSafeLR"
  [lrIsSafeDesc, capRef, qRef]
```

Figure 4.17: The `iMods` definition

knowledge should be structured and organized within the softifact, it follows that those are all we should need to generate our SRS. As an aside, the complete recipe for the SRS specification of GlassBR[6] including the system information object is contained in a single file in fewer than 370 lines of code.[7] Teasing our results: the generated SRS is a 44-page Portable Document Format (PDF) file (with the generated LaTeX source being 1,742 lines[8]). This is partially due to our ability to strip away all of the common boilerplate text that we need not worry about right now, as that will be a problem for the generator (Section 4.4). As it stands we have created the recipe language to represent what we truly want out of a recipe: a simplified, unnecessary-duplication-avoiding, fully traceable representation of our target softifact.

With the combination of system information and the SRS recipe we can move on to creating our generator for the finished SRS document. As the other softifact

[6]Not including chunks referenced from within our knowledge bases and core Drasil infrastructure, as they can be shared, reused, and are not necessarily specific to only this system.

[7]Including many comments, whitespace, and import statements.

[8]At time of writing.

recipes were developed in tandem with the SRS, they follow a similar structure and in the interest of brevity we will skip the breakdown of each recipe. It is left to the reader to investigate them by diving into our examples via the Drasil github. However, there is one softifact that is significantly different, and as such, we will go into more depth in discussing in the following section: the executable code recipe. Said recipe allows us to not only specify knowledge and the way we wish it structured, but also implementation choices that we would like to make in the final generated source code.

4.3.2 Example: Executable Code Recipe for GlassBR

As discussed in the previous section, the executable code recipe is fundamentally different from the documentation-heavy SRS recipe. While the SRS recipe focuses on organizing and projecting knowledge for human consumption, the executable code recipe must specify the structure and content required for generating working source code. This includes not only the relevant knowledge chunks, but also the implementation choices (modules, functions, and other details) necessary for code generation.

The executable-code recipe for GlassBR somewhat mirrors the document recipe in structure but emphasizes implementation choices: module layout, data representation, target languages, optional features, and constraint-handling strategies. Practically, we supply the modules and computational relationships (the mathematical execution order), target languages (ex. Python, C++, Java), modularity (single file or separated modules), data bundling (structs/classes), and logging/documentation options.

The recipe for GlassBR's executable code constructed using the `CodeSpec` DSL,

Body.hs

```
fullSI :: SystemInformation
fullSI = fillcdbSRS mkSRS si
```

Choices.hs

```
code :: CodeSpec
code = codeSpec fullSI choices allMods

choices :: Choices
choices = defaultChoices {
  lang = [Python, Cpp, CSharp, Java, Swift],
  architecture = makeArchit (Modular Separated) Program,
  dataInfo = makeData Bundled Inline Const,
  optFeats = makeOptFeats
    (makeDocConfig [CommentFunc, CommentClass, CommentMod] Quiet Hide)
    (makeLogConfig [LogVar, LogFunc] "log.txt")
    [SampleInput "../../datafiles/glassbr/sampleInput.txt", ReadME],
  srsConstraints = makeConstraints Exception Exception
}
```

Figure 4.18: The Recipe for GlassBR's Executable Code

is shown in Figure 4.18. Here, the `codeSpec` function takes three arguments:

- `fullSI`: the system information object, which aggregates all the knowledge chunks relevant to GlassBR (as described in the SRS example).
- `choices`: a record specifying code generation options, such as target languages, architecture, data handling, and auxiliary files. The example `choices` definition for GlassBR can be seen in Figure 4.18 and the object itself will be explained in more detail later in this section.
- `allMods`: a list of modules to be generated, each defined in terms of the knowledge chunks and functions they encapsulate.

The `CodeSpec` DSL itself is defined in Figure 4.19, and a subsection of its key fields are summarized in Table 4.2. As mentioned above, we use a helper function `codeSpec` to extract the appropriate fields from the system information, choices, and module list retaining full traceability and avoiding unnecessary manual duplication.

```
41 -- | Code specifications. Holds information needed to generate code.
42 data CodeSpec where
43   CodeSpec :: (HasName a) => {
44   -- | Program name.
45   pName :: Name,
46   -- | Authors.
47   authors :: [a],
48   -- | All inputs.
49   inputs :: [Input],
50   -- | Explicit inputs (values to be supplied by a file).
51   extInputs :: [Input],
52   -- | Derived inputs (each calculated from explicit inputs in a single step).
53   derivedInputs :: [Derived],
54   -- | All outputs.
55   outputs :: [Output],
56   -- | List of files that must be in same directory for running the executable.
57   configFiles :: [FilePath],
58   -- | Mathematical definitions, ordered so that they form a path from inputs to
59   -- outputs.
60   execOrder :: [Def],
61   -- | Map from 'UID's to constraints for all constrained chunks used in the problem.
62   cMap :: ConstraintCEMap,
63   -- | List of all constants used in the problem.
64   constants :: [Const],
65   -- | Map containing all constants used in the problem.
66   constMap :: ConstantMap,
67   -- | Additional modules required in the generated code, which Drasil cannot yet
68   -- automatically define.
69   mods :: [Mod],  -- medium hack
70   -- | The database of all chunks used in the problem.
71   sysinfodb :: ChunkDB
72   } -> CodeSpec
```

Figure 4.19: The Recipe Language for Executable Code (CodeSpec)

Table 4.2: CodeSpec object breakdown

Property	**Chunk(s) Referenced**
pName	Program name
authors	List of authors
inputs	All input variables
outputs	All output variables
configFiles	Required configuration files
execOrder	Mathematical definitions ordered such that they form a path from inputs to outputs
cMap	Constraints on variables
constants	Constants used in the program
mods	List of additional code modules to generate
sysinfodb	Database of all knowledge chunks for the given problem

```
readTableMod :: Mod
readTableMod = packmod "ReadTable"
  "Provides a function for reading glass ASTM data" [] [readTable]

readTable :: Func
readTable = funcData "read_table"
  "Reads glass ASTM data from a file with the given file name"
  [ singleLine (repeated [quantvar zVector]) ',',
    multiLine (repeated (map quantvar [xMatrix, yMatrix])) ','
  ]
```

Figure 4.20: The `readTableMod` module definition

As with the SRS recipe, the code recipe references knowledge chunks defined elsewhere. For example, the list of input variables is defined in `Unitals.hs` just as they were for the SRS. Similarly, the modules to be generated are defined in `ModuleDefs.hs` and make reference to other chunks (both system specific and generic) from our knowledge-base. Looking into the modules, we can see that the `readTableMod` module, for example, is defined as shown in Figure 4.20. This module provides a function for reading ASTM International glass data from a file, encapsulating both the knowledge of the data format and the implementation logic required for code generation.

While the encapsulation of knowledge in chunks is interesting, we have seen it before in the SRS recipe example. What is novel to the code recipe is the `Choices` object. As mentioned above, it contains the outcome of very important implementation-level choices that must be made to generate the executable code. The definition for the `Choices` object can be seen in Figure 4.21 and a breakdown of each property can be found in Table 4.3.

With our understanding of the `Choices` object, we can now look back at the GlassBR example (Figure 4.18) and understand what specific implementation choices have been made. First off, the `lang` property has been set so the generated code will be

```
data Choices = Choices {
  -- | Target languages.
  -- Choosing multiple means program will be generated in multiple languages.
  lang :: [Lang],
  -- | Architecture of the program, include modularity and implementation type
  architecture :: Architecture,
  -- | Data structure and represent
  dataInfo :: DataInfo,
  -- | Maps for 'Drasil concepts' to 'code concepts' or 'Space' to a 'CodeType
  maps :: Maps,
  -- | Setting for Softifacts that can be added to the program or left it out
  optFeats :: OptionalFeatures,
  -- | Constraint violation behaviour. Exception or Warning.
  srsConstraints :: Constraints,
  -- | List of external libraries what to utilize
  extLibs :: [ExtLib]
}
```

Figure 4.21: The `Choices` object definition

Table 4.3: Choices object breakdown

Property	
lang	List of target languages
architecture	Architecture of the generated code. Includes modularity (whether to split into modules or generate a single flat file) and implementation type (library to be consumed or standalone program)
dataInfo	Data structure and representation choices. (Ex. bundle inputs together into a class/struct, define constants inline, use the languages constant mechanism, etc.)
maps	Mapping of Drasil concepts and mathematical spaces to code concepts and types in the target language(s) (ex. One could map the concept of π with the language's built-in π and the $\mathbb{R}$ space to single/double precision floating point numbers)
optFeats	Choices for optional features including documentation (comments and verbosity), logging, and auxiliary files.
srsConstraints	Constraint violation behaviour. Can be used to specify whether to throw a warning or exception for physical/software constraints independently.
extLibs	List of external libraries to use (ex. for solving specific classes of mathematics problems). These libraries are external to Drasil and give the option for users to link to optimized, well-established libraries rather than reimplementing them from scratch in Drasil.

output in five different programming languages: Python, C++, C#, Java, and Swift. This is a good demonstration of Drasil's ability to target multiple platforms from a single knowledge base. The `architecture` is configured as modular, with input-related functions separated into their own modules (`Modular Separated`), and the implementation type is set to `Program`, meaning the generated code will be a complete, runnable application rather than just a library.

For data representation, the `dataInfo` field specifies that inputs should be bundled together (for example, as a class or struct), constants should be defined inline within the code, and the language's constant mechanism should be used where possible. The `optFeats` field enables comprehensive documentation by including function, class, and module-level comments, but sets the documentation verbosity to quiet and hides date fields in the generated comments. Logging is also enabled for both variable assignments and function calls, with all logs directed to a file named `log.txt`. Additionally, the auxiliary files generated include a sample input file (pointing to a real data file) and a `README`, supporting both usability and reproducibility.

Finally, the `srsConstraints` field is set so that any violation of software or physical constraints will result in an exception, ensuring that errors are caught and handled strictly during execution. This configuration ensures that the generated GlassBR code is well-documented, maintainable, and ready for use in a variety of environments.

By structuring the executable code recipe in this way, Drasil ensures that all generated code is fully traceable to the underlying knowledge base and any external libraries. Any change in the knowledge chunks or their relationships is automatically reflected in the generated code, just as it is in the documentation. This approach minimizes duplication, maximizes consistency, and enables reliable code generation

for scientific computing applications.

4.3.3 Operational Summary

Having established how Drasil captures, organizes, and operationalizes knowledge through both documentation and executable code recipes, we are now poised to explore the final step in Drasil's operation: turning these structured specifications into tangible softifacts. The next piece of the story will show how Drasil takes these recipes and, through a unified process, produces the diverse softifacts required by different stakeholders. In the following section, we delve into the mechanisms and philosophy behind Drasil's generation and rendering, showing how the framework brings together all the ingredients we've discussed to deliver consistent, traceable, and maintainable softifacts.

4.4 Cooking It All Up: Generation and Rendering

The previous sections have detailed how Drasil captures domain knowledge as chunks, organizes it via a robust hierarchy, and structures it into recipes that specify the desired artifacts. In this section, we focus on the final stages of the generation pipeline: *rendering* and *assembly*. These stages are responsible for transforming the intermediate representations (recipes and their referenced chunks) into tangible, concrete, consumable softifacts, such as documents and executable code.

4.4.1 Rendering: From Abstract Structure to Concrete Formats

At its core, Drasil treats recipes as "little programs". Rendering in Drasil refers to the process by which these abstract, language-agnostic structures defined in recipes are transformed into specific output formats suitable for end-users and stakeholders. This stage bridges the gap between the high-level, reusable knowledge representations and the diverse forms required for practical use (e.g., LaTeX, Hypertext Markup Language (HTML), PDF, or source code in various programming languages). Drasil achieves this through a set of dedicated *printers* and, for code, through the use of the Generic Object-Oriented Language (GOOL)[12, 16]. Each printer is designed to interpret the structured data produced by the recipe and knowledge projection stages and emit output in the desired format, handling both content and layout[9].

Document Rendering: Printers and Output Engines

For documentation artifacts (such as SRS or Module Guide), Drasil provides specialized printers for each supported format:

- `genTeX` in `Language.Drasil.TeX.Print` — renders the document as LaTeX source, suitable for PDF generation.
- `genHTML` in `Language.Drasil.HTML.Print` — produces HTML for web-based consumption.
- `genJSON` in `Language.Drasil.JSON.Print` — outputs a machine-readable JavaScript

[9] To some respect, depending on the desired output format. For instance, LaTeX handles the majority of the finished softifact's layout where Drasil generates the content

```
41 -- | Generates a LaTeX document.
42 genTeX :: L.Document -> PrintingInformation -> TP.Doc
43 genTeX doc@(L.Document _ _ toC _) sm =
44   runPrint (buildStd sm toC $ I.makeDocument sm $ L.checkToC doc) Text
```

Figure 4.22: `genTeX` function for LaTeX rendering

Object Notation (JSON) representation, supporting further processing or integration.

Each printer traverses the recipe structure, visiting each section and sub-section, and calls formatting routines appropriate to the target format. For example, a Data Definition chunk will be rendered as a LaTeX table row, an HTML table entry, or a JSON object, depending on the printer invoked.

Example: SRS to LaTeX

As a concrete illustration, consider the printing of the GlassBR SRS to LaTeX. The process begins with the invocation of the `genTeX` function, which recursively walks the document tree produced by the recipe, emitting LaTeX code for sections, equations, and tables. Figure 4.22 shows the entry point to this process.

Code Rendering: GOOL and Target Language Emission

For executable code artifacts, Drasil uses GOOL [12, 16, 44] as an intermediate layer. GOOL provides a collection of type-safe combinators and abstractions for representing object-oriented and procedural constructs in a language-agnostic manner. The code recipe (ex. a `CodeSpec`) is first translated into a GOOL abstract syntax tree (AST), which is then emitted as source code in one or more target languages. The design and implementation of GOOL itself are treated in the cited works; for our purposes,

the important point is that it provides an intermediate representation between the recipe and the emitted code.

In brief, the code emission process follows a straightforward translation process: The code recipe and its referenced chunks are traversed to construct a GOOL AST, representing classes, functions, variables, and control flow; The GOOL backend is invoked for each specified target language, translating the AST into concrete source code (ex. Python, C++, Java, C#); Auxiliary files (such as README, Doxygen config, and sample input files) are generated via dedicated routines.

Example: Emitting GlassBR Code in Python

For GlassBR, suppose the code recipe specifies Python as a target. The generation process calls `generateCode` in `Language.Drasil.Code.Imperative.Generator`, which constructs the GOOL AST and then emits Python code by invoking the Python backend. The resulting code includes all functions, data structures, and comments specified in the recipe and choices object, as well as any automatically generated traceability links (ex. comments referencing requirements or data definitions). Figure 4.23 shows the relevant code path.

4.4.2 Assembly: Producing the Final Softifact Set

Once rendering is complete, the generated document or code sections must be assembled into coherent artifacts ready for distribution and use. Assembly includes writing the primary artifacts to disk (e.g., `.tex`, `.html`, `.py`, and `.cpp` files), generating auxiliary files (e.g., Makefiles, Cascading Style Sheets (CSS), README files, and example input files), linking cross-references, and ensuring the integrity of tables, figures, and

```haskell
-- | Generates a package with the given 'DrasilState'. The passed
-- un-representation functions determine which target language the package will
-- be generated in.
generateCode :: (OOProg progRepr, PackageSym packRepr) => Lang ->
  (progRepr (Program progRepr) -> ProgData) -> (packRepr (Package packRepr) ->
  PackData) -> DrasilState -> IO ()
generateCode l unReprProg unReprPack g = do
  workingDir <- getCurrentDirectory
  createDirectoryIfMissing False (getDir l)
  setCurrentDirectory (getDir l)
  let (pckg, ds) = runState (genPackage unReprProg) g
      code = makeCode (progMods $ packProg $ unReprPack pckg)
        ([ad "designLog.txt" (ds ^. designLog) | not $ isEmpty $
          ds ^. designLog] ++ packAux (unReprPack pckg))
  createCodeFiles code
  setCurrentDirectory workingDir
```

Figure 4.23: generateCode and genPackage for code rendering

```haskell
-- document information and printing information. Then generates the document file.
prntDoc' :: String -> String -> Format -> Document -> PrintingInformation -> IO ()
prntDoc' dt' fn format body' sm = do
  createDirectoryIfMissing True dt'
  outh <- openFile (dt' ++ "/" ++ fn ++ getExt format) WriteMode
```

Figure 4.24: `prntDoc'` function for document output

code modules. The remainder of this section illustrates these steps in practice.

Example: SRS Assembly

After rendering the GlassBR SRS to LaTeX via the `genTeX` function seen in Figure 4.22, the `prntDoc'` function, Figure 4.24, writes the source file, while `prntMake`, Figure 4.25, generates a Makefile for automated PDF compilation. The resulting directory contains a fully cross-referenced, typeset-ready SRS and all supporting files.

```haskell
-- | Helper for writing the Makefile(s).
prntMake :: DocSpec -> IO ()
prntMake ds@(DocSpec (DC dt _) _) =
  do outh <- openFile (show dt ++ "/PDF/Makefile") WriteMode
     hPutStrLn outh $ render $ genMake [ds]
     hClose outh
```

Figure 4.25: `prntMake` function for document output

```haskell
-- | Creates the requested 'Code' by producing files.
createCodeFiles :: Code -> IO () -- [(FilePath, Doc)] -> IO ()
createCodeFiles (Code cs) = mapM_ createCodeFile cs

-- | Helper that uses pairs of 'Code' to create a file written with the given
    document at the given 'FilePath'.
createCodeFile :: (FilePath, Doc) -> IO ()
createCodeFile (path, code) = do
  createDirectoryIfMissing True (takeDirectory path)
  h <- openFile path WriteMode
  hPutStrLn h (render code)
  hClose h
```

Figure 4.26: createCodeFiles for code output

Example: Code Assembly

Once the code has been rendered through the GOOL pipeline into target-language text, the next stage is to assemble the resulting files into a coherent project structure. In this case, `createCodeFiles` (Figure 4.26) is called to write each generated source file and all auxiliary documentation to disk using consistent directory and naming conventions. For multi-language builds, this process is repeated for each target language, ensuring consistency across all output.

In essence, this stage performs for code what the SRS assembly stage does for documentation: it gathers the individually generated fragments and organizes them into a complete, ready-to-use product. The result comprises all files necessary for the generated system source code, build instructions, and supporting assets. They are generated and ready for compilation or integration into a development environment. Because these outputs are directly derived from the same knowledge base that informed the SRS, Drasil ensures that the implementation remains consistent with the documentation.

```
130 --DD7--
131
132 dimLLEq :: Expr
133 dimLLEq = mulRe (sy demand) (square (mulRe (sy plateLen) (sy plateWidth)))
134   $/ mulRe (mulRe (sy modElas) (sy minThick $^ exactDbl 4)) (sy gTF)
135
136 dimLLQD :: SimpleQDef
137 dimLLQD = mkQuantDef dimlessLoad dimLLEq
138
139 dimLL :: DataDefinition
140 dimLL = ddE dimLLQD [dRef astm2009, dRefInfo campidelli $ Equation [7]] Nothing "
        dimlessLoad"
141   [qRef, aGrtrThanB, stdVals [modElas], hRef, gtfRef]
```

Figure 4.27: Definition of $\hat{q}$ as a chunk in the GlassBR knowledge base

4.4.3 A GlassBR Example: From Chunk Definition to Final SRS Rendering

To illustrate the rendering and assembly process, we follow a single chunk through each step of generation in the GlassBR SRS. Earlier, we described how the SRS recipe for GlassBR is constructed by specifying the document structure and referencing the necessary knowledge chunks through the system information object (see Section 4.3.1). Now we observe the journey of the dimensionless load chunk ($\hat{q}$), from its definition in the knowledge base to its appearance as typeset-ready LaTeX formatted text in the generated SRS. The following subsection then traces the same chunk into the generated source code to show how the same captured knowledge supports both renderings.

Step 1: Knowledge Capture

The definition of $\hat{q}$ is captured in the Drasil knowledge base as a chunk (Figure 4.27). This chunk contains all necessary information: a unique identifier, symbol, natural language description, units, and the defining mathematical expression.

Step 2: Recipe Specification

In this step, the previously defined $\hat{q}$ chunk is referenced and included in the system information and recipe for the GlassBR SRS. Here, $\hat{q}$ is added to the list of Data Definitions (can be seen as `GB.dataDefs` in Figure 4.16), which will later be used to populate the relevant sections (Table of Symbols, Data Definitions, Traceability Matrix, etc.) of the document. This step is about specifying what knowledge is relevant and where it should appear, not about how it is rendered.

Step 3: Generation and Rendering

When the generator is invoked to produce the SRS (ex. as a LaTeX document), it traverses the recipe, visiting each section. For the Data Definitions section, it projects the full definition of $\hat{q}$, including its symbol, description, units, and defining equation. For the Table of Symbols, it projects only the symbol, a brief description, and the units. The rendering engine formats these projections according to the conventions of the target output.

During rendering, the SRS generator traverses the recipe and, for each chunk referenced, projects and formats it for the target output. For the Data Definitions section, it projects the full definition of $\hat{q}$, including its symbol, description, units, and defining equation (Figure 4.28). For the Table of Symbols, it projects only the symbol, a brief description, and the units. The rendering engine formats these projections according to the conventions of the target output (as a LaTeX table or section).

```
\vspace{\baselineskip}
\noindent
\begin{minipage}{\textwidth}
\begin{tabular}{>{\raggedright}p{0.13\textwidth}>{\raggedright\arraybackslash}p{0.82\
    textwidth}}
\toprule \textbf{Refname} & \textbf{DD:dimlessLoad}
\phantomsection
\label{DD:dimlessLoad}
\\ \midrule \\
Label & Dimensionless load

\\ \midrule \\
Symbol & $\hat{q}$

\\ \midrule \\
Units & Unitless

\\ \midrule \\
Equation & \begin{displaymath}
           \hat{q}=\frac{q \left(a b\right)^{2}}{E h^{4} \mathit{GTF}}
           \end{displaymath}
\\ \midrule \\
Description & \begin{symbDescription}
              \item{$\hat{q}$ is the dimensionless load (Unitless)}
              \item{$q$ is the applied load (demand) (${\text{Pa}}$)}
              \item{$a$ is the plate length (long dimension) (${\text{m}}$)}
              \item{$b$ is the plate width (short dimension) (${\text{m}}$)}
              \item{$E$ is the modulus of elasticity of glass (${\text{Pa}}$)}
              \item{$h$ is the minimum thickness (${\text{m}}$)}
              \item{$\mathit{GTF}$ is the glass type factor (Unitless)}
              \end{symbDescription}
\\ \midrule \\
Notes & $q$ is the 3 second duration equivalent pressure, as given in \hyperref[DD:
    calofDemand]{DD:calofDemand}.

        $a$ and $b$ are the dimensions of the plate, where ($a\geq{}b$).

        $E$ comes from \hyperref[assumpSV]{A:standardValues}.

        $h$ is defined in \hyperref[DD:minThick]{DD:minThick} and is based on the
    nominal thicknesses.

        $\mathit{GTF}$ is defined in \hyperref[DD:gTF]{DD:gTF}.

\\ \midrule \\
Source & \cite{astm2009} and \cite[(Eq. 7)]{campidelli}

\\ \midrule \\
RefBy & \hyperref[DD:stressDistFac]{DD:stressDistFac}

\\ \bottomrule
\end{tabular}
\end{minipage}
```

Figure 4.28: Raw generated LaTeXof the $\hat{q}$ Data Definition

Step 4: Assembly and Output

Finally, the rendered sections are assembled into the complete SRS artifact. Cross-references are automatically generated, so that, for example, references to $\hat{q}$ in Instance Models or Requirements link back to its definition in the Data Definitions and/or Table of Symbols. Auxiliary materials, such as the Table of Units and Table of Abbreviations, if required, are generated by traversing the relevant lists in the system information object, ensuring consistency and completeness. The completed document is output as a `.tex` file with $\hat{q}$ fully integrated and accessible alongside all other relevant auxiliary files (ex. `.bib` file for citations). A makefile is also generated to enable the user to easily create a human-readable PDF file from the LaTeX source and auxiliary files.

4.4.4 Multiple Renderings from a Single Source: From Chunk to Code

As a continuation of the previous sectionâĂŹs example, we now follow the dimensionless load $\hat{q}$ from its knowledge base definition through to its realization in the generated source code. This example concretely demonstrates how DrasilâĂŹs single-source-of-truth approach ensures that mathematical concepts are faithfully and consistently reflected in both documentation and code artifacts.

Referencing $\hat{q}$ in the Code Recipe

Recall that $\hat{q}$ was previously defined as a chunk containing its symbol, natural language description, units, and mathematical formula (see Figure 4.27). In the process

```
34 ## \brief Calculates dimensionless load
35 # \param inParams structure holding the input values
36 # \param q applied load (demand): 3 second duration equivalent pressure (Pa)
37 # \return dimensionless load
38 def func_q_hat(inParams, q):
39 
40     return q * (inParams.a * inParams.b) ** 2.0 / (7.17e10 * inParams.h ** 4.0 *
       inParams.GTF)
```

Figure 4.29: $\hat{q}$ as defined in the generated python code

of generating executable code, this chunk is referenced in the system information object (`fullSI`) and included in the `CodeSpec` recipe for GlassBR(Figure 4.18). The recipe dictates which variables and formulae are required in the target program, and provides implementation-level choices such as type mapping and function structure.

Rendering via GOOL

During code generation, Drasil translates the recipe and referenced chunks into a GOOL intermediate representation. The $\hat{q}$ chunk, for example, is mapped to a function definition within the GOOL abstract syntax tree (AST), preserving its formula and metadata. This intermediate form enables Drasil to emit code in multiple target languages without duplicating logic. As GOOL is not the subject of this thesis, we will elide the GOOL-specific details and continue down the code generation pipeline.

Generated Source Code

In the final step, the GOOL backend emits concrete code for $\hat{q}$ in each selected language. For example, in Python, $\hat{q}$ appears as a function computing its value from its dependencies(Figure 4.29) just as we've seen before in Figure 3.8. The generated code maintains traceability through naming conventions and comments drawn from the original chunk. The final python output can be seen in Figure 4.29.

Discussion

This example demonstrates how DrasilâĂŹs architecture ensures that any update to the definition or formula of $\hat{q}$ is automatically propagated to the generated code, just as it is to documentation artifacts. The use of a knowledge-centric pipeline, combined with language-agnostic intermediate representations, guarantees consistency and traceability throughout the software system. Changes need only be made once, and are reflected everywhere $\hat{q}$ appears, reducing maintenance cost and risk of error.

4.4.5 Rendering Summary

In summary, rendering and assembly are the final, crucial steps that operationalize Drasil's knowledge-centric approach and translate the abstract representations produced by our chunks and recipes into practical, high-quality softifacts. The examples from GlassBR demonstrate the practical effectiveness of this approach for both documentation and code.

The next section will discuss how iteration and refinement were used in the creation of Drasil, and how the framework enables rapid evolution of both knowledge and artifacts.

4.5 Seasoning to Taste: Iteration and Refinement

With the understanding of Drasil's structure and implementation, we can now discuss continuous improvement. The development of Drasil did not follow anything even remotely close to the waterfall model of software development. If anything its development began with a simple set of ideas that created a basis for refinement.

That is to say: it *barely* worked.

We approached the development knowing the scope was far too large to plan out all the details ahead of time (in any reasonable time-frame), and made conscious choices on compromises we were willing to accept to get a minimum viable framework and continuously improve upon it, similar to agile software development practices [1, 14, 23, 73]. As such, and as we have eluded to throughout this and the previous chapter, we took a very practical approach to development by starting with a known-good state (our case studies), generalizing as much as possible (as seen in Chapter 3) to distill what we needed to understand to generate software, and updating the framework as our assumptions were challenged or found wanting. Finding holes is a lot easier when you constantly step in them.

Many of our updates involved modifying implementation details (such as Chunk types, combinators, and adding entirely new concepts) when we started looking at the various case study domains and determining how we'd distill and capture the knowledge necessary to reproduce those case studies.

Throughout this process of iterative refinement, we inevitably found a number of places to improve upon the existing case studies and applied those changes to our known-good state. We found errors and undocumented assumptions in the case studies that caused us to rework and improve them for the future. Crucially, moving the case studies into Drasil made those mistakes trivial to locate and correct. A representative example is the SSP symbol inconsistency (Issue #348 on the GitHub issue tracker). The original SSP SRS used the pair S_i and P_i in one place and later replaced P_i with τ_i. This blurred the distinction between shear force and shear stress, since τ_i should denote shear stress rather than resistive shear force. τ_i was also omitted

from the Table of Symbols. To reiterate, that inconsistency existed in the original case study and was not a generator bug but an error in the source document we had accepted as our starting point. Once the SSP material was encoded as chunks in Drasil the fix was minimal: ensure we had encoded the correct definitions of S_i, P_i, and τ_i and used them in the appropriate equations. Then the correction automatically propagated to all generated artifacts (Table of Symbols, cross-references, and any code that referenced the symbol).

We also noticed a unit mismatch during the investigation of that issue, which sparked some confusion until we determined there was an unspecified assumption of 1m depth “into the page" for one of the equations. That assumption was later captured and encoded into the specific knowledge for that family of software systems as well. Overall the effort to fix both of these related issues was essentially a focused edit to a minuscule subset of the captured knowledge, not a widespread rewrite.

This pattern repeats across many issues: we would find a mismatch between a case study and our Drasil output, or some as-yet unencoded knowledge (ex. implicit assumptions) that made the generated artifacts inconsistent or unclear, update the chunk(s) that encoded the offending knowledge, and re-generate. That workflow was the dominant mode of refinement of the original case studies and their Drasil implementations. It kept fixes small, localized, and low cost while producing broad, immediate benefits in every artifact that depended on the corrected knowledge.

As Drasil encodes both semantics (definitions, units, relations) and presentation choices (how a chunk is projected), correcting semantics upstream was also used to eliminate downstream inconsistencies with, often, minimal engineering effort. One example of this is changing our Chunk hierarchy to include derived-unit chunks. After

we determined the need for `Unital` chunks which tied units to captured knowledge, it was obvious that the units themselves would need to be able to represent their relationships, particularly for those derived from standard SI units (ex. m/s^2). This involved the addition of new combinators for creating `UnitDefn` chunks (examples of some of these combinators have already been shown in Figure 4.7).

In short, we learned how to improve Drasil by trying to reproduce existing case studies, encoding what we needed as chunks, and then correcting or extending the chunks when mismatches were discovered. Fixes were either small edits to the knowledge base (cheap to apply, and automatically, consistently propagated to all generated outputs) or an extension of the way we chose to capture knowledge (new chunk types or combinators to encode previously un-encodable knowledge or relationships between chunks).

4.6 Summary

In this chapter we have described the architecture and implementation choices that make Drasil practical: a knowledge-centric pipeline built from reusable, typed "chunks", lightweight DSLs for expressions and natural language fragments, and a recipe language that codifies how knowledge is projected into concrete artifacts. The design is intentionally pragmatic: we capture what is needed to reproduce real case studies, provide well-typed building blocks for assembling that knowledge, and offer generators (printers and the GOOL back end) that render the same source into multiple, consistent outputs. The result is a single source of truth that spans documentation and executable code.

By separating the concerns of knowledge capture, structuring, rendering, and

assembly, Drasil ensures that softifacts remain maintainable, consistent, traceable, and free of unnecessary manual duplication from their high-level specification down to their concrete instances. All of which increases confidence in the resulting software systems.

The chunk hierarchy and its classification system give us the levers we need to balance expressivity and reuse. Simple chunks (NamedChunk, ConceptChunk) let us represent terminology and definitions; richer chunks (UnitalChunk, UnitDefn, DataDefn, InstanceModel) let us attach units, symbols, and executable expressions; and the Expr / Sentence DSLs give us language-agnostic ways to express mathematics and narrative. Recipes then select and arrange those chunks for particular stakeholders so that the same underlying knowledge can be rendered as an SRS, a module guide, or multi-language source code with full traceability between pieces.

The iterative approach to Drasil's development ensured modularized components that improve extensibility and maintainability of the system as a whole, as well as continuous refinement and expansion of the system's capabilities. A key practical lesson is that encoding case studies into Drasil exposes errors and implicit assumptions in the original artifacts but makes them trivial to fix. Fixes are usually local edits to the knowledge base and those edits automatically propagate through all generated artifacts. This demonstrates the principal payoff of the single-source, chunk-based approach: small, cheap changes at the source yield broad, consistent improvements everywhere.

Drasil is not a universal solution âĂŤ it is a tool targeted at domains where the models are well understood, long-lived, and shared across multiple stakeholders. Within that scope, Drasil shifts effort from repetitive editing of many artifacts to

careful modeling of domain knowledge. The framework we have presented allows that modeling to pay dividends immediately (consistent documents and code) and continuously (streamlined evolution through iteration and refinement).

Chapter 5

Results

In this chapter we discuss our observations following the reimplementation of our case studies using Drasil. Content here is primarily illustrative: concrete examples, issue identifiers, and sample artifacts that document how the single-source approach affected consistency, traceability, and reimplementation effort. Formal, controlled experiments are future work (see Chapter 7.3.8). For interpretation and implications of these observations, see Chapter 6.

5.1 Value in the Mundane

This section highlights routine, practical benefits that emerged from the reimplementations. Each subsection focuses on a specific, often-overlooked advantage — consistency, traceability, and reproducibility — and shows how these “mundane" improvements materially reduce maintenance effort and increase confidence in generated artifacts.

5.1.1 Consistency by Construction

Drasil's single-source knowledge representation produced artifacts that are consistent by construction. Small, localized edits to domain knowledge or recipes propagated deterministically to all generated outputs (documentation and code). During reimplementations, this property reduced the number of manual synchronization tasks normally required when maintaining separate artifacts. In practice, consistency by construction meant that small corrections such as renaming a quantity to use a single canonical identifier, fixing a typographical error in a definition or table entry, or revising the wording of an assumption were made once in the knowledge base and then appeared immediately and uniformly across the SRS, reference tables, and generated code after regeneration. Rather than checking each artifact individually, regeneration ensured that every occurrence reflected the same correction.

Additionally, all domain knowledge must be explicitly encoded, which prevents dangling references or undefined symbols as observed in a number of early passes over the original case study documentation (for example, issue #348 in the issue tracker which will be discussed in more detail in Section 5.2.1 and Section 6.1.3).

5.1.2 Traceability and Maintainability

Every generated element (formulae, symbols, tables, code identifiers) can be traced back to a named knowledge chunk or recipe. The generation pipeline produces traceability matrices and reference-material tables as part of normal output, enabling explicit provenance for items that are typically undocumented. This traceability materially reduced the cognitive effort required to locate the origin of a definition or relation during maintenance tasks and refactoring: it was trivial to identify the

source to edit rather than hunting through multiple artifacts.

For example, in the GlassBR case study (see issue #349), a unit conversion was embedded implicitly in the original documentation. With Drasil, it was straightforward to use the generated traceability information to identify the exact chunk from which the formula was produced. That made it clear where the conversion logic belonged: instead of leaving the division by 1000 implicit in the rendered equation, we made the conversion explicit in the source chunk itself. After regeneration, the corrected expression appeared consistently in the SRS and in the source code computations derived from that chunk, demonstrating more explicitly how traceability supports single-point maintenance.

5.1.3 Reproducibility and Determinism

Artifact generation in Drasil is fully deterministic: regenerating the same softifacts from the same knowledge base and recipe set produces byte-for-byte equivalent outputs when compared to the then-current stable versions. These outputs were confirmed using a diff tool that runs as part of an automated regression test suite.

Furthermore, repeatability checks executed during the case study reimplementations confirmed that regenerating softifacts after environment-consistent builds did not introduce nondeterministic differences, supporting the stronger notion of reproducibility discussed in Section 2.2.3. When the knowledge base and recipe set are preserved, the same softifacts can be regenerated exactly while keeping the assumptions and design decisions used to create them explicit. This means that Drasil contributes more than repeatable execution of an existing workflow: it preserves the structured knowledge needed to recreate documentation and code without depending on tacit

steps or manual reconstruction. In that sense, the reproducibility benefit observed in the case studies was not only that outputs could be regenerated unchanged, but that the knowledge required to regenerate them remained available, inspectable, and reusable.

5.2 Observations: Case Studies and Reimplementations

We reimplemented each of our case studies using Drasil and recorded measurable outcomes for generated softifacts and the knowledge source that produced them.

For the GlassBR case study, the Drasil implementation generated a multi-page SRS (44 pages in the final PDF) and produced working implementations in multiple target languages. In the repository snapshot used for this study, the GlassBR case study itself comprised 1,651 lines of code (LOC) of handwritten, example-specific Haskell built on top of approximately 44,269 lines of shared Drasil infrastructure, and generated 1,778 lines of SRS LaTeX, 4,461 lines of SRS HTML, and 7,625 lines of source code.

At the time of writing, the six case studies considered here were built on top of approximately 44,269 lines of shared Drasil infrastructure Haskell. All line counts reported below were measured from the repository snapshot current at the time of writing and may increase or decrease as the codebase and generated artifacts evolve.

Comparable results were observed for other reimplemented case studies: each system combined a modest-sized handwritten example layer with larger generated documentation artifacts and, where present in the stable snapshot, generated code.

Table 5.1: Snapshot statistics for Drasil case study reimplementations at the time of writing

Case Study	SRS Pages	Example-Specific Haskell LOC†	Generated SRS TeX LOC†	Generated SRS HTML LOC†	Generated Code LOC†‡
GlassBR	44	1651	1778	4461	7625
SWHS	50	2273	1949	6354	–
NoPCM	32	674	1209	3494	1760
SSP	84	3036	2993	9459	–
Projectile	31	1377	1244	3214	4768
GamePhysics	46	2029	1841	4767	–

† All line counts were measured by an automated script over the current repository snapshot at the time of writing and may increase or decrease as the codebase, generators, and emitted artifacts evolve.
‡ A dash indicates that no corresponding generated source tree was present in the checked-in stable snapshot for that case study.

See Table 5.1 for a summary. Across the six case studies, the handwritten example-specific layer totals 11,040 lines of Haskell, while the generated artifacts total 11,014 lines of SRS LaTeX, 31,749 lines of SRS HTML, and 14,153 lines of source code. Generated code for each system followed the same pattern: language-agnostic knowledge encoded as chunks and recipes was emitted to multiple target languages with consistent identifiers and traceability annotations.

Generated code and documentation remained consistent after edits to the knowledge base, with regeneration producing updated artifacts across all targets. For example, the GlassBR unit-conversion correction from issue #349 was made once in the source knowledge and then propagated to the SRS and generated code after regeneration, while the NoPCM variant was produced from the SWHS knowledge base by excluding PCM-related chunks rather than rewriting the specification. These examples are discussed further in Section 5.2.1, Section 5.2.2, and in the interpretive discussion in Section 6.1.2.

5.2.1 Quality and Bug Detectability

Because all artifacts are derived from the same knowledge, a defect in the knowledge base manifested consistently across every generated artifact. This property made defects more visible ("pervasive bugs"), which in turn aided discovery and correction. During reimplementations, categories of problems that were revealed included inconsistent symbol naming, implicit numeric constants, and unstated domain assumptions. Fixing those issues at the knowledge level eliminated their occurrences across documentation and code simultaneously.

An instructive, concrete example is issue #348 from the issue tracker. In the original SSP artifacts several closely related symbols (for shear stress and associated forces) were used inconsistently across sections and tables. When the SSP content was encoded as Drasil chunks these inconsistencies became immediately visible: the same semantic item was projected with conflicting identifiers, and cross-references did not align (see Sections 4.5 and 6.1.1 for more details). Correcting the underlying chunk in the knowledge base removed the inconsistencies from all generated artifacts (SRS and code) after regeneration. During the investigation of the same issue, it was also discovered that one of the terms had an implicit unit vector "into the page" that was not documented in the original artifacts. Encoding the unit vector explicitly in Drasil made that assumption visible, allowing it to be documented properly in the generated SRS and ensuring that the generated code reflected the correct physical interpretation.

A similar example arose in the GlassBR case study (see issue #349), where an implicit division by 1000 for unit conversion was embedded in the original documentation. Drasil's explicit chunk encoding surfaced this hidden assumption, enabling

a single-point correction that propagated to all generated artifacts. Similar issues surfaced throughout the reimplementations, including inconsistent symbol names, missing symbol definitions, implicit unit conversions, undocumented unit vectors, mismatched units across tables and equations, and unstated domain assumptions. Taken together, these recurring issue types reinforced that centralized and formalized knowledge capture makes such problems easier to detect, localize, and correct.

5.2.2 Design for Change and Software Families

The single-source approach enabled rapid generation of software-family variants. Where artifacts differed only by parameterization or small domain choices (for example, SWHS with PCM vs NoPCM variants; refer to Figure 3.12 for a brief reminder of the variations), encoding those choices in the knowledge base allowed whole-family updates by editing a small set of chunks. This capability materially reduced the effort to produce consistent variants of the same artifact family.

Table 5.2: Key differences between SWHS and NoPCM Drasil implementations

Chunk/Concept	**SWHS**	**NoPCM**	**Notes**
PCM properties	Yes	No	Only in SWHS
Latent heat	Yes	No	Only in SWHS
Sensible heat	Yes	Yes	Present in both
Energy balance	Yes	Yes	PCM terms omitted in NoPCM
PCM input parameters	Yes	No	Only in SWHS
PCM assumptions	Yes	No	Only in SWHS
Tank/water properties	Yes	Yes	Shared
Recipes	Modified	Modified	NoPCM recipe omits PCM-related content

For example, producing the NoPCM variant from SWHS required removing all reference to PCM by excluding the PCM-related chunks. The majority of the knowledge base and recipes were reused without change, demonstrating the efficiency of the single-source approach for software families. Table 5.2 summarizes the key differences between the SWHS and NoPCM Drasil implementations, highlighting which chunks and concepts were unique to each variant and which were shared.

5.3 Usability and Adoption

From the reimplementation work it became clear that Drasil's expressiveness comes with an onboarding cost. Contributors familiar with the codebase could encode and modify domain knowledge effectively; new users required a nontrivial ramp-up period of up to two weeks of mentoring to understand the chunk combinators and recipe patterns. Common challenges included understanding how to build new chunks, particularly while referencing other chunks, and how to find existing chunks of the appropriate type in the knowledge base.

Feedback collected during onboarding, along with the observations listed above, indicated that improved search tools and documentation would significantly reduce ramp-up time. Nevertheless, undergraduate contributors and short-term students were able to make meaningful contributions after initial mentoring.

5.4 Summary and Takeaways

The combined results from the case studies show that the primary benefits of the Drasil approach are increased consistency, strong provenance for generated artifacts,

and deterministic reproducibility. These benefits manifested in lower ongoing synchronization effort, higher visibility of domain assumptions and defects, and faster family-wide changes when those domain choices were encoded explicitly. The observations collected during the reimplementations supports these claims while also documenting the practical cost of initial knowledge capture and onboarding.

Chapter 6

Discussion

This chapter interprets the results reported in Chapter 5. We synthesise the practical benefits observed — automation, traceability, and rapid adaptation — evaluate costs such as onboarding and modelling effort, and outline limitations and directions for formal validation. We refer back to observations in Chapter 5 and cite illustrative examples.

6.1 Interpreting the Results

This section organises the principal outcomes from Chapter 5 into interpretation themes that mirror the observations: consistency, traceability, reproducibility, and the tradeoffs that follow from centralizing knowledge.

6.1.1 Consistency by Construction

Drasil projects a single knowledge base into diverse artifacts. The case studies demonstrate that local changes to domain knowledge yield system-wide corrections. Practically, this reduces the maintenance surface area: a single canonical edit replaces repeated, artifact-level fixes. The primary tradeoff is increased upfront modelling effort and the need for disciplined knowledge capture.

Disciplined knowledge capture also forces clarification of implicit assumptions that would otherwise lead to inconsistencies. For example, in the SSP case study (Issue #348, referenced in Sections 4.5 and 5.2.1), the original artifacts used the closely related symbols S_i, P_i, and τ_i inconsistently across equations and tables. In particular, later equations replaced P_i with τ_i, even though τ_i denotes shear stress rather than resistive shear force and was omitted from the Table of Symbols. Encoding the symbols as Drasil chunks while reimplementing the original manually written SSP case study in Drasil made the inconsistency explicit and enforced a single canonical name and identifier, eliminating the inconsistencies across all generated artifacts.

Similarly, as we are forced to encode all domain knowledge explicitly in Drasil, it is impossible to end up in a situation where we have undefined symbols or equations that reference missing quantities. A concrete example of this appeared in the same issue (#348): the symbol τ_i was used in equations in the original SRS, but did not appear in the table of symbols or have a formal definition. When encoding the knowledge in Drasil, we had to define these symbols explicitly as chunks, which ensured that they were properly documented in the generated SRS and correctly referenced in the generated code.

A further benefit of this approach is that several supporting sections, such as the

many tables in the Reference Materials section of the SRS as well as the traceability matrices, are generated automatically from the structure of the knowledge in use. These sections, which are often neglected or inconsistently maintained in traditional workflows (as per the issue mentioned above), are kept up to date “for free" as a consequence of Drasil’s single-source knowledge capture, further reinforcing consistency and reducing manual effort.

6.1.2 Traceability and Refactoring

Centralized knowledge makes provenance explicit: formulae, symbols, and identifiers trace back to named chunks or recipes. That provenance eases refactoring because changes localized to chunks produce immediate updates across downstream artifacts. The recipe DSL allows reorganizing output structures (documentation sections, code modules, etc.) without manual edits to each target artifact.

A key advantage of Drasil’s architecture is the separation of concerns between knowledge and presentation. The structure of generated artifacts can be modified (“on the fly") by changing recipes, without altering the underlying knowledge base. For example, generated code can be adapted to include features like logging by updating code generation recipes alone. This enhances maintainability, traceability, and adaptability of the generated artifacts.

Drasil also supports the systematic evolution of software families. By separating domain knowledge from artifact-specific recipes, Drasil enables controlled introduction of variation points. New family members can be generated by modifying recipes or selectively extending the knowledge base, rather than duplicating and manually editing artifacts. This reduces the risk of divergence and inconsistency among related

products.

A concrete example is seen in the development of the SWHS and NoPCM case studies (see Section 5.2.2 for related results). Both are members of the same software family, sharing core physical concepts and requirements but differing in specific details (i.e. SWHS included water and a phase-change material in the heating tank, while NoPCM only included water). Because the fundamental knowledge was already encoded for SWHS, creating NoPCM required only targeted removal of PCM-related chunks from the existing recipes. This reuse minimized duplication, ensured consistency across both projects, and allowed rapid adaptation of documentation and code for NoPCM. The shared knowledge base made it straightforward to propagate improvements or corrections on the commonalities between family members, further demonstrating the advantages of Drasil's approach for software families.

When adapting a requirements specification for a new variant, contributors can introduce new chunks or adjust parameters in the knowledge base, while reusing existing recipes to maintain structure and traceability. Alternatively, recipes can be specialized to produce tailored documentation or code modules for specific family members, all while referencing the same core knowledge. This design for change streamlines the creation and maintenance of software families, and supports rapid, reliable adaptation to evolving requirements.

6.1.3 Reproducibility, Provenance, and Error Visibility

Artifact generation was deterministic in our case studies, but the more important point is that Drasil makes reproducibility auditable. Each artifact is generated from a specific knowledge base together with a specific recipe, so the provenance of an output

includes not only what was generated, but which family member and configuration produced it. In the SWHS/NoPCM family, for example, the combination of the shared knowledge base and the recipe's choice to include or exclude PCM-related chunks determines whether a generated SRS, table, figure, or source file belongs to SWHS or NoPCM. That information is part of the artifact's generation path rather than tacit build knowledge.

This matters for reproducibility because reproducing an artifact requires more than rerunning an opaque workflow. If the relevant revision of the knowledge base and the corresponding recipe are preserved, then the exact family member that produced a given artifact can be identified and regenerated. In principle, version control can pin both inputs to a particular repository state, making it possible to reproduce the exact byte-for-byte output rather than merely a similar artifact produced by a related system. At the same time, defects in the knowledge base appear across all generated artifacts, which increases visibility and simplifies root-cause correction.

Recall Issue #348 from SSP mentioned in Section 5.2.1. To summarize, the issue was that some closely related symbols had their uses mixed up in several places in the original case study. In more detail, the SSP inconsistencies manifested in three concrete ways:

1. Multiple names and short identifiers for the same physical quantity across equations and tables.

2. Implicit or mismatched units appearing in different tables.

3. References and captions that used variant labels, breaking cross-references.

Encoding the SSP concepts as chunks forced a single canonical name, an explicit

unit declaration, and a consistent label for each concept that could not get switched to a different symbol.

Using Drasil, the remediation followed a small, local workflow: identify the semantic chunk representing the quantity, unify the identifier and unit within that chunk, and regenerate. The result was immediate and verifiable: the SRS, tables, and generated code all used the same identifier and unit, table cross-references resolved, and no further manual edits were necessary. This example illustrates two benefits of centralized knowledge: it made the inconsistent use of P_i and τ_i, the missing definition of τ_i, and the resulting unit mismatch immediately apparent, and it enabled concise, single-point fixes.

6.2 Human Factors and Usability

Experienced contributors adapted quickly to Drasil's patterns, but new users found the knowledge-capture DSL challenging. In practice, the difficulty was not usually understanding the domain concept itself, but understanding how to encode that concept in Drasil's abstractions. For example, a contributor may know they need to represent a quantity such as a heat transfer coefficient, but still need to determine which kind of chunk to create, what supporting knowledge it should reference, and where related chunks already exist in the knowledge base. This extra layer of representational work creates a real onboarding cost and helps explain why mentoring was needed during the reimplementations. The broader implication is that Drasil's current authoring model remains better suited to experienced contributors than to new users, which limits accessibility and broader adoption. Improved authoring support is therefore less a convenience than a prerequisite for scaling the approach.

6.3 Limitations and Directions for Validation

The current evidence is largely qualitative and anecdotal. Formal, controlled experiments are required to measure productivity gains, maintenance cost reduction, and error rates across larger teams. Until such studies are performed, claims about absolute gains should be framed as preliminary.

Future validation should include controlled studies measuring onboarding metrics (time to first meaningful edit, amount of mentoring required), maintenance effort, and error rates across teams of varying experience. Experimental designs could compare Drasil-based workflows to traditional manual approaches, quantifying differences in initial productivity, long-term maintenance, and defect reduction. These metrics will be essential for substantiating the qualitative benefits observed in this thesis. Feedback from such studies will also guide the development of improved authoring tools and onboarding resources, helping to lower the initial ramp-up cost and make Drasil accessible to a broader range of users.

Scalability concerns also merit attention: as the knowledge base grows, improved tools for discovery, efficient projection, and modularization will be essential to maintain developer productivity.

A related limitation lies in the use of chunks themselves as the primary representation for knowledge. Chunks work best when knowledge can be decomposed into explicit, named, and reusable units with relatively stable relationships. This makes them well suited to definitions, quantities, equations, units, and other structured domain concepts, but less natural for knowledge that is tacit, highly contextual, temporally evolving, or strongly dependent on complex system state. In such cases,

deciding how to factor the knowledge into chunks and how to represent the relationships between them can require substantial modelling effort, and some information may be awkward to encode using the current chunk hierarchy. Relatedly, many dependencies must still be specified explicitly rather than inferred automatically, which increases authoring burden and limits scalability as the knowledge base grows.

Drasil's current implementation is also limited primarily to scientific computing software that follows an input $\rightarrow$ process $\rightarrow$ output (IPO) paradigm. This constraint arises because Drasil currently works best in domains where the relevant knowledge is well-understood [13], structured, and amenable to explicit capture. This does not mean that such software has simple or fixed requirements: scientific computing problems often require substantial experimentation and iteration during model development and refinement. Rather, Drasil is well suited to these settings because that iterative work can still be grounded in domain knowledge, assumptions, definitions, and relations that can be made explicit and projected into multiple artifacts.

In contrast, domains with less structured or more dynamic requirements (such as interactive systems, event-driven applications, or software with complex stateful behaviors) present significant challenges for knowledge capture and automated artifact generation. The underlying information in these domains may be harder to formalize, stabilize, or reuse, limiting the effectiveness of Drasil's current approach. Extending Drasil to support these broader classes of software will therefore require advances in more flexible modeling, richer knowledge representation, and recipe design.

6.4 Lessons Learned

1. Many dangerous inconsistencies in legacy artifacts trace back to tacit assumptions and implicit knowledge during requirements capture; making those assumptions explicit in chunks forces clarification and improves both maintainability and traceability.

2. Drasil's architecture shifts the correctness burden from artifact authors to knowledge-base modelers: this trades repeated artifact-level maintenance for concentrated, knowledge-level review.

3. Rapid case-study reimplementation and ease of propagation suggest the approach can scale across domains given improved onboarding and validation, provided the domain knowledge can be effectively captured.

4. The recipe DSL enables flexible artifact structuring and adaptation without modifying the underlying knowledge, supporting maintainability and evolution.

5. Centralized knowledge capture increases error visibility by making defects pervasive across all generated artifacts, simplifying root-cause analysis and correction.

6. The single-source approach facilitates rapid generation of software-family variants by enabling controlled introduction of variation points in the knowledge base and recipes, reducing duplication and ensuring consistency across related products.

7. Improved authoring tools and onboarding processes are critical to lower the initial ramp-up cost and make Drasil accessible to a broader range of users.

8. Formal, controlled studies are needed to quantify productivity gains, maintenance cost reductions, and error rates to validate the qualitative observations reported here.

9. Deterministic, automated artifact generation supports scientific reproducibility by ensuring that repeated builds from the same knowledge base yield identical outputs, reducing the risk of accidental divergence and increasing confidence in results.

10. Centralized, structured knowledge enables the automatic generation and consistent maintenance of supporting materials (such as traceability matrices and reference tables) that are often neglected or inconsistently updated in traditional workflows.

11. The development and reporting of quantitative metrics (such as onboarding time, regeneration timings, and maintenance effort) are essential for guiding future improvements and validating the practical impact of the approach.

12. Drasil's extensible architecture allows it to adapt to new requirements and scientific advances by incorporating richer or more abstract domain knowledge, future-proofing the system as knowledge capture mechanisms evolve.

6.5 Summary and Takeaways

Drasil's single-source approach improves consistency, traceability, and reproducibility by centralizing domain knowledge and projecting it into all softifacts. Across the case studies, this reduced manual duplication, made errors easier to detect and correct,

and supported efficient adaptation to new project variants.

These benefits come with a clear tradeoff: contributors must invest more effort up front to encode knowledge explicitly and learn the modeling abstractions. The practical value of the approach therefore depends on whether those initial costs are offset by downstream gains in maintenance, reuse, and reliability.

The results in Chapter 5 suggest that this tradeoff is often favourable for scientific software, especially where softifacts share substantial domain knowledge. The next step is to validate that claim more rigorously through quantitative studies and provide improved authoring support.

Chapter 7

Conclusion

This thesis set out to address the persistent challenges of consistency, traceability, and reproducibility in scientific software and documentation. Maintainability is treated here as a consequence of those properties: once domain knowledge lives in a single place, changes can be propagated through regeneration rather than repeated manual edits. The introduction established the need for a principled, automated approach to knowledge capture and artifact generation, motivating the development and evaluation of Drasil.

The results chapter provided concrete evidence that Drasil's single-source architecture enforces consistency by construction, improves traceability through explicit provenance, and enhances reproducibility via deterministic artifact generation. Here, deterministic means that repeated runs from the same knowledge base and recipe set produce the same artifacts, unlike the stochastic variation associated with LLM-style generation. Case studies demonstrated that local changes to domain knowledge propagate automatically across all generated outputs, reducing manual effort and the

risk of inconsistencies. The results also highlighted practical benefits such as the automatic maintenance of supporting materials and the ease of creating new software family members by reusing and extending the knowledge base.

The discussion chapter interpreted these findings, emphasizing that Drasil's approach shifts the burden of correctness to the knowledge base, clarifies implicit assumptions, and enables rapid, reliable adaptation to new requirements. It also acknowledged tradeoffs, such as the increased upfront modeling and onboarding effort, and identified the need for improved tooling and formal validation.

The primary contribution of this thesis is the demonstration that a centralized, recipe-driven knowledge base can systematically address many of the recurring problems in scientific software engineering. By explicitly capturing domain knowledge and automating artifact generation, Drasil provides a scalable, extensible foundation for reliable and reproducible scientific software, while improving maintainability through explicit provenance, single-point maintenance, and software-family reuse. The lessons learned and directions for future work outlined here position Drasil as both a practical tool and a platform for further research in automated knowledge capture and software generation.

7.1 Summary of Main Contributions

This thesis contributes Drasil: a framework involving a centralized, recipe-driven knowledge base for generating software and documentation from a single source of domain knowledge. It operationalizes our understanding of software engineering principles—design for change, separation of concerns (specifically with respect to the underlying knowledge versus presentation), among others (as discussed in Sections 5.2.2

and Chapter 6)—into concrete mechanisms for capture, projection, and regeneration. By explicitly capturing domain knowledge and automating artifact generation, Drasil preserves consistency by construction, exposes provenance, and supports deterministic regeneration, as shown in Chapter 5. Those case studies also demonstrate that local changes propagate across generated outputs and that the knowledge base can be extended and reused to produce software-family variants and supporting materials with less manual effort. Taken together, these contributions offer a practical basis for reproducible scientific software.

7.2 Limitations and Open Questions

While Drasil demonstrates significant progress toward automating knowledge capture and artifact generation, several important limitations remain, many of which were examined more fully in Chapter 6. First, the evidence presented in this thesis is still primarily qualitative, so the benefits observed in the case studies should be understood as promising rather than definitively quantified (see Section 6.3). Formal validation is still needed to establish how well these results generalize across teams, projects, and scientific domains.

Second, Drasil currently works best in domains where the relevant knowledge is well-understood [13], structured, and amenable to explicit capture. In its present form, this largely means scientific computing software that fits an input $\rightarrow$ process $\rightarrow$ output paradigm. Whether the same knowledge-centric approach can be extended effectively to more dynamic, interactive, or less formalized systems remains an open question (see Section 6.3).

Finally, the work of building and maintaining the knowledge base still demands

substantial modelling effort and domain familiarity. As discussed in Section 6.2, this creates practical challenges for onboarding, tool usability, and long-term scalability (see also Section 6.3) as the body of captured knowledge grows. These limitations do not undermine the value of the approach, but they do define the current boundaries of its applicability and motivate the future directions outlined next.

7.3 Future Work

Building on the foundation established in this thesis, several avenues for future work are apparent. As Drasil continues to evolve, its ongoing development will be shaped by both practical experience and feedback from real-world adoption. The directions outlined below reflect the need to broaden Drasil's applicability, improve its usability, and deepen its integration into scientific software engineering workflows. The following subsections highlight key areas where further research and development can have the greatest impact.

7.3.1 Expanding the Knowledge Base and Knowledge Capture Mechanisms

As Drasil is adopted for new projects, the encoded chunk database will naturally expand. This gradual growth is essential for increasing the breadth and depth of reusable domain knowledge, which in turn enhances the quality and variety of generated artifacts. Each new project contributes additional chunks, models, and recipes, making it easier to support related domains and reducing the effort required for future knowledge capture. Over time, a richer knowledge base will enable Drasil to generate

more comprehensive documentation and software, facilitate cross-project reuse, and improve consistency across scientific computing applications.

Similarly, designing more flexible and extensible knowledge capture mechanisms is essential for broadening Drasil's applicability. Currently, the types of information encoded as chunks are fairly primitive, often limited to basic definitions, equations, and simple relationships. Future work should focus on advancing the sophistication of chunks to enable Drasil to generate higher-quality artifacts and address more nuanced requirements. Additionally, developing new methods and interfaces for knowledge capture (such as guided authoring tools, semi-automated chunk inference, and support for alternative modeling paradigms) will allow domain experts to contribute knowledge more efficiently and facilitate extensibility into new scientific fields.

7.3.2 Inference Improvements

A key direction for future work is improving Drasil's ability to infer knowledge chunks and their relationships directly from recipes and other high-level specifications. Currently, much of the chunk structure and dependency information must be defined explicitly, which can be tedious and error-prone as projects scale. For example, if a recipe references an equation and the participating quantities are already known, Drasil could infer a missing unit annotation for a derived chunk, or detect that the equation is dimensionally inconsistent and flag that to the author. Automating chunk inference would reduce the need for manual list definitions and make knowledge capture more efficient and less repetitive. Achieving this will require advances in both recipe design and the underlying inference passes. Progress in this area will streamline the authoring process, lower the barrier to entry for new users, and further enhance

Drasil's scalability and maintainability.

7.3.3 Typed Expression Language

Developing a robust typed expression language within Drasil is a promising direction for future work. Currently, expressions are represented in a way that limits the ability to perform deep sanity checks or enforce domain- specific constraints at the knowledge level. Introducing a typed expression language would enable more rigorous validation of mathematical models, formulae, and relationships before code or documentation is generated. This would help catch errors early, ensure consistency across artifacts, and support advanced features such as type-driven code generation and automated reasoning about units and dimensions. Ultimately, a typed expression language would increase the reliability and expressiveness of Drasil's knowledge base, making it easier to support complex scientific domains and reduce the risk of subtle errors in generated artifacts.

7.3.4 Usability Enhancements

A major direction for future work is raising the level at which users interact with Drasil. As discussed in Section 6.2, the current knowledge-capture process places substantial demands on contributors, who must understand not only the domain concepts they wish to encode, but also the internal chunk and recipe abstractions used to represent them. Future work should therefore focus on interfaces and workflows that allow users to work more directly with domain knowledge itself, while leaving more of the structural and representational bookkeeping to the system.

This could include guided authoring environments, stronger support for integrated

development environments (IDEs), and tools for discovering, reusing, and navigating existing knowledge in the database. The goal is not simply to make its knowledge-centric approach more accessible to domain experts who are not already deeply familiar with the framework. Progress in this area would help reduce onboarding costs, improve the maintainability of the knowledge base as it grows, and make the framework more practical for broader adoption.

7.3.5 Output Formats and Artifact Types

Expanding the range of output formats and artifact types is essential for making Drasil applicable to a wider array of audiences and use cases. Future work should focus on supporting additional programming languages (by expanding the existing GOOL backend) and document formats, such as docx or other widely used file types.

Jupyter notebook support has already been added since the period covered by this thesis, demonstrating that Drasil can be extended beyond its earlier set of generated artifacts.[1] Future work can build on that progress by introducing additional artifact types, such as journal papers, and by broadening support for other forms of technical communication. These enhancements will enable Drasil to serve the diverse needs of scientific computing practitioners, educators, and researchers, and will help bridge the gap between code, documentation, and knowledge dissemination.

7.3.6 Natural Language Variants

Supporting multiple natural language variants and context-sensitive artifact generation will significantly enhance Drasil's adaptability and accessibility. This capability

[1] Jupyter notebook generation was added by other members of the Drasil team after the main period of work discussed in this thesis.

is especially important for international and interdisciplinary teams, where documentation and software artifacts may need to be produced in different languages tailored to specific audiences. Future work should focus on developing mechanisms for encoding and selecting language variants within the knowledge base, as well as strategies for generating artifacts that reflect cultural and contextual differences. These improvements will broaden Drasil's reach and facilitate collaboration across diverse scientific communities.

7.3.7 Test Case Generation

Developing more robust automated test case generation is an important direction for future work, both for improving confidence in generated artifacts and for strengthening Drasil's support for verification activities. A useful first step toward this goal is a better understanding of testing artifacts themselves: which kinds of tests, relations, and supporting terminology should be captured and how they should be represented. Crawford's analysis of software testing terminology provides an initial foundation for this line of work by systematically cataloguing test approaches and exposing inconsistencies in how testing concepts are defined and related [18]. Building on that kind of groundwork, future work could explore how test knowledge should be captured in Drasil so that comprehensive and meaningful test cases can be generated from both domain knowledge and software requirements. Progress in this area would help verify artifact integrity and support the adoption of Drasil in safety-critical and high-assurance scientific computing projects.

7.3.8 Experiments and Validation

Building on the limitations discussed in Section 6.3, future work should carry out controlled empirical studies to quantify Drasil's effects on onboarding, maintenance, and defect detection across teams and domains. The goal is not only to substantiate the qualitative benefits reported in this thesis, but also to identify where the approach scales well and where additional tooling or modeling support is needed.

Such studies should compare Drasil-based workflows to traditional manual approaches and provide the quantitative evidence needed to assess the framework's practical benefits, limitations, and generalizability. In turn, their results can guide improvements to authoring support, onboarding, and scalability, while also strengthening the case for broader adoption within the scientific computing and software engineering communities.

7.3.9 Extending Beyond Input → Process → Output

Extending Drasil beyond the input → process → output paradigm represents a significant challenge and opportunity for future work. Many scientific and engineering applications involve interactive, event-driven, or stateful behaviors that do not fit neatly into the current modeling framework. Addressing these domains will require advances in flexible modeling, chunk representation, and recipe design, enabling the capture and projection of more complex knowledge structures. Progress in this direction will broaden Drasil's applicability to a wider range of software systems, including those with dynamic requirements and less formalized processes, and will help realize the vision of fully automated, knowledge-centric artifact generation across scientific computing.

As Drasil continues to develop, ongoing research and collaboration will be essential to realizing its full potential and advancing the practice of automated, knowledge-centric scientific software engineering.

7.4 Broader Impact and Personal Reflection

The development of Drasil represents a step toward more principled and automated scientific software engineering. More broadly, it suggests that knowledge-centric methods can play a larger role in how scientific software is documented, maintained, and communicated across communities of researchers, developers, and other stakeholders. In that sense, Drasil is not only a tool for generating artifacts, but also an example of how software engineering practice can be reorganized around explicit, reusable knowledge.

Drasil's evolution has been shaped by its iterative development process, where feedback from case studies and practical use continually informed refinement of both the knowledge base and framework itself. This iterative approach has enabled Drasil to adapt to new requirements, address unforeseen challenges, and incrementally improve its capabilities while maintaining validated, working case study software projects.

On a personal level, this work has highlighted the importance of interdisciplinary thinking and the value of bridging gaps between software engineering theory and practical implementation. The challenges encountered throughout the design and development of Drasil have reinforced my understanding of the need for flexibility, collaboration, and ongoing refinement. The experience has been both intellectually rewarding and formative, shaping my perspective on the future of scientific software.

7.5 Closing Remarks

This thesis has demonstrated the feasibility and benefits of a knowledge-centric approach to scientific software engineering. By operationalizing software engineering principles and automating the generation of software and documentation artifacts, Drasil offers a scalable solution to persistent challenges of consistency, traceability, and reproducibility, while improving maintainability through centralized knowledge and regeneration. The work presented here lays the groundwork for further advances in knowledge capture and automated artifact generation, and invites continued research and collaboration to realize the full potential of this approach.

As the scientific computing community evolves, tools like Drasil may play an increasingly important role in supporting reliable, reproducible, and adaptable software systems. The lessons learned from this work highlight the value of centralized knowledge, systematic artifact generation, and the importance of ongoing refinement. Continued investment in these principles will help shape the future of scientific software engineering, ensuring that software remains trustworthy, maintainable, and responsive to new challenges.

Appendix A

HGHC SRS

HGHC Software Requirements Specification

The following appendix contains the full HGHC SRS document.

SRS for h_g and h_c

Spencer Smith

October 21, 2014

Table of Units

Throughout this document SI (Système International d'Unités) is employed as the unit system. In addition to the basic units, several derived units are employed as described below. For each unit, the symbol is given followed by a description of the unit with the SI name in parentheses.

m	- for length (metre)
kg	- for mass (kilogram)
s	- for time (second)
K	- for temperature (kelvin)
^oC	- for temperature (centigrade)
J	- for energy (joule, J=$\frac{\text{kgm}^2}{\text{s}^2}$)
cal	- for energy (calorie, cal ≈ 4.2 $\frac{\text{kgm}^2}{\text{s}^2}$)
mol	- for amount of substance (mole)
W	- for power (watt, W=$\frac{\text{kgm}^2}{\text{s}^3}$)

Table of Symbols

The table that follows summarizes the symbols used in this document along with their units. The choice of symbols was made with the goal of being consistent with the nuclear physics literature and that used in the FP manual. The SI units are listed in brackets following the definition of the symbol.

h_c	- convective heat transfer coefficient between clad and coolant ($\frac{\text{kW}}{\text{m}^2\text{C}}$)
h_g	- effective heat transfer coefficient between clad and fuel surface ($\frac{\text{kW}}{\text{m}^2\text{C}}$)

1 Data Definitions

Number	**DD1**
Label	h_g
Units	$ML^0t^{-3}T^{-1}$
SI equivalent	$\frac{\text{kW}}{\text{m}^2\text{°C}}$
Equation	$h_g = \frac{2k_c h_p}{2k_c + \tau_c h_p}$
Description	h_g is the gap conductance τ_c is the clad thickness h_p is initial gap film conductance k_c is the clad conductivity NOTE: Equation taken from the code
Sources	source code

Number	**DD2**
Label	h_c
Units	$ML^0t^{-3}T^{-1}$
SI equivalent	$\frac{\text{kW}}{\text{m}^2\text{°C}}$
Equation	$h_c = \frac{2k_c h_b}{2k_c + \tau_c h_b}$
Description	h_c is the effective heat transfer coefficient between the clad and the coolant τ_c is the clad thickness h_b is initial coolant film conductance k_c is the clad conductivity NOTE: Equation taken from the code
Sources	source code

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